US 20250275973A1

(19) **United States**

(12) **Patent Application Publication**

MENZER et al.

(10) **Pub. No.: US 2025/0275973 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **USE OF ALC1 INHIBITORS AND SYNERGY WITH PARPI**

(71) Applicant: **EISBACH BIO GMBH**, Martinsried (DE)

(72) Inventors: **William M. MENZER**, Martinsried (DE); **Katharina SAHIRI**, Martinsried (DE); **Adrian SCHOMBURG**, Martinsried (DE); **Andreas LADURNER**, Martinsried (DE); **Peter SENNHENN**, Martinsried (DE)

(21) Appl. No.: **18/862,525**

(22) PCT Filed: **May 2, 2023**

(86) PCT No.: **PCT/EP2023/061584**

§ 371 (c)(1),
(2) Date: **Nov. 1, 2024**

(30) **Foreign Application Priority Data**

May 2, 2022 (EP) 22171217.7

Publication Classification

(51) **Int. Cl.**

A61K 31/517	(2006.01)
A61K 31/454	(2006.01)
A61K 31/502	(2006.01)
A61K 31/5025	(2006.01)
A61P 35/00	(2006.01)

(52) **U.S. Cl.**

CPC ***A61K 31/517*** (2013.01); ***A61K 31/454*** (2013.01); ***A61K 31/502*** (2013.01); ***A61K 31/5025*** (2013.01); ***A61P 35/00*** (2018.01)

(57) **ABSTRACT**

The present invention relates to the use of small molecule compounds that allosterically inhibit ALC1 (CHD1L) and which induce the trapping of PARP1, PARP2 and/or PARP3 on chromatin or at DNA damage sites for the treatment of proliferative diseases. Disruption of the chromatin remodeling forces of ALC1 through these agents enables a highly selective therapy for targeting the DNA damage functions of PARP enzymes in several proliferative diseases, notably BRCA-deficient cancers. Via inhibition of the enzymatic activity, the compounds engage the synthetic lethality between mutations in HRD pathways, including BRCA1/2 and ALC1. By trapping PARP enzymes, inhibitors of ALC1 potentiate the cancer cell killing properties of PARP inhibitors, enable therapeutic approaches where ALC1 is amplified as an oncogene, therapeutically make it possible to overcome PARP inhibitor resistance mechanisms and enable an alternative approach to the treatment of germline or acquired BRCA1/BRCA2 deficiency, including tumors defined by "BRCAness" or other changes in DNA repair networks.

Figure 1

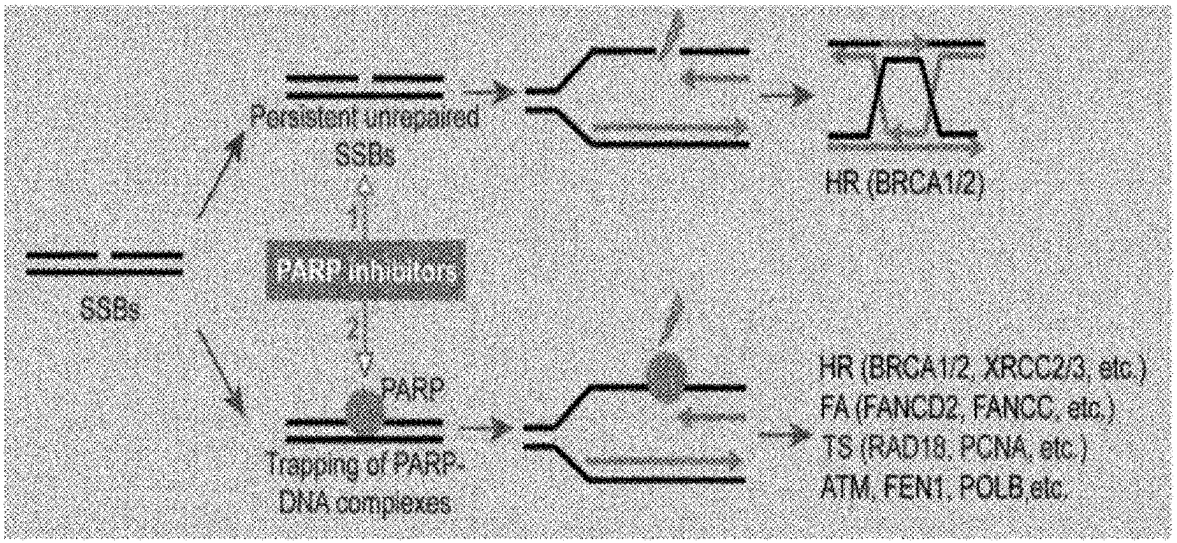


Figure 2

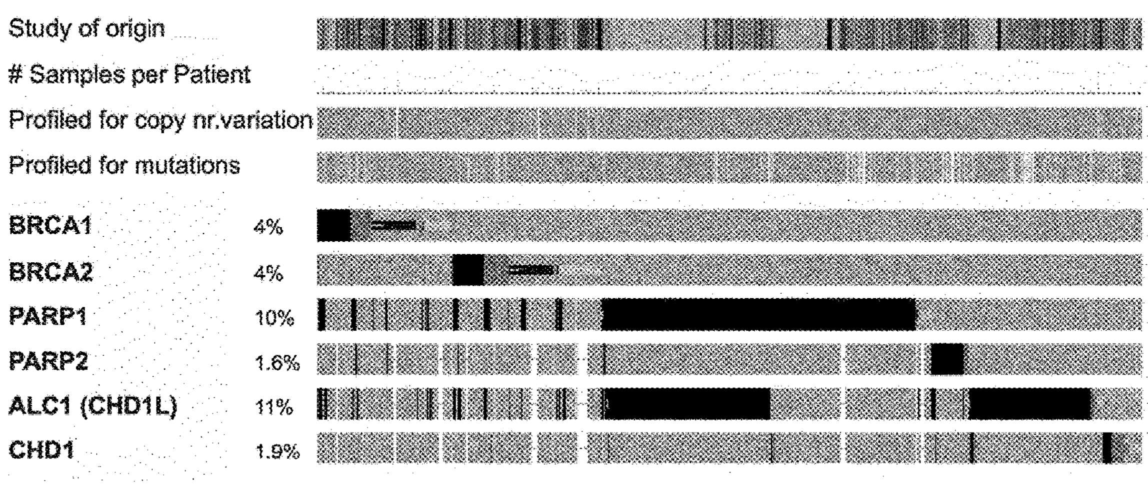
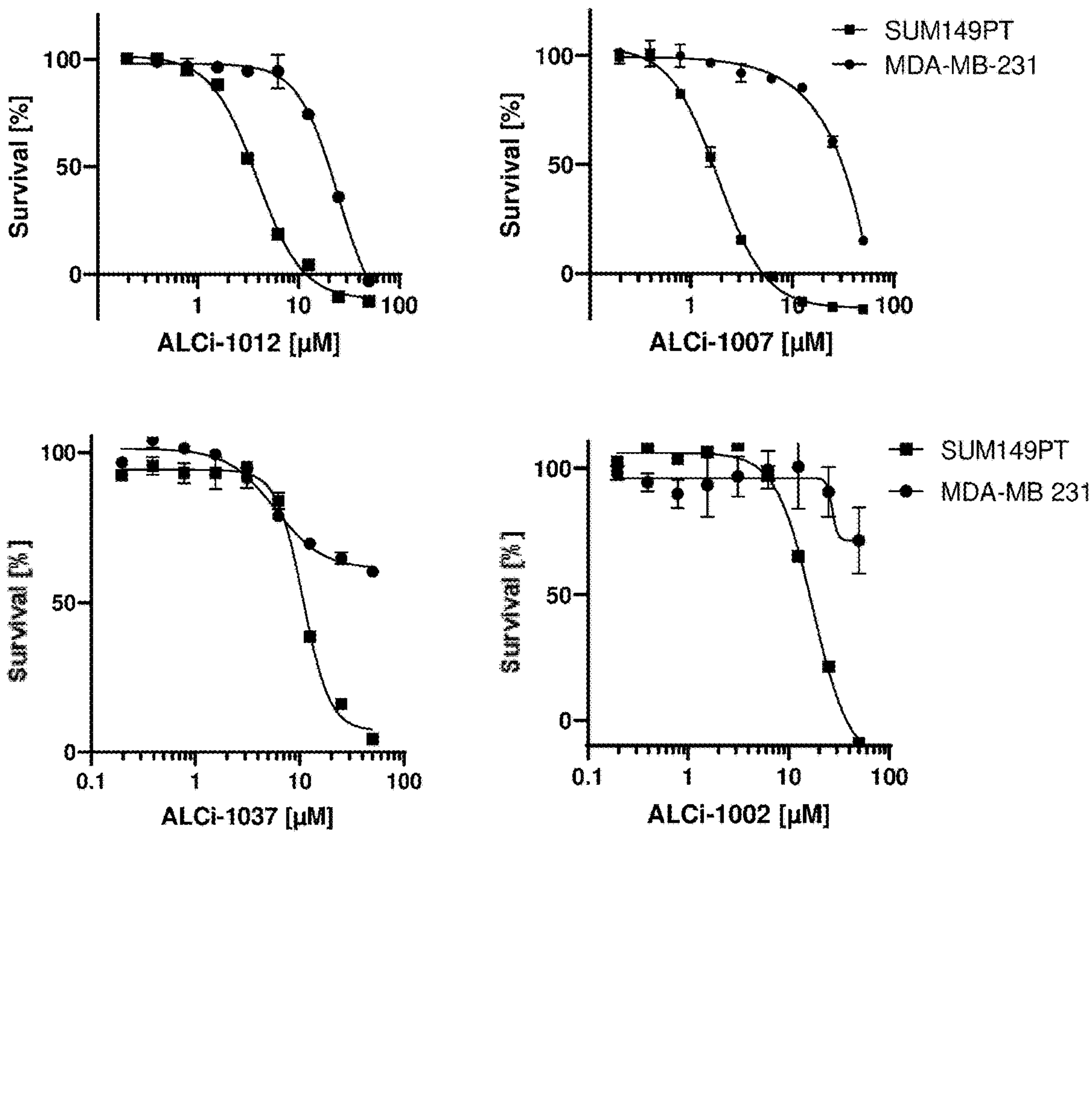


Figure 3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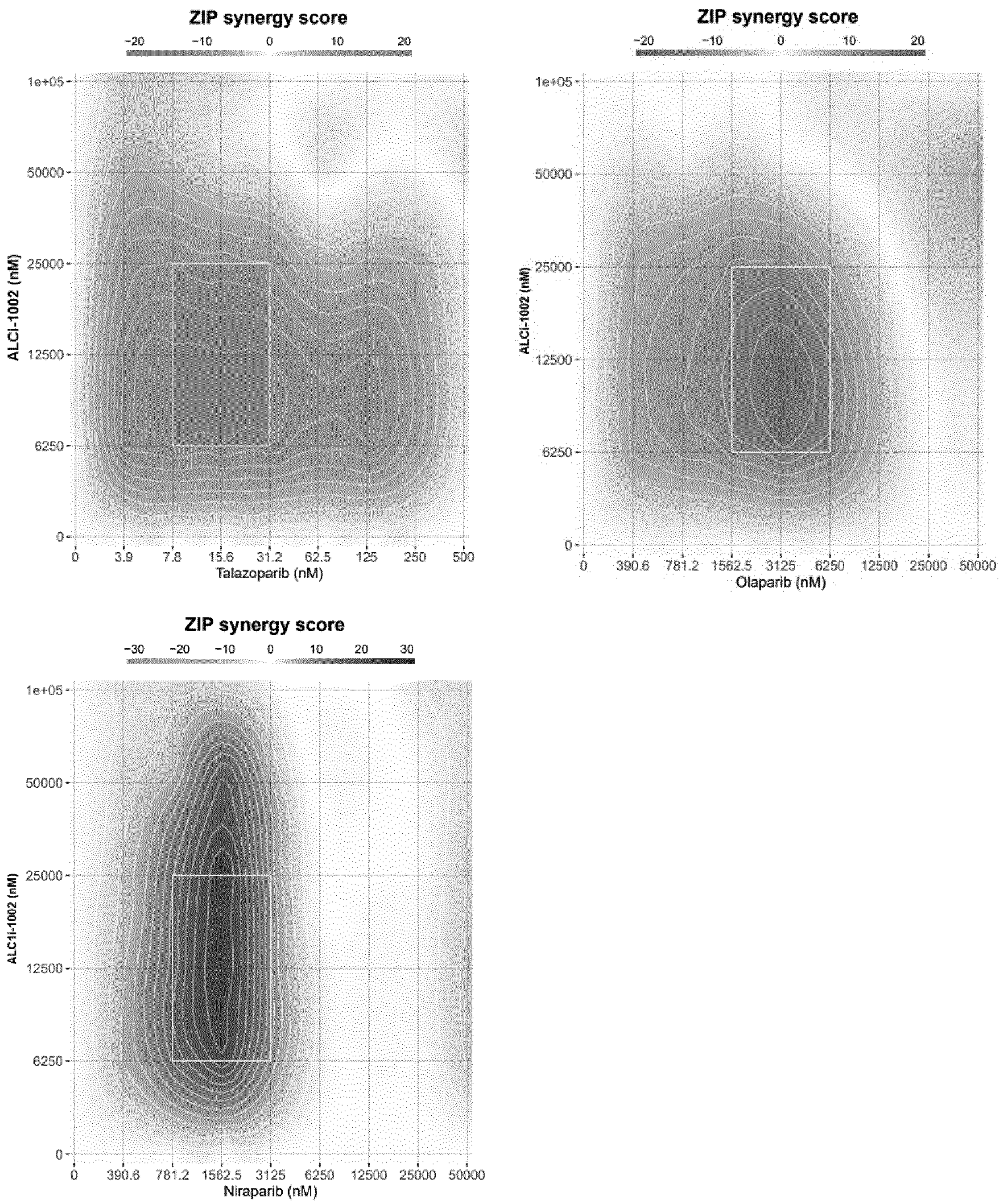
ALCi-1007 [μM]
ALCi-1037 [μM]
ALCi-1002 [μM]
MDA-MB 231

Figure 4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10
20
1e+05
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25000
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6250
0
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0
3.9
7.8
15.6
31.2
62.5
125
250
500
Talazoparib (nM)
ZIP synergy score
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-10
0
10
20
1e+05
50000
25000
12500
6250
0
ALCI-1002 (nM)
0
390.6
781.2
1562.5
3125
6250
12500
25000
50000
Olaparib (nM)
ZIP synergy score
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-20
-10
0
10
20
30
1e+05
50000
25000
12500
6250
0
ALC1i-1002 (nM)
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6250
12500
25000
50000
Niraparib (nM)

Figure 5

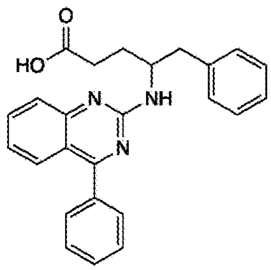
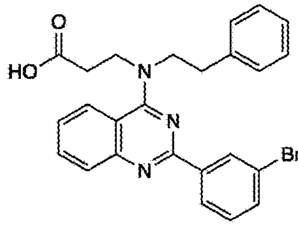
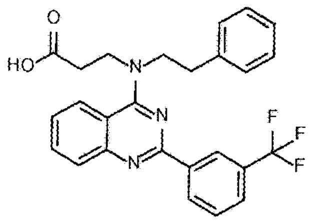
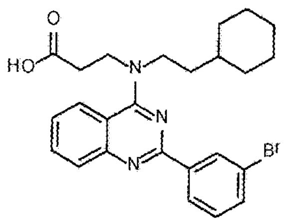
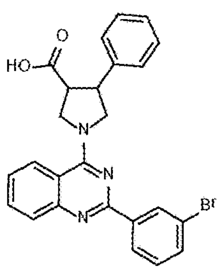
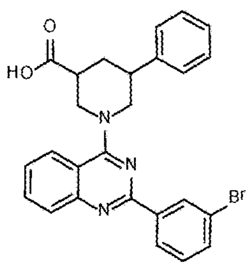
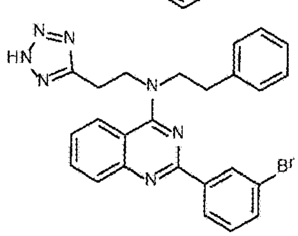
Structure	Molecule Name
	ALCi-1001
	ALCi-1002
	ALCi-1003
	ALCi-1004
	ALCi-1005
	ALCi-1006
	ALCi-1007

Figure 5 (continued)

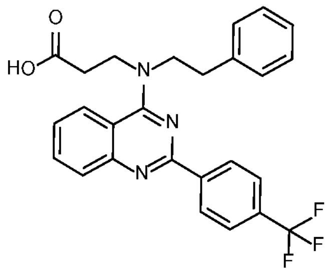
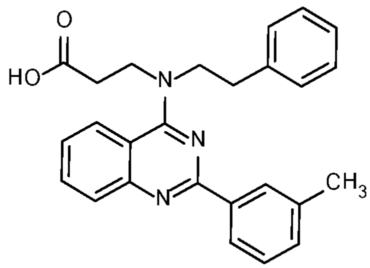
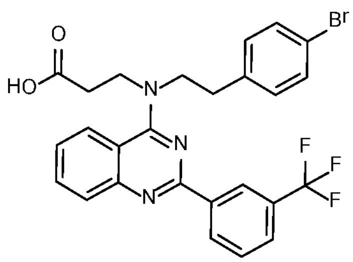
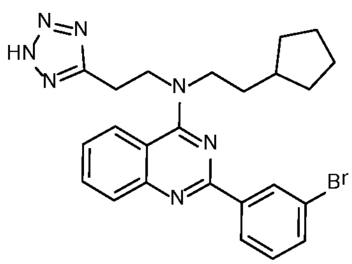
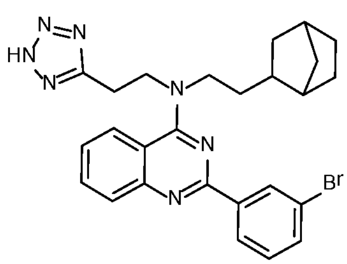
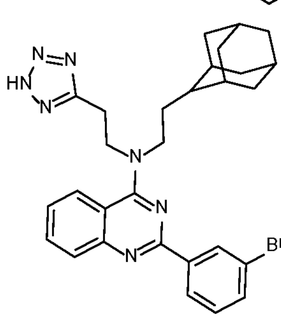
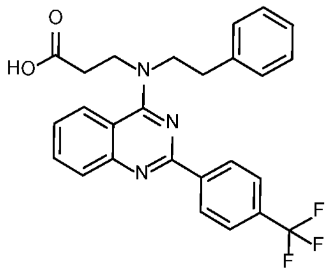
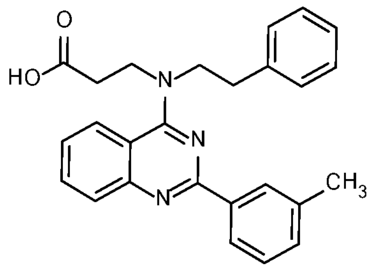
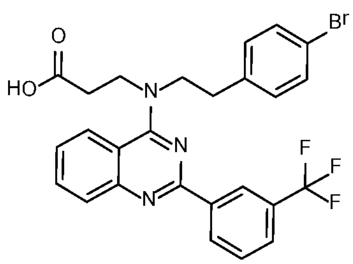
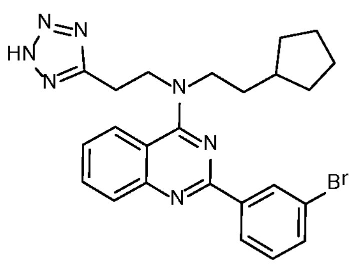
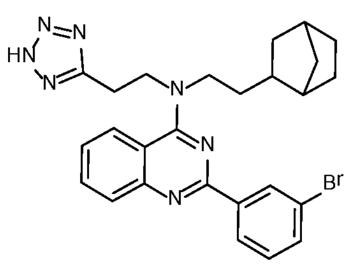
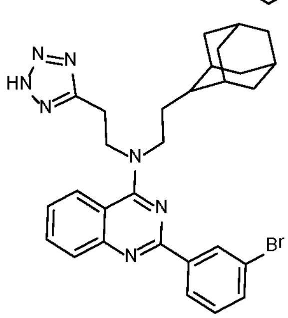
Structure	Molecule Name
	ALCi-1008
	ALCi-1009
	ALCi-1010
	ALCi-1011
	ALCi-1012
	ALCi-1013

Figure 5 (continued)

Structure	Molecule Name
	ALCi-1008
	ALCi-1009
	ALCi-1010
	ALCi-1011
	ALCi-1012
	ALCi-1013

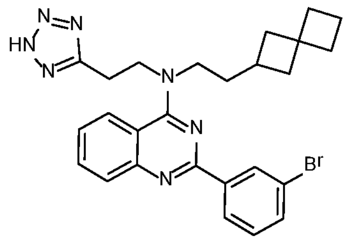
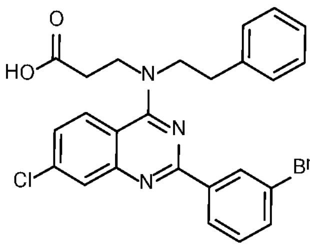
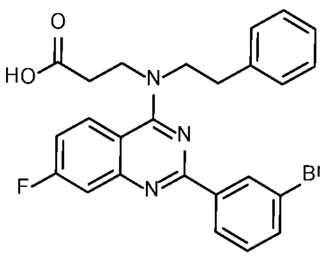
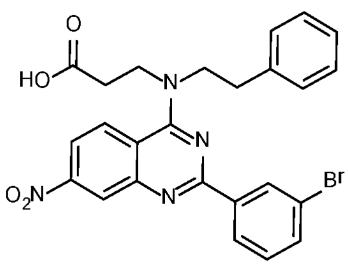
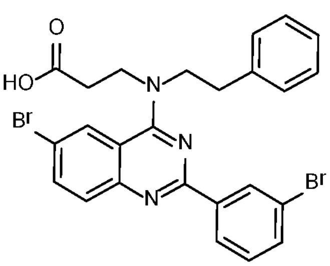
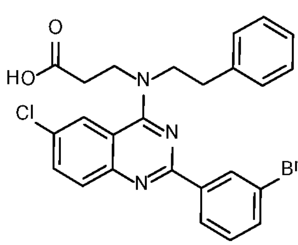
Structure	Molecule Name
	ALCi-1014
	ALCi-1015
	ALCi-1016
	ALCi-1017
	ALCi-1018
	ALCi-1019

Figure 5 (continued)

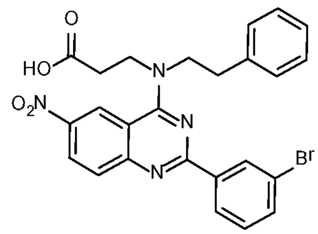
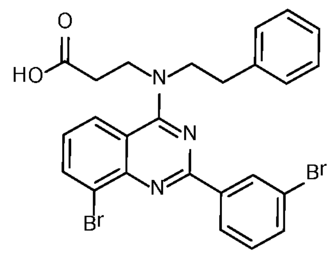
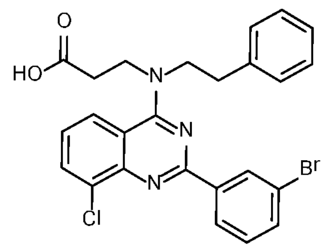
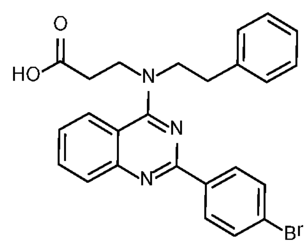
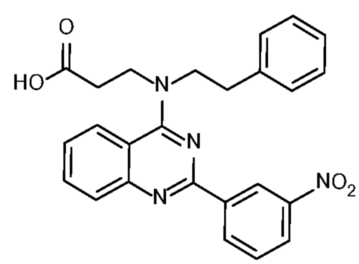
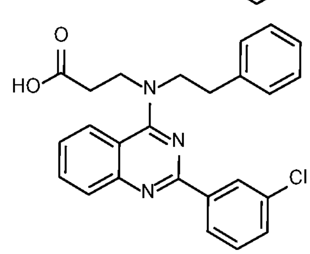
Structure	Molecule Name
	ALCi-1020
	ALCi-1021
	ALCi-1022
	ALCi-1023
	ALCi-1024
	ALCi-1025

Figure 5 (continued)

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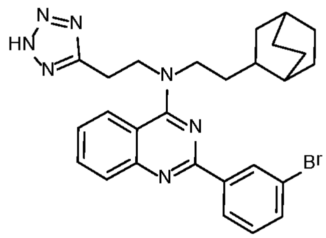
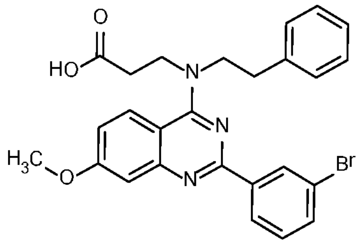
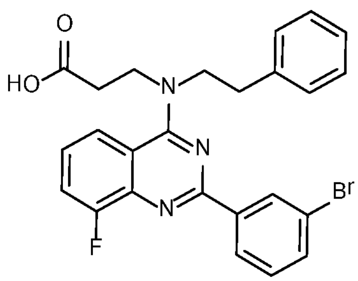
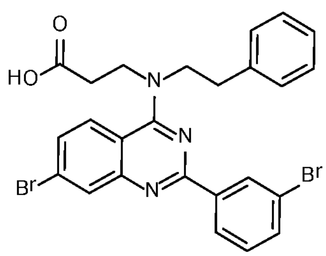
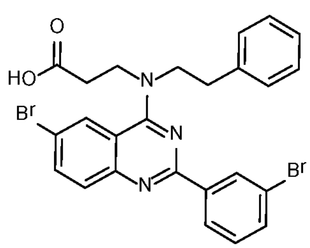
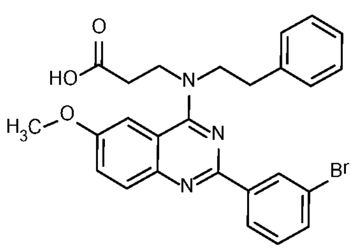
	ALCi-1026
	ALCi-1027
	ALCi-1028
	ALCi-1029
	ALCi-1030
	ALCi-1031

Figure 5 (continued)

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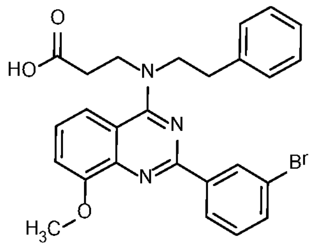
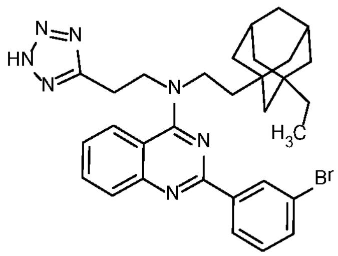
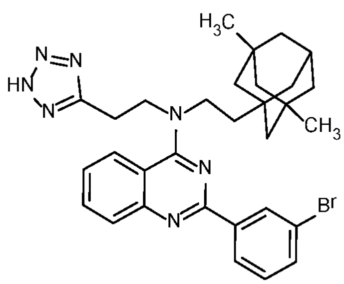
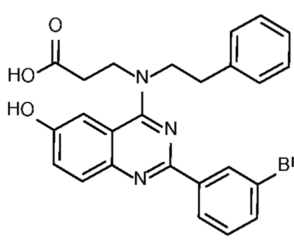
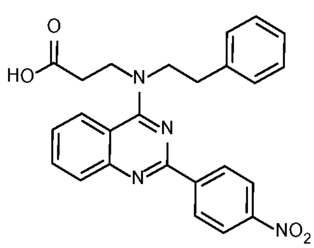
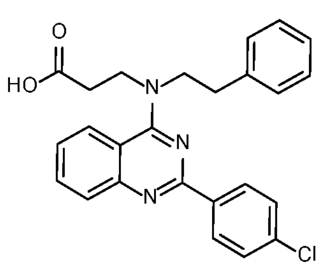
	ALCi-1032
	ALCi-1033
	ALCi-1034
	ALCi-1035
	ALCi-1036
	ALCi-1037

Figure 5 (continued)

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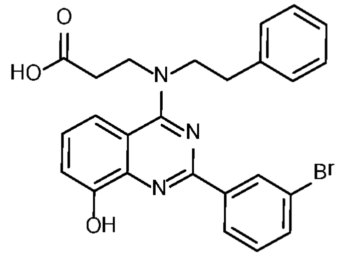
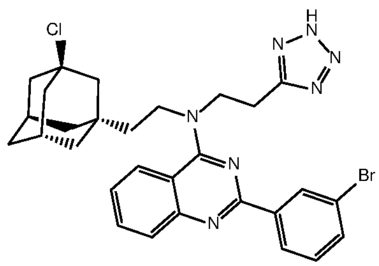
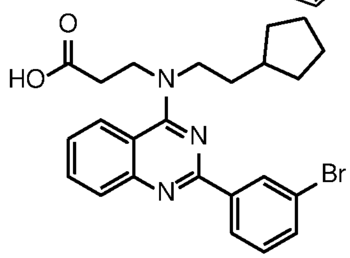
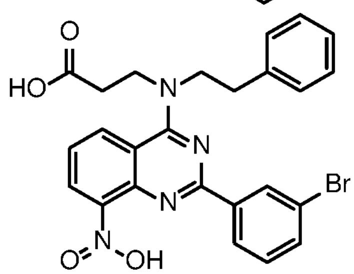
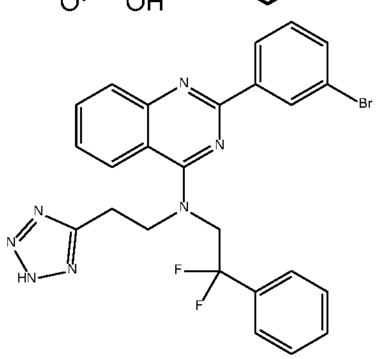
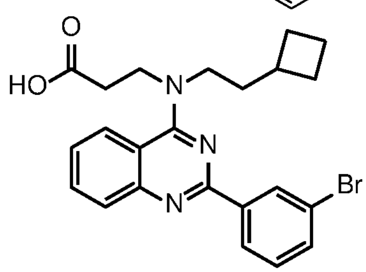
	ALCi-1038
	ALCi-1039
	ALCi-1040
	ALCi-1041
	ALCi-1042
	ALCi-1043

Figure 5 (continued)

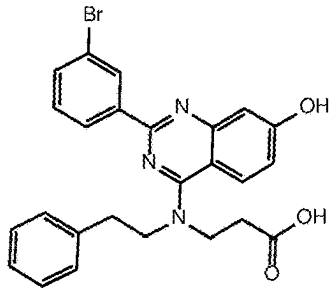
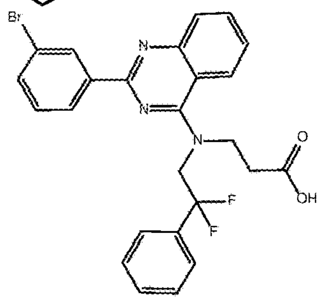
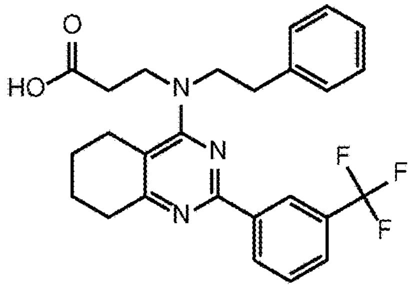
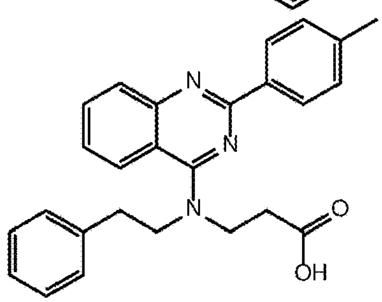
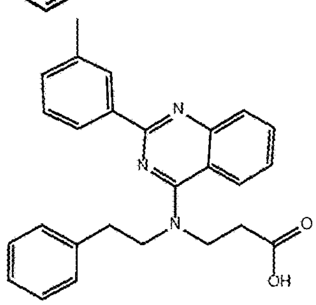
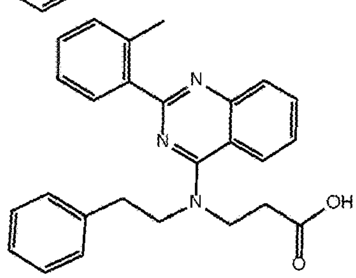
Structure	Molecule Name
	ALCi-1044
	ALCi-1045
	ALCi-1046
	ALCi-1047
	ALCi-1048
	ALCi-1049

Figure 5 (continued)

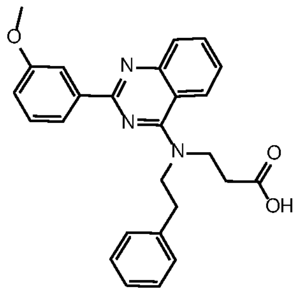
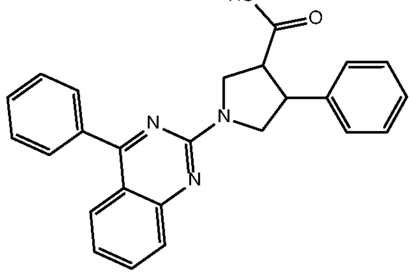
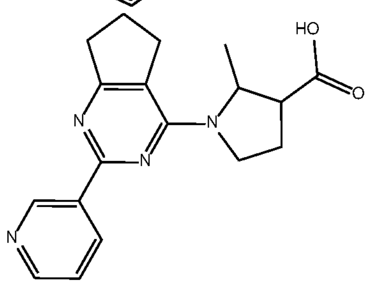
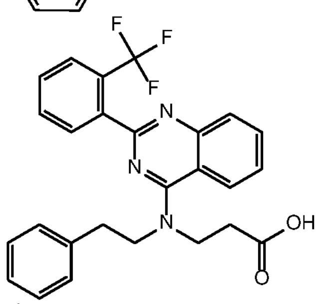
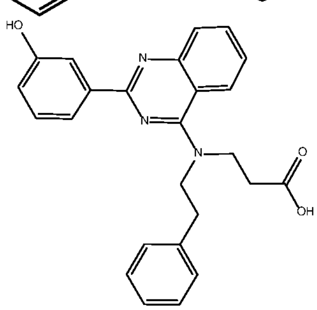
Structure	Molecule Name
	ALCi-1050
	ALCi-1051
	ALCi-1052
	ALCi-1054
	ALCi-1055

Figure 6

Compound Code	Nucleosome Remodeling Inhibition IC_{50}
ALCi-1001	+
ALCi-1002	+++
ALCi-1003	+++
ALCi-1004	+++
ALCi-1005	+++
ALCi-1006	+++
ALCi-1007	+++
ALCi-1008	+++
ALCi-1009	++
ALCi-1010	+++
ALCi-1011	+++
ALCi-1012	+++
ALCi-1013	+++
ALCi-1014	+++
ALCi-1015	+++
ALCi-1016	+++
ALCi-1017	+++
ALCi-1018	+++
ALCi-1019	+++
ALCi-1020	+++
ALCi-1021	+++
ALCi-1022	+++
ALCi-1023	+++
ALCi-1024	++
ALCi-1025	++
ALCi-1026	+++
ALCi-1027	+++
ALCi-1028	+++
ALCi-1029	+++
ALCi-1030	+++
ALCi-1031	+++
ALCi-1032	++
ALCi-1033	+++
ALCi-1034	+++
ALCi-1035	+++
ALCi-1036	++
ALCi-1037	+++
ALCi-1038	+++
ALCi-1039	+++
ALCi-1040	+++
ALCi-1041	+++
ALCi-1042	+++
ALCi-1043	+++
ALCi-1044	+++

Compound Code	Nucleosome Remodeling Inhibition IC_{50}
ALCi-1045	+++
ALCi-1046	+++
ALCi-1047	++
ALCi-1048	++
ALCi-1049	++
ALCi-1050	++
ALCi-1051	+
ALCi-1052	+
ALCi-1054	+
ALCi-1055	+

Figure 7

Compound Code	MDA-MB-231 (WT)	SUM-149PT (-)
ALCi-1001	-	++
ALCi-1002	+	++
ALCi-1003	-	++
ALCi-1004	+	++
ALCi-1005	-	++
ALCi-1006	+	++
ALCi-1007	+	+++
ALCi-1008	+++	+++
ALCi-1009	-	++
ALCi-1010	+	+++
ALCi-1011	+	+++
ALCi-1012	++	+++
ALCi-1013	++	+++
ALCi-1014	+	+++
ALCi-1015	+	+++
ALCi-1016	+	+++
ALCi-1017	+	+++
ALCi-1018	-	++
ALCi-1019	-	++
ALCi-1020	+	++
ALCi-1021	-	++
ALCi-1022	-	++
ALCi-1023	+	++
ALCi-1024	-	++
ALCi-1025	-	++
ALCi-1026	+	+++
ALCi-1027	+	++
ALCi-1028	-	++
ALCi-1029	+	+++
ALCi-1030	-	+++
ALCi-1031	-	+++
ALCi-1032	-	+
ALCi-1033	++	+++
ALCi-1034	++	+++
ALCi-1035	+	+++
ALCi-1036	+++	+++
ALCi-1037	-	++
ALCi-1038	-	++
ALCi-1039	++	+++
ALCi-1040	+	++
ALCi-1041	-	++
ALCi-1042	-	+++
ALCi-1043	-	++
ALCi-1044	-	+

Compound Code	MDA-MB-231 (WT)	SUM-149PT (-)
ALCi-1045	+	+
ALCi-1046	-	++
ALCi-1047	-	++
ALCi-1048	-	++
ALCi-1049	-	++
ALCi-1050	-	++
ALCi-1051	+	+
ALCi-1052	-	-
ALCi-1054	-	+
ALCi-1055	ND	ND

ND = not determined

Figure 8

Compound Code	Supplier
ALCi-1001	Enamine
ALCi-1002	ChemDiv
ALCi-1003	Intonation Research Laboratories
ALCi-1004	Intonation Research Laboratories
ALCi-1005	Intonation Research Laboratories
ALCi-1006	Intonation Research Laboratories
ALCi-1007	Intonation Research Laboratories
ALCi-1008	Intonation Research Laboratories
ALCi-1009	Intonation Research Laboratories
ALCi-1010	Intonation Research Laboratories
ALCi-1011	Intonation Research Laboratories
ALCi-1012	Intonation Research Laboratories
ALCi-1013	Intonation Research Laboratories
ALCi-1014	Intonation Research Laboratories
ALCi-1015	Intonation Research Laboratories
ALCi-1016	Intonation Research Laboratories
ALCi-1017	Intonation Research Laboratories
ALCi-1018	Intonation Research Laboratories
ALCi-1019	Intonation Research Laboratories
ALCi-1020	Intonation Research Laboratories
ALCi-1021	Intonation Research Laboratories
ALCi-1022	Intonation Research Laboratories
ALCi-1023	Intonation Research Laboratories
ALCi-1024	Intonation Research Laboratories
ALCi-1025	Intonation Research Laboratories
ALCi-1026	Intonation Research Laboratories
ALCi-1027	Intonation Research Laboratories
ALCi-1028	Intonation Research Laboratories
ALCi-1029	Intonation Research Laboratories
ALCi-1030	Intonation Research Laboratories
ALCi-1031	Intonation Research Laboratories

Compound Code	Supplier
ALCi-1033	Intonation Research Laboratories
ALCi-1034	Intonation Research Laboratories
ALCi-1035	Intonation Research Laboratories
ALCi-1036	Intonation Research Laboratories
ALCi-1037	Intonation Research Laboratories
ALCi-1038	Intonation Research Laboratories
ALCi-1039	Intonation Research Laboratories
ALCi-1040	Intonation Research Laboratories
ALCi-1041	Intonation Research Laboratories
ALCi-1042	Intonation Research Laboratories
ALCi-1043	Intonation Research Laboratories
ALCi-1044	Intonation Research Laboratories
ALCi-1045	Intonation Research Laboratories
ALCi-1046	Intonation Research Laboratories
ALCi-1047	Intonation Research Laboratories
ALCi-1048	Intonation Research Laboratories
ALCi-1049	Intonation Research Laboratories
ALCi-1050	Intonation Research Laboratories
ALCi-1051	Intonation Research Laboratories
ALCi-1052	FCH Group
ALCi-1054	Intonation Research Laboratories
ALCi-1055	Intonation Research Laboratories

ALCi-1032 Intonation Research Laboratories

Figure 9

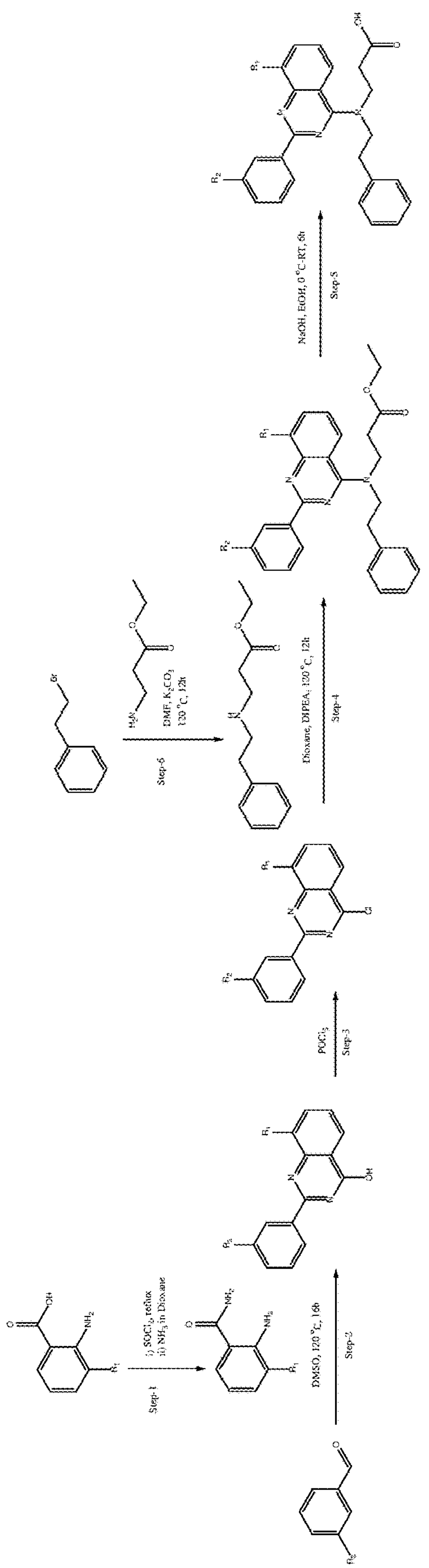


Figure 10

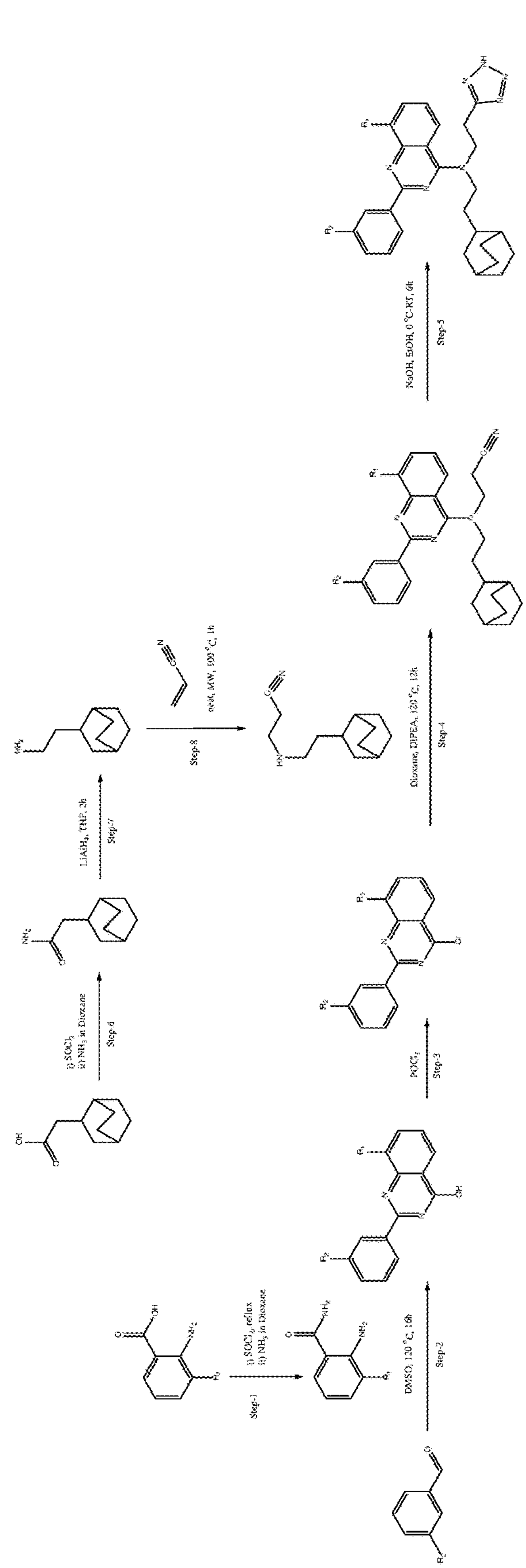


Figure 11

Compound Code	% Inhibition at 250µM	Compound Code	% Inhibition at 250µM
ALCi-1001	54.5	ALCi-1044	**100**
ALCi-1002	**100**	ALCi-1045	**100**
ALCi-1003	91.6	ALCi-1046	**100**
ALCi-1004	94.1	ALCi-1047	**100**
ALCi-1005	**100**	ALCi-1048	**100**
ALCi-1006	**100**	ALCi-1049	**100**
ALCi-1007	95.1	ALCi-1050	**100**
ALCi-1008	96.5	ALCi-1051	**100**
ALCi-1009	**100**	ALCi-1052	64.9
ALCi-1010	95.4	ALCi-1054	93
ALCi-1011	99.7	ALCi-1055	96.8
ALCi-1012	91.6		
ALCi-1013	**100**		
ALCi-1014	94.26		
ALCi-1015	**100**		
ALCi-1016	98		
ALCi-1017	**100**		
ALCi-1018	**100**		
ALCi-1019	**100**		
ALCi-1020	**100**		
ALCi-1021	**100**		
ALCi-1022	98.8		
ALCi-1023	**100**		
ALCi-1024	**100**		
ALCi-1025	**100**		
ALCi-1026	92.5		
ALCi-1027	91.4		
ALCi-1028	**100**		
ALCi-1029	**100**		
ALCi-1030	**100**		
ALCi-1031	**100**		
ALCi-1032	**100**		
ALCi-1033	**100**		
ALCi-1034	**100**		
ALCi-1035	**100**		
ALCi-1036	**100**		
ALCi-1037	**100**		
ALCi-1038	**100**		
ALCi-1039	**100**		
ALCi-1040	97.1		
ALCi-1041	97.3		
ALCi-1042	**100**		
ALCi-1043	**100**		

The assay produced inhibition percentages above 100% in some experiments. Percentages >100% have been set to exactly 100% (in bold type).

Figure 12

cell line	Entity	BRCA mutation	HRD gene deletions	EC50 ALCi-1002 (96h)	EC50 ALCi 1002 (11d)
22rv1	Prostate	BRCA2	ARID1A, NOTCH2, FANCE, JAK1, MTOR	25.00	16.00
BXPC3	Prostate	-	-	30.00	37.00
Caco2	colorectal	-	-	99.00	n.a
Capan1	Pancreas	BRCA2	FANCA, SMAD4	99.00	n.a
DLD1	Colorectal	-	-	22.00	17.00
DLD1 KO	Colorectal	BRCA2	-	22.00	21.00
DU145	Prostate	BRCA1 + 2	NOTCH2	34.00	>50
HCC1428	Breast	-	-	99.00	n.a
HCC1937	Breast	BRCA1	PTEN	24.00	n.a
HCT116	Colorectal	BRCA2	FANCA, SMARCA4	25.00	n.a
HepG2	Liver	-	-	25.00	>50
HUH7	Liver	-	FANCC	n.a	17.00
LNCAP	Prostate	-	ARID1A, NOTCH2, ATM, CHD1L, PTEN, JAK1,	11.00	n.a
MDA MB 231	Breast	-	-	99.00	36.00
MDA MB 436	Breast	BRCA1	-	33.00	34.00
PC3	Prostate	-	PTEN	35.00	9.70
PSN1	Pancreas	-	SMAD4	22.00	17.00
sum 149 pt	Breast	BRCA1	-	9.00	7.80
SW620	Colorectal	-	ATRX	99.00	n.a
T47D	Breast	-	ARID1A	5.30	n.a
U2OS	bone	-	-	30.00	n.a

Figure 13

Gene Name (synonyms)	Activity	Chromosome location	Accession number
Base excision repair (BER)	**DNA glycosylases: major altered base released**	Top of Page	
UNG	U	12q24.11	NM_080911
SMUG1	U	12q13.13	NM_014311
MBD4	U or T opposite G at CpG sequences	3q21.3	NM_003925
TDG	U, T or ethenoC opposite G	12q23.3	NM_003211
OGG1	8-oxoG opposite C	3p25.3	NM_016821
MUTYH (MYH)	A opposite 8-oxoG	1p34.1	NM_012222
NTHL1 (NTH1)	Ring-saturated or fragmented pyrimidines	16p13.3	NM_002528
MPG	3-meA, ethenoA, hypoxanthine	16p13.3	NM_002434
NEIL1	Removes thymine glycol	15q24.2	NM_024608
NEIL2	Removes oxidative products of pyrimidines	8p23.1	NM_145043
NEIL3	Removes oxidative products of pyrimidines	4q34	NM_018248
Other BER and strand break joining factors	Top of Page		
APEX1 (APE1)	AP endonuclease	14q11.2	NM_001641
APEX2	AP endonuclease	Xp11.21	NM_014481
LIG3	DNA Ligase III	17q12	NM_013975
XRCC1	LIG3 accessory factor	19q13.31	NM_006297
PNKP	Converts some DNA breaks to ligatable ends	19q13.33	NM_007254
APLF	Accessory factor for DNA end-joining	2p13.3	NM_173545
HMCES	Reacts with AP sites	3q21.3	NM_020187
Poly(ADP-ribose) polymerase (PARP) enzymes that bind to DNA	Top of Page		
PARP1 (ADPRT)	Protects strand interruptions	1q42.12	NM_001618
PARP2 (ADPRTL2)	PARP-like enzyme	14q11.2	NM_005484
PARP3 (ADPRTL3)	PARP-like enzyme	3p21.1	NM_001003931
PARG	Poly(ADP-ribose) glycohydrolase	10q11.23	NM_003631
PARPBP	Binds PARP and modulates recombination	12q23.2	NM_001319988
Direct reversal of damage	Top of Page		
MGMT	O6-meG alkyltransferase	10q26.3	NM_002412
ALKBH2 (ABH2)	1-meA dioxygenase	12q24.11	NM_001001655
ALKBH3 (DEPC1)	1-meA dioxygenase	11p11.2	NM_139178
Repair of DNA-protein crosslinks	Top of Page		
TDP1	Removes 3'-tyrosylphosphate and 3'-phosphoglycolate from DNA; human disorder SCAN1	14q32.11	NM_018319

TDP2 (TTRAP)	5'- and 3'-tyrosyl DNA phosphodiesterase	6p22.3	NM_016614
SPRTN (Spartan)	Reads ubiquitylation	1q42.12-q43	NM_032018
Mismatch excision repair (MMR)	Top of Page		
	Mismatch (MSH2-MSH6) and loop (MSH2-MSH3) recognition		
MSH2	MSH2, MSH3, MSH6	2p21	NM_000251
MSH3	5q14.1	NM_002439	
MSH6	2p16.3	NM_000179	
	MutL homologs, forming heterodimer		
MLH1	MLH1, PMS2	3p22.3	NM_000249
PMS2	7p22.1	NM_000535	
	MutS homologs specialized for meiosis		
MSH4	MSH4, MSH5	1p31.1	NM_002440
MSH5	6p21.33	NM_002441	
	MutL homologs of unknown function		
MLH3	MLH3, PMS1, PMS2L3	14q24.3	NM_014381
PMS1	2q32.2	NM_000534	
PMS2P3 (PMS2L3)	7q11.23	NM_005395	
HFM1	Helicase in meiotic-crossover formation	1p22.2	NM_001017975
Nucleotide excision repair (NER)	(XP = xeroderma pigmentosum)	Top of Page	
	Binds DNA distortions		
XPC	XPC, RAD23B, CETN2	3p25.1	NM_004628
RAD23B	9q31.2	NM_002874	
CETN2	Xq28	NM_004344	
RAD23A	Substitutes for RAD23B	19p13.13	NM_005053
XPA	Binds damaged DNA in preincision complex	9q22.33	NM_000380
	Complex defective in XP group E		
DDB1	DDB1, DDB2	11q12.2	NM_001923
DDB2 (XPE)	11p11.2	NM_000107	
	Binds DNA in preincision complex		
RPA1	RPA1, RPA2, RPA3	17p13.3	NM_002945
RPA2	1p35.3	NM_002946	
RPA3	7p21.3	NM_002947	

TFIIH	*Catalyzes unwinding in preincision complex*	Top of Page	
ERCC3 (XPB)	3' to 5' DNA helicase	2q14.3	NM_000122
ERCC2 (XPD)	5' to 3' DNA helicase	19q13.32	NM_000400
GTF2H1	Core TFIIH subunit p62	11p15.1	NM_005316
GTF2H2	Core TFIIH subunit p44	5q13.2	NM_001515
GTF2H3	Core TFIIH subunit p34	12q24.31	NM_001516
GTF2H4	Core TFIIH subunit p52	6p21.33	NM_001517
GTF2H5 (TTDA)	Core TFIIH subunit p8	6p25.3	NM_207118
GTF2E2	TFIIE subunit, mutated in TTD	8p12	NM_002095
	Kinase subunits of TFIIH		
CDK7	CDK7, CCNH, MNAT1	5q13.2	NM_001799
CCNH	5q14.3	NM_001239	
MNAT1	14q23.1	NM_002431	
ERCC5 (XPG)	3' incision	13q33.1	NM_000123
ERCC1	5' incision DNA binding subunit	19q13.32	NM_001983
ERCC4 (XPF)	5' incision catalytic subunit	16p13.12	NM_005236
LIG1	DNA ligase	19q13.32	NM_000234
NER-related	Top of Page		
	Cockayne syndrome and UV-Sensitive Syndrome; Needed for transcription-coupled NER		
ERCC8 (CSA)	ERCC8, ERCC6, UVSSA	5q12.1	NM_000082
ERCC6 (CSB)	10q11.23	NM_000124	
UVSSA (KIAA1530)	4p16.3	NM_020894	
XAB2 (HCNP)	Pre-mRNA splicing; Associates with PRPF19, CSA, CSB	19p13.2	NM_020196
MMS19	Iron-sulfur cluster loading and transport	10q24.1	NM_022362
Homologous recombination	Top of Page		
RAD51	Homologous pairing	15q15.1	NM_002875
RAD51B	Rad51 homolog	14q24.1	NM_002877
RAD51D	Rad51 homolog	17q12	NM_002878
HELQ (HEL308)	DNA helicase in RAD51 paralog complex	4q21.23	NM_133636
SWI5	Accessory factor for loading RAD51	9q34.11	NM_001040011
	Shu subunits, RAD51 recruitment		
SWSAP1	SWSAP1, ZSWIM7, SPIDR, PDS5B	19p13.2	NM_175871
ZSWIM7 (SWS1)	17p12	NM_001042697	

SPIDR	8q11.21	NM_001080394	
PDS5B	13q13.1	NM_015032	
DMC1	Rad51 homolog, meiosis	22q13.1	NM_007068
	DNA break and crosslink repair		
XRCC2	XRCC2, XRCC3	7q36.1	NM_005431
XRCC3	14q32.33	NM_005432	
	Accessory factors for recombination		
RAD52	RAD52, RAD54L, RAD54B	12p13.33	NM_134424
RAD54L	1p34.1	NM_003579	
RAD54B	8q22.1	NM_012415	
BRCA1	Accessory factor for transcription and recombination, E3 Ubiquitin ligase	17q21.31	NM_007294
BARD1	BRCA1-associated	2q35	NM_000465
ABRAXAS1	In BRCA1 A complex	4q21.23	NM_139076
PAXIP1 (PTIP)	MDC1 paralog in 53BP1 pathway	7q36.2	NM_007349
	Recruit cohesion during HR		
SMC5	SMC5, SMC6	9q21.12	NM_015110
SMC6	2p24.2	NM_001142286	
	Suppressing end-resection		
SHLD1	SHLD1, SHLD2 , SHLD3	20p12.3	NM_001303477
SHLD2 (FAM35A)	10q23.2	NM_001330112	
SHLD3	5q12.3	NM_001365341	
SEM1 (SHFM1) (DSS1)	BRCA2 associated	7q21.3	NM_006304
RAD50	ATPase in complex with MRE11A, NBS1	5q23.3	NM_005732
MRE11A	3' exonuclease, defective in ATLD (ataxia-telangiectasia-like disorder)	11q21	NM_005590
NBN (NBS1)	Mutated in Nijmegen breakage syndrome	8q21.3	NM_002485
RBBP8 (CtIP)	Promotes DNA end resection	18q11.2	NM_002894
	Subunits of structure-specific DNA nuclease		
MUS81	MUS81, EME1, EME2	11q13.1	NM_025128
EME1 (MMS4L)	17q21.33	NM_152463	

EME2	16p13.3	NM_001257370	
SLX1A (GIYD1)	Subunit of SLX1-SLX4 structure-specific nuclease, two identical tandem genes in the human genome SLX1A , SLX1B	16p11.2	NM_001014999
SLX1B (GIYD2)	16p11.2	NM_024044	
GEN1	Nuclease cleaving Holliday junctions	2p24.2	NM_182625
Fanconi anemia	Tolerance and repair of DNA crosslinks and other adducts in DNA	Top of Page	
FANCA	FANCA	16q24.3	NM_000135
FANCB	FANCB	Xp22.31	NM_152633
FANCC	FANCC	9q22.32	NM_000136
BRCA2 (FANCD1)	Cooperation with RAD51, essential function	13q13.1	NM_000059
FANCD2	Target for monoubiquitination	3p25.3	NM_033084
FANCE	FANCE	6p21.31	NM_021922
FANCF	FANCF	11p14.3	NM_022725
FANCG (XRCC9)	FANCG	9p13.3	NM_004629
FANCI (KIAA1794)	target for monoubiquitination	15q26.1	NM_018193
BRIP1 (FANCJ)	DNA helicase, BRCA1-interacting	17q23	NM_032043
FANCL	FANCL	2p16.1	NM_018062
FANCM	helicase / translocase	14q21.3	NM_020937
PALB2 (FANCN)	Co-localizes with BRCA2 (FANCD1)	16p12.1	NM_024675
RAD51C (FANCO)	Rad51 homolog , FANCO	17q23.2	NM_002876
SLX4(FANCP)	nuclease subunit / scaffold SLX4 , FANCP	16p13.3	NM_032444
FAAP20 (C1orf86)	FANCA - associated	1p36.33	NM_182533
FAAP24 (C19orf40)	FAAP24	19q13.11	NM_152266
FAAP100	Part of FA core complex	17q25.3	NM_025161
UBE2T (FANCT)	E2 ligase for FANCL	1q32.1	NM_014176
Non-homologous end-joining	Top of Page		
XRCC6 (Ku70)	DNA end binding subunit	22q13.2	NM_001469
XRCC5 (Ku80)	DNA end binding subunit	2q35	NM_021141
PRKDC	DNA-dependent protein kinase catalytic subunit	8q11.21	NM_006904
LIG4	Ligase	13q33.3	NM_002312
XRCC4	Ligase accessory factor	5q14.2	NM_003401
DCLRE1C (Artemis)	Nuclease	10p13	NM_022487
NHEJ1 (XLF, Cernunnos)	End-joining factor	2q35	NM_024782
Modulation of nucleotide pools	Top of Page		

NUDT1 (MTH1)	8-oxoGTPase	7p22.3	NM_002452
DUT	dUTPase	15q21.1	NM_001948
RRM2B (p53R2)	p53-inducible ribonucleotide reductase small subunit 2 homolog	8q22.3	NM_015713
PARK7 (DJ-1)	Guanine glycation repair	1p36.23	NM_007262
DNPH1	Hydrolase for 5-hydroxymethyl deoxyuridine	6p21.1	NM_006443
NUDT15 (MTH2)	Hydrolysis of thiopurines?	13q14.2	NM_018283
NUDT18 (MTH3)	Hydrolysis of 8-hydroxypurine diphosphates	8p21.3	NM_024815
DNA polymerases	Top of Page		
POLA1	DNA synthesis at resected ends	Xp22.1-p21.3	NM_001330360
POLB	BER in nuclear DNA	8p11.21	NM_002690
	Pol delta subunits		
	POLD1 (catalytic subunit), POLD2 (Also subunit of pol zeta),		
POLD1	POLD3 (Also subunit of pol zeta), POLD4 (auxiliary subunit)	19q13.33	NM_002691
POLD2	7p13	NM_001127218	
POLD3	11q13.4	NM_006591	
POLD4	11q13.2	NM_021173	
	Pol epsilon subunits		
POLE (POLE1)	POLE1, POLE2, POLE3, POLE4	12q24.33	NM_006231
POLE2	14q21.3	NM_002692	
POLE3	9q32	NM_017443	
POLE4	2p12	NM_019896	
REV3L (POLZ)	DNA pol zeta catalytic subunit, essential function	6q21	NM_002912
MAD2L2 (REV7)	DNA pol zeta and shieldin subunit	1p36.22	NM_006341
REV1 (REV1L)	dCMP transferase	2q11.2	NM_016316
POLG	Mitochondrial DNA repair and replication	15q26.1	NM_002693
POLH	xeroderma pigmentosum (XP) variant	6p21.1	NM_006502
POLI (RAD30B)	Lesion bypass	18q21.2	NM_007195
POLQ	TMEJ (alt-EJ)	3q13.33	NM_199420
POLK (DINB1)	Lesion bypass and NER	5q13.3	NM_016218
POLL	Gap-filling during non-homologous end-joining	10q24.32	NM_013274
POLM	Gap filling during non-homologous end-joining	7p13	NM_013284
POLN (POL4P)	Lesion bypass / recombination?	4p16.3	NM_181808
PRIMPOL	Primase and DNA-directed polymerase	4q35.1	NM_152683
DNTT	Terminal transferase	10q24.1	NM_004088

Editing and processing nucleases	Top of Page		
FEN1 (DNase IV)	5' nuclease	11q12.2	NM_004111
FAN1 (MTMR15)	5' nuclease interacting with FANCD2	15q13.2	NM_014967
TREX1	3' exonuclease	3p21.31	NM_033629
TREX2	3' exonuclease	Xq28	NM_080701
EXO1 (HEX1)	5' exonuclease	1q43	NM_003686
APTX (aprataxin)	Processing of DNA single-strand interruptions	9p21.1	NM_175073
SPO11	endonuclease	20q13.32	NM_012444
ENDOV	incision 3' of hypoxanthine and uracil in DNA and inosine in RNA	17q25.3	NM_173627
DNA2	Seckel Syndrome 8	10q21.3	NM_001080449
DCLRE1A (SNM1A)	5'-3' exonuclease, DNA crosslink repair	10q25.3	NM_014881
DCLRE1B (SNM1B)	5' - 3' exonuclease, APOLLO	1p13.2	NM_022836
EXO5	exonuclease	1p34.2	NM_001346946
Ubiquitination and modification	Top of Page		
UBE2A (RAD6A)	Ubiquitin-conjugating enzyme	Xq24-q25	NM_003336
UBE2B (RAD6B)	Ubiquitin-conjugating enzyme	5q31.1	NM_003337
RAD18	E3 ubiquitin ligase	3p25.3	NM_020165
SHPRH	E3 ubiquitin ligase, SWI/SNF related, homolog of *S. cerevisiae* Rad5	6q24.3	NM_001042683
HLTF (SMARCA3)	E3 ubiquitin ligase, SWI/SNF related, homolog of S. cerevisiae Rad5	3q25.1-q26.1	NM_003071
RNF168	E3 ubiquitin ligase for DSB repair; ATM-like and RIDDLE syndrome	3q29	NM_152617
RNF8	E3 ubiquitin ligase for DSB repair	6p21.2	NM_003958
RNF4	E3 ubiquitin ligase	4p16.3	NM_001185009
UBE2V2 (MMS2)	Ubiquitin-conjugating complex UBE2V2, UBE2N	8q11.21	NM_003350
UBE2N (UBC13)	12q22	NM_003348	
USP1	Ubiquitin-specific protease for FANCD2, PCNA	1p31.3	NM_003368
WDR48	Necessary for USP1 activity	3p22.2	NM_020839
HERC2	Control of several DNA repair factors	15q13.1	NM_004667
Chromatin Structure and Modification	Top of Page		
H2AX (H2AFX)	Histone, phosphorylated after DNA damage	11q23.3	NM_002105
CHAF1A (CAF1)	Chromatin assembly factor	19p13.3	NM_005483
SETMAR (METNASE)	DNA damage-associated histone methylase and nuclease	3p26	NM_006515
ATRX	Chromatin remodeling, transcription factor	Xq21.1	NM_000489

Genes defective in diseases associated with sensitivity to DNA damaging agents	Top of Page		
BLM	Bloom syndrome helicase	15q26.1	NM_000057
RMI1	In BLM-TOP3A complex	9q21.32	NM_001358291
TOP3A	Topoisomerase IIIa	17p11.2	NM_004618
WRN	Werner syndrome helicase / 3' - exonuclease	8p12	NM_000553
RECQL4	Rothmund-Thompson syndrome	8q24.3	NM_004260
ATM	ataxia telangiectasia	11q22.3	NM_000051
MPLKIP (TTDN1)	non-photosensitive form of trichothiodystrophy	7p14	NM_138701
Other identified genes with known or suspected DNA repair function	Top of Page		
RPA4	Similar to RPA2	Xp21.33	NM_013347
PRPF19 (PSO4)	Pre-mRNA splicing; DNA crosslink repair; binding to SETMAR	11q12.2	NM_014502
RECQL (RECQ1)	DNA helicase	12p12.1	NM_002907
RECQL5	DNA helicase	17q25.1	NM_001003715
RDM1 (RAD52B)	Similar to RAD52	17q12	NM_145654
NABP2 (SSB1)	Single-stranded DNA binding protein (ATM / MRN pathway)	12q13.3	NM_024068
Other conserved DNA damage response genes	Top of Page		
ATR	ATM- and PI-3K-like essential kinase	3q23	NM_001184
ATRIP	ATR-interacting protein	3p21.31	NM_130384
MDC1	Mediator of DNA damage checkpoint	6p21.3	NM_014641
PCNA	Sliding clamp for pol delta and pol epsilon	20p12.3	NM_002592
	subunits of PCNA-like sensor of damaged DNA		
RAD1	RAD1, RAD9, HUS1	5p13.2	NM_002853
RAD9A	11q13.2	NM_004584	
HUS1	7p12.3	NM_004507	
RAD17 (RAD24)	RFC-like DNA damage sensor	5q13.2	NM_002873
	Effector kinases		
CHEK1	CHEK1, CHEK2	11q24.2	NM_001274
CHEK2	22q12.1	NM_007194	
TP53	Regulation of the cell cycle	17p13.1	NM_000546
TP53BP1 (53BP1)	chromatin-binding checkpoint protein	15q15-q21	NM_001141980
RIF1	suppressor of 5'-end-resection	2q23.3	NM_001177665
TOPBP1	DNA damage checkpoint control	3q22.1	NM_007027
CLK2	S-phase check point and biological clock protein	1q21	NM_003993
PER1	S-phase check point and biological clock protein	17p12	NM_002616

Figure 14:

HR-related genes

53BP1	CDKN2B	FGF23	KDM5C	PALB2 (FANCN)	RMI1	TP53
AADAT	CDKN2C	FGF3	KDM6A	PARP1	RMI1 b	TP53BP1
ADM	CEBPA	FGF4	KDR	PARP2	RMI2	TP53I3
AHSA1	CENPA	FGF5	KEAP1	PARP3	RMI2 b	TP63
ALDH3B1	CENPE	FGF6	KEL	PARP4	RNF43	TRAF2
ALDH6A1	CENPJ	FGF9	KIT	PARPBP	ROS1	TRAF7
ALG8	CHAF1A	FGFR1	KLF4	PAX5	RPA1	TRAPPC13
ANLN	CHEK1	FGFR2	KLHL6	PAXX	RPA1 b	TSC1
ARID1a	CHEK2	FGFR3	KMT2A (MLL)	PBRM1	RPA2	TSC2
ARID1b	CIC	FGFR4	KMT2B	PCNA	RPA3	TSHR
ARSD	COL2A1	FH	KMT2C (MLL3)	PDCD1 (PD-1)	RPA4	TTK
ASF1B	COP1	FLCN	KMT2D (MLL2)	PDCD1LG2 (PD-L2)	RPS6KA4	TYRO3
ASPM	CREBBP	FLT1	KNSTRN	PDGFB	RPS6KB1	U2AF1
ATM	CRKL	FLT3	KRAS	PDGFRA	RPS6KB2	VEGFA
ATP10B	CRLF2	FLT4	LATS1	PDGFRB	RPTOR	VHL
ATR	CSF1R	FOXA1	LATS2	PDK1	RSPO1	VTCN1
ATRIP b	CSF3R	FOXL2	LCK	PER1	RSPO2	WRN
ATRX	CTCF	FOXO1	LIG4	PGD	RUNX1	WT1
AURKB	CTLA4	FOXP1	LMO1	PGR	RUNX1T1 (ETO)	XIAP
BAP1	CTNNA1	FRS2	LRP1B	PHF6	RYBP	XPO1
BARD1	CTNNB1	FTO	LYN	PHOX2B	SDC4	XRCC2
BBOX1	CUL3	FUBP1	MAD2L2	PIK3C2B	SDHA	XRCC3
BIRC2	CUL4A	FYN	MAGOH	PIK3C2G	SDHAF2	XRCC4
BIRC3	CUL4B	GABRA6	MALT1	PIK3C3	SDHB	XRCC5
BLM	CUX1	GADD45A	MAP2K1 (MEK1)	PIK3CA	SDHC	XRCC6
BMPR1A	CXCR4	GATA1	MAP2K2 (MEK2)	PIK3CB	SDHD	YAP1
BRAF	CYLD	GATA2	MAP2K4	PIK3CD	SESN1	YES1
BRCA1	CYP2C19	GATA3	MAP2K7	PIK3CG	SETBP1	ZFHX3
BRCA1 (FANCS)	DAXX	GATA4	MAP3K1	PIK3R1	SETD2	ZNF217
BRCA2	DCLRE1C	GATA6	MAP3K13	PIK3R2	SF3B1	ZNF703
BRCA2 (FANCD1)	DCUN1D1	GEN1	MAP3K14	PIK3R3	SGK1	ZRSR2

BRD4	DDB1	GID4 (C17orf39)	MAP3K4	PIM1	SH2B3	
BRI3BP	DDR1	GLI1	MAPK1	PLCG1	SH2D1A	
BRIP1	DDR2	GNA11	MAPK3	PLCG2	SHLD1	
BRIP1 (FANCJ)	DDX3X	GNA13	MAPK8	PLK2	SHLD2	
BRIP1 b	DICER1	GNAQ	MAX	PMAIP1	SHPRH	
BTG1	DIS3	GNAS	MCL1	PML	SHQ1	
BTG2	DLL3	GPS2	MDC1	PMS1	SLC34A2	
BTK	DMC1 b	GRIN2A	MDM2	PMS2	SLIT2	
BUB1	DNA2	GRM3	MDM4	PNKP	SLX4	
C10orf119	DNAJB1	GSK3B	MED12	PNRC1	SMAD2	
C10orf73	DNMT1	H1-2	MEF2B	POLA1	SMAD3	
C11orf82	DNMT3A	H2AX	MEN1	POLD1	SMAD4	
C13orf3	DNMT3B	H2BC5	MERTK	POLE	SMARCA2	
C14orf145	DOT1L	H3-3A	MET	POLQ	SMARCA4	
C15orf42	E2F3	H3-3B	MGA	PPARG	SMARCAD1	
C16orf59	EED	H3-4	MGMT	PPM1D	SMARCB1	
C16orf68	EGFL7	H3-5	MITF	PPP1R3A	SMARCD1	
C17orf41	EGFR	H3C1	MLH1	PPP2R1A	SMC1A	
C1orf112	EIF1AX	H3C10	MLH3	PPP2R2A	SMC3	
C1orf54	EIF4A2	H3C11	MPL	PPP4R4	SMC5	
C1QTNF6	EIF4E	H3C12	MRE11	PPP6C	SMC6	
C20orf19	ELF3	H3C13	MRE11A	PRC1	SMO	
C20orf82	ELOC (TCEB1)	H3C2	MRE11A (MRE11)	PRDM1	SNCAIP	
C4orf34	EML4	H3C3	MRN complex	PREX2	SOCS1	
C5orf38	EMSY (C11orf30)	H3C4	MSH2	PRG4	SOS1	
C6orf48	ENO1	H3C6	MSH3	PRKAR1A	SOX10	
CALR	EP300	H3C7	MSH6	PRKCI	SOX17	
CARD11	EPCAM	H3C8	MST1	PRKDC	SOX2	
CASP8	EPHA2	HDAC2	MST1R	PRKN	SOX9	
CBFB	EPHA3	HDAC9	MTAP	PTCH1	SPEN	
CBL	EPHA5	HELQ	MTOR	PTEN	SPOP	
CBLB	EPHA7	HERC2	MUTYH	PTK2	SRC	
CCDC138	EPHB1	HFM1	MXD4	PTPN11	SRSF2	
CCDC92	ERBB2	HGF	MYB	PTPRB	SSBP1	
CCN6	ERBB3	HLA-A	MYC	PTPRD	STAG2	
CCNA2	ERBB4	HNF1A	MYCL (MYCL1)	PTPRS	STAT3	
CCNB1	ERCC1	HOXB13	MYCN	PTPRT	STAT4	
CCNB1IP1	ERCC2	HRAS	MYD88	QKI	STAT5A	

CCND1	ERCC3	HSD3B1	MYOD1	RAB35	STAT5B	
CCND2	ERCC4	HSP90AA 1	NBN	RAC1	STK11	
CCND3	ERCC5	HSP90AB1	NBN (NBS1)	RAD21	STK19	
CCNE1	ERCC6	ICOSLG	NCOA3	RAD50	STK40	
CD274 (PD-L1)	ERG	ID3	NCOR1	RAD51	SUFU	
CD276	ERRFI1	IDH1	NEGR1	RAD51AP1	SUZ12	
CD68	ESR1	IDH2	NF1	RAD51B	SYK	
CD74	ETV1	IFNGR1	NF2	RAD51C	TBC1D4	
CD79A	ETV4	IGF1	NFE2L2	RAD51D	TBX3	
CD79B	ETV5	IGF1R	NFKBIA	RAD52	TCF3 (E2A)	
CD8A	ETV6	IGF2	NHEJ1	RAD54L	TCF7L2	
CDC20	EWSR1	IKBKE	NKX2-1	RAF1	TEK	
CDC27	EXO1	IKZF1	NKX3-1	RARA	TENT5C	
CDC6	EZH2	IL10	NOTCH1	RASA1	TERC	
CDC7	EZR	IL7R	NOTCH2	RB1	TERT	
CDC73	FADD	INHA	NOTCH3	RBM10	TET1	
CDCA2	FANCA	INHBA	NOTCH4	RECQL4	TET2	
CDCA3	FANCC	INO80	NPM1	REL	TFE3	
CDCA5	FANCD2	INPP4A	NRAS	RET	TGFB1	
CDCA7	FANCE	INPP4B	NSD1	REV3L	TGFBR1	
CDCA8	FANCF	INSR	NSD3	RFC1	TGFBR2	
CDH1	FANCG	IRF2	NT5C2	RFC2	TMEM127	
CDK12	FANCI	IRF4	NTRK1	RFC3	TMPRSS2	
CDK2	FANCL	IRS1	NTRK2	RFC4	TNF	
CDK4	FANCM	IRS2	NTRK3	RFC5	TNFAIP3	
CDK6	FAS (TNFRSF6)	JAK1	NUP93	RHEB	TNFRSF14	
CDK8	FAT1	JAK2	NUTM1	RHOA	TOP1	
CDKN1A	FBXW7	JAK3	PAK1	RICTOR	TOP2A	
CDKN1B	FGF10	JUN	PAK3	RIF1	TOP3A	
CDKN1C	FGF14	KAT6A (MYST3)	PAK5	RIF1 b	TOP3A b	
CDKN2A	FGF19	KDM5A	PALB2	RIT1	TOPBP1	

USE OF ALC1 INHIBITORS AND SYNERGY WITH PARPI

[0001] The present invention relates to small molecule compounds that allosterically inhibit ALC1 (CHD1L) and are synergistic the inhibition of the PARP family of enzymes. These inhibitors possess anti-proliferative activity in cancers that acquired a BRCA1 or BRCA2 deficiency. Disruption of the chromatin remodeling forces of ALC1 through these agents enables a highly selective therapy for targeting the DNA damage functions of PARP enzymes in several proliferative diseases, notably BRCA-deficient cancers. Via inhibition of the enzymatic activity, the compounds engage the synthetic lethality between BRCA1/2 and ALC1. By being synergistic with PARP enzyme inhibitors, ALC1 inhibitors potentiate the cancer cell killing properties of these PARP inhibitors, enabling therapeutic approaches where ALC1 is amplified as an oncogene. Additionally, these ALC1 inhibitors make it possible to overcome PARP inhibitor resistance mechanisms and enable an alternative approach to the treatment of germline or acquired BRCA1/BRCA2 deficiency, including tumors defined by "BRCAness" or other changes in DNA repair networks.

BACKGROUND OF THE INVENTION

[0002] For recognition of single-strand and double-strand breaks (SSBs/DSBs), the nuclear Poly-ADP-ribose polymerase (PARP) enzymes are early key factors in the DNA damage response (DDR). Using the metabolite NAD^+, PARP-1 and -2 add poly-ADP-ribose (PAR) chains to chromatin components and to factors belonging to the DDR, while PARP-3 targets chromatin components via mono-ADP-ribosylation. PARPs get recruited to DNA lesions by recognizing specifically altered, DNA-damage induced structures, which activates their PARylation activity, which in turn regulates their activity and the activity of other DDR and chromatin proteins, facilitating the DDR (Ray Chaudhuri and Nussenzweig, 2017).

[0003] This catalytic activity can be inhibited by NAD^+ analogues and has become of particular interest and clinically useful in genetically-defined cancers. Notably, by targeting synthetic lethality in the context of BRCA1 or BRCA2 deficiency, so-called PARP inhibitors (PARPi) are used to treat homologous-recombination (HR)-deficient and other cancers. This is thought to occur by lowering PARP activity and/or by biochemically "trapping" of PARP-1/2/3 on chromatin (Murai et al., 2012, 2014). While "trapping" remains molecularly ill-defined, the term defines an enhanced recruitment, association and/or retention of PARP-1/2/3 enzymes on damaged chromatin, typically induced by treatment of PARP-1 enzyme with PARP inhibitors (PARPi), or a decrease in the release of PARP-1/2/3 enzymes following their initial recruitment, which leads to a prolonged retention. This biochemically manifests itself in an enhanced steady-state association/retention/binding ("trapping") of the PARP enzymes with damaged genome regions/loci.

[0004] Due to the clinical advent of PARPi, PARP-1 has emerged as a powerful target for an increasing list of cancers, including in combination with immuno-oncology therapies, such as the Lynparza/Keytruda trials, to give all but one of many examples. Moreover, first-line PARPi therapies and applications in contexts outside of germline BRCA-1/2 mutations are becoming possible.

[0005] Importantly, clinical PARPi compounds all bind essentially the same location at the catalytic center of the active site by blocking the binding of the substrate NAD^+, thus preventing poly(ADP-ribose) synthesis, largely by virtue of their structural similarity to nicotinamide, a moiety of the NAD^+ nucleotide.

[0006] Yet, PARPi exhibit greatly different clinical efficacy in tumor killing and patient outcomes in the clinic. One fundamental difference in the action of these PARPi is that they promote highly distinct levels of PARP-1/-2 trapping on chromatin. It is currently generally thought that the most powerful and clinically effective PARPi trap PARP-1 at the site of a DNA break much more strongly than clinically less useful PARPi. Upon PARP trapping, DNA lesions become more cytotoxic, especially in mutant tumor cells with genetic or epigenetic deficiencies in the repair of DNA strand breaks, such as functionally HR-deficient BRCA-1/2 mutant tumor cells. Further, it is currently not clear what relative or dominant contributions are carried out by PARP-1 vs. PARP-2 vs. PARP-3 enzymes, since all enzymes are involved in sensing DNA strand breaks and recruit to DNA damage sites, while PARP-1 and PARP-2 both promote PARylation of chromatin factors, while all existing clinical PARPi molecules barely distinguish between the two related PAR polymerase enzymes PARP-1 and PARP-2. In turn, PARP trapping is thought to lead to DNA replication stress, genomic instability and cell death in cancer cells (Lord and Ashworth, 2012). Enhanced trapping of PARP1, for example, is thought to lead to an increased ability to kill cancer cells, especially cancers with defective DNA repair pathways (Zandarashvili et al., 2020). FIG. **1** shows the cytotoxic mechanism of PARP trapping via PARPi.

[0007] "PARP trapping" is thus described as an (enhanced) association of PARP-1 and/or PARP-2 and/or PARP-3 with chromatin in living cells. Using PARPi, the allosteric mechanism that contributes to binding of PARP-1 to DNA can be disrupted (Zandarashvili et al., 2020), with some PARPi contributing to retention and others facilitating pro-release mechanisms based on in vitro PARP-1, PARPi and DNA interactions (Zandarashvili et al., 2020). U2OS cells treated with talazoparib show an enhanced retention of GFP-tagged PARP1 and PARP2 at induced DNA lesions, whereas cells treated with veliparib reveal overall less PARP1/PARP2 recruitment to the DNA lesions.

[0008] Either PARP-1 or PARP-2 are necessary for a sufficient recruitment of SSB repair proteins to the damage site. Via PARylation, chromatin remodeling or histone-modifying enzymes are activated at DNA damage sites, which leads to changes in chromatin compaction (Luijsterburg et al., 2016; Mehrotra et al., 2011; Sellou et al., 2016; Smeenk et al., 2013; Timinszky et al., 2009).

[0009] Activation of PARP-1 and/or PARP-2 enzymes leads to the recruitment of many proteins, notably also specific chromatin remodeling enzymes, notably including the macrodomain-containing nucleosome remodeler ALC1 (CHD1L) (Ahel et al., 2009; Gottschalk et al., 2009; Lehmann et al., 2017; Singh et al., 2017). Macrodomains generally bind ADP-ribose, oligo-ADP-ribose and poly-ADP-ribose (Karras et al., 2005), thus proteins containing macrodomains respond and recruit to PARP activation sites on the genome, including during DNA damage and with relevance for cancer. Importantly, PAR or oligo-ADP-ribose binding to the macrodomains of ALC1 robustly turns on chromatin remodeling activity (Ahel et al., 2009; Gottschalk et al., 2009; Lehmann et al., 2017; Singh et al., 2017), revealing ALC1 as an allosterically-regulated chromatin remodeling enzyme, the first of its kind, and one of the very few enzymes whose catalytic activity is directly regulated by PAR. Additionally, ALC1 is a validated oncogene and is often genetically amplified together with PARPI in BRCA1/2-deficient ovarian and breast cancer samples (see FIG. **2**). ALC1 inhibitors, such as small molecules inhibiting the ATPase function and or nucleosome remodeling functions of ALC1, could potentiate the effect PARPi, lead to enhanced cancer cell killing and/or reduce off-target effects and thus lessen cellular toxicity in non-cancer cells. The fact that altering the expression level of ALC1 (by a CRISPR-based knockout) impacts the sensitivity of cancer cells to PARP inhibitors, also opens up the opportunity that altering the activity levels of ALC1 could overcome PARP inhibitor resistance, since removing ALC1 robustly potentiates the PARPi Olaparib to a level that may be sufficient to circumvent PARPi resistance, such as upon (but not limited to) reversion of the BRCA-deficiency status (e.g. by internal deletions or through loss of epigenetic BRCA1/2 gene silencing). Since the sensitivity of cancer cells to PARP inhibitors is generally accepted to (also) stem from the ability of PARPi to trap PARP1 on chromatin (and likely also trapping PARP2 enzymes—this has not been formally established in the field), we hypothesized that small molecule ALC1 inhibitors may mediate PARPi sensitization. Specifically, we hypothesized that inhibition of ALC1 with small molecules would promote DNA damage accumulation and cancer cell killing.

[0010] Since ALC1 is upregulated in several tumors and is a validated oncogene in hepatocellular carcinoma (Cheng et al., 2013; Li et al., 2019; Su et al., 2014), inhibition of ALC1 via small molecule inhibitors could drive robust changes in DNA damage responses.

[0011] The present inventors hypothesized that manipulation of ALC1 activity via small molecules could induce strong anti-proliferative effects and in addition be sufficient to bypass a low or high level of PARPi resistance. Thus, in addition to a potential use as a monotherapy, combined ALC1 and PARP-1/2 inhibition could be exploited to refine PARP-targeted therapies in oncology.

[0012] This hypothesis led to a part of the present invention that ALC1 manipulation via small molecule inhibitors impacts the response to DNA damage. These compounds, designed to inhibit the enzymatic activity of the ATP-dependent chromatin remodeler ALC1 (CHD1L), will potentiate accumulation of DNA damage and thus to mediate synthetic lethality upon BRCA deficiency or more generally, in HRD high tumors, since PARP inhibition has been well documented to be synthetic lethal with loss of BRCA-1/2 tumor suppressor function. Based on preclinical and clinical evidence, ALC1 inhibition could therefore serve as an additional therapeutic approach for cancers that have intact HR pathways, and/or allow the targeting of ALC1-amplified tumors by disrupting ALC1's oncogene function. Also, since the ALC1 gene is a key mediator of PARP-chromatin rearrangements upon induced DNA damage (Sellou et al., 2016), small molecules targeting ALC1 activity may impact the nuclear, DNA-damage relevant functions of PARP-1/2/3 on chromatin rearrangements without impacting other roles of PARP-1/2/3 inside or outside of the cell's nucleus or without impacting non-DNA-damage induced PARP enzymes, which may result in a "second-generation PARPi" with reduced off-target effects and/or reduced side-effects.

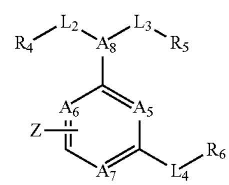
[0013] Given the above-described relevance of ALC1 for various proliferative diseases, in particular those that are BRCA1/2 deficient, the present invention provides a novel class of compounds to treat or ameliorate tumor diseases and in particular tumor diseases characterized by increased activity of ALC1, e.g. due to increased expression.

[0014] Furthermore, the present inventors determined that by using a combination of PARPi and an ALC1 inhibitor, preferably the allosteric ALC1 inhibitors of the present invention, the effect of PARPis can surprisingly be enhanced. Thus, the present invention provides inter alia (i) an efficient therapy of tumors that are HRD deficient and hence sensitive to PARPi, (ii) mediate PARPi sensitization, (iii) bypass PARPi resistance, (iv) and/or allow the reduction of the amount of PARPi that is administered.

SUMMARY OF THE INVENTION

[0015] In a first aspect, the present invention is directed at an inhibitor of ALC1 (ALC1i) according to formula (I)

Formula I



[0016] wherein

[0017] A5 and A8 are each independently selected from N or CH;

[0018] A6 is selected from N or CH, or when A6 takes part in the annulated carbo- or heterocycle Z, then A6 is C;

[0019] A7 is selected from N or CH, or when A7 takes part in the annulated carbo- or heterocycle Z, then A7 is C;

[0020] L2 is selected from the group consisting of —CH_2—R4, —CF_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4, —O—R4, —NH—R4, —N═R4;

[0021] L3 is selected from the group consisting of CH_2—R5, —CF_2—R5, —CH_2—CH_2—R5, —CH_2—CF_2—R5, —CH_2—CH_2—CH_2—R5, —O—R5, —NH—R5, —N═R5;

[0022] or

[0023] L2 and L3 together with the A8 to which they are connected form a 5- or 6-membered heterocycle substituted by R4 and/or R5;

[0024] L4 is CH_2, —CF_2—, CH_2—CH_2, CH_2—CH_2—CH_2, O, N, and NH, or is absent;

[0025] Z is a 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three, (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0026] and can be annulated to the central core or connected via a covalent bond

[0027] R4 is 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three, (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —CF_3, Me, Et, —OMe, and —SMe, or R4 is hydrogen, methyl, or COOH;

[0028] R5 is a 4-, 5-, 6-, 7-, 8-, 9- or 10-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe;

[0029] R6 is a 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe;

[0030] or R6 is H;

[0031] or

[0032] when A7 takes part in the annulated carbo- or heterocycle Z then A5 and A6 are independently selected from —N or —CH and A8 is selected from —N, —CH, —CH_2—N, —CH_2—CH, or —NH—CH;

[0033] or pharmaceutically acceptable salt thereof,

[0034] for treating or ameliorating a proliferative disease in a patient.

[0035] In a second aspect the present invention relates to a bifunctional compound comprising the ALC1i for use as specified for the first aspect of the invention and a compound which recruits E3 ubiquitin ligase to ALC1(E3 recruiter), wherein the ALC1i and the E3 recruiter are covalently linked, optionally through a linker.

[0036] In a third aspect the present invention relates to a pharmaceutical composition comprising the ALC1i for use of the first aspect of the invention or the bifunctional compound of the second aspect of the invention and one or more pharmaceutically acceptable excipients.

[0037] In a fourth aspect the present invention relates to an ALC1i for use of the first aspect of the invention or the bifunctional compound of the second aspect of the invention for use in treating or ameliorating a proliferative disease, wherein the treating and ameliorating the proliferative disease comprises the administration of said ALC1i or of said bifunctional compound and the administration of a Poly (ADP-ribose)-Polymerase inhibitor (PARPi).

[0038] In a fifth aspect the present invention relates to a PARPi for use in treating or ameliorating a proliferative disease in a patient, wherein the treating and ameliorating the proliferative disease comprises the administration of said PARPi and the administration of the ALC1i for use of the first aspect of the invention or of the bifunctional compound of the second aspect of the invention.

[0039] In a sixth aspect the present invention relates to a kit of parts comprising separately packaged a PARPi and an ALC1i for use of the first aspect of the invention or the bifunctional compound of the second aspect of the invention or a composition comprising a PARPi and an ALC1i, preferably with instructions for use to treat or ameliorate a proliferative disease.

DESCRIPTION OF THE FIGURES

[0040] In the following, the content of the Figures comprised in this specification is described. In this context please also refer to the detailed description of the invention above and/or below.

[0041] FIG. **1**: Cytotoxic mechanisms of PARPi on DNA repair pathways leading to PARP trapping. Upper pathway shows interference with DNA repair of single strand breaks (SSBs) via DNA replication fork damage leading to repair via the homologous recombination (HR) mechanism. Lower pathway shows trapping of PARP1/2 proteins on damaged DNA, leading to replication fork damage utilizing additional repair pathways including Fanconi pathway (FA), template switching (TS), ATM, FENI (replicative flap endonuclease) and DNA polymerase f (Murai et al., 2012, S. 5591).

[0042] FIG. **2**: The chromatin remodeler ALC1 (CHD1L) is frequently co-amplified with the PARP1 gene in human ovarian and breast cancer samples. Genomic alterations of ALC1 (CHD1L), PARPI, PARP2, BRCA1, BRCA2 and the most closely related chromatin remodeler CHD1 among the genomes of 10792 breast, fallopian and ovarian cancer patients (OncoPrint analysis conducted on Aug. 12, 2020 at the cBioPortal—www.cbioportal.org). The percentage numbers indicate the percentage of alterations in a particular gene for all genomes where the specific gene has been profiled. Gene amplifications (black), deep gene deletions (dark grey) are highlighted. Compared to CHD1, ALC1 is often amplified together with PARP1 (ALC1-PARP1: Odd's ratio (OR)=1.581, q-value=<0.001; CHD1-PARP1: OR=–0.125, q-value=0.250). Neither ALC1, nor PARP1 exhibit a deep deletion or frequent mutation when the tumors exhibit a deep deletion or mutation in the BRCA1 or BRCA2 tumor suppressor genes (ALC1-BRCA1/2: OR=<–3/-2.178, q-value=<0.001; PARP1-BRCA1/2: OR=<–3, q-value=<0.001).

[0043] FIG. **3**: 96 hour SRB assay of BRCA positive and BRCA negative cells treated with ALCi. MDA-MB-231 cells (BRCA1/2 wildtype) and SUM-149-PT cells (BRCA1 negative) cells were seeded into 96-well plates and treated with titrations of different ALCi starting at 50 μM. The cells were cultured at 37° C., CO_2 5% for 4 days, fixed with 10% TCA and stained with sulforhodamine staining to analyse cell survival. The data was normalized to DMSO controls indicating 100% survival. Inhibitor vs. response curves with variable slope (four parameters) were fitted using GraphPad Prism. Standard deviation bars are shown for two technical replicates.

[0044] FIG. **4**: PARPi co-treatment Cell proliferation assay of pancreatic cancer cells. PSN-1 cells were seeded into 96-well plates and treated with a 2-D titration of PARPi vs. ALC1i. The cells were cultured at 37° C., CO_2 5% for 4 days, fixed with 10% TCA and stained with Sulforhodamine staining to analyze cell survival. Single agent titration curves are shown on the left, a dose response matrix is shown in the middle and a synergy score calculation is shown on the right. Treatment with ALC1i-1002 and Talazoparib, Olaparib, or Niraparib shows most synergsistic area scores of 14.28, 13.27, and 15.41 respectively. ZIP synergy scores over 0.0 are an indication of strong synergy.

[0045] FIG. **5**: Structures and associated compound codes of ALC1 inhibitors.

[0046] FIG. **6**: Nucleosome remodeling inhibition. IC_{50}s (μM) of ALC1 inhibitors in a FRET based nucleosome remodelling assay. IC_{50} values are presented by the following symbols: +++: IC_{50}<25 μM; ++: IC_{50}=25-100 μM; +: IC_{50}=>100-250 μM; and '-': compound is not active, i.e. never observe inhibition >50%.

[0047] FIG. **7**: Cell proliferation inhibition EC50s (μM) of ALC1 inhibitors in a 4 day cell proliferation assay with an

SRB based readout. EC_{50} values are presented by the following symbols: +++: EC_{50}<10 μM; ++: EC_{50}=10-25 μM; +: EC_{50}=>25-50 μM; and '-': compounds is not active, i.e. never observe inhibition >50%.

[0048] FIG. **8**: A table containing the supplier used for each of the inhibitors.

[0049] FIG. **9**: A general synthesis scheme for inhibitors according to formula I.

[0050] FIG. **10**: A general synthesis scheme for inhibitors according to formula I.

[0051] FIG. **11**: A table containing the percentage inhibition of ALC1 in the FRET-based nucleosome sliding assay at a compound concentration of 250 μM for each of the inhibitors. For some compounds, the concentration of 250 μM had to be lowered due to solubility problems as indicated in the legend at the bottom of the table.

[0052] FIG. **12**: EC50 values of 96 hour SRB assay and 11 days colony formation assay of HR proficient (HRP) and HR deficient (HRD) cells treated with ALCi. Different cancer cell lines with a variation of deleterious mutations (downloaded from https://depmap.org/portal/) in the homologous recombination repair pathways (HR) were seeded into 96-well plates and treated with titrations of different ALCi starting at 50 μM. The cells were cultured at 37° C., CO_2 5% for 4 days (96h) or 11 days (colony formation), fixed with 10% TCA and stained with sulforhodamine staining to analyse cell survival. The data was normalized to DMSO controls indicating 100% survival. Inhibitor vs. response curves with variable slope (four parameters) were fitted using GraphPad Prism. Standard deviation bars are shown for two technical replicates.

[0053] FIG. **13**: List of genes involved in DNA Damage Response (DDR) (Human DNA Repair Genes, n.d.) cited in Wood R D, Mitchell M, & Lindahl T Mutation Research, 2005, in Science, 2001, in the reference book DNA Repair and Mutagenesis, 2nd edition, 2006, and in Nature Reviews Cancer, 2011 (modified by R. Wood and M. Lowery on Wednesday 10 Jun. 2020). FIG. **14**: list of genes related to the homologous recombination repair pathway (HR). Summary of genes from Toh & Ngeow, 2021; Kim et al., 2021; Yamamoto & Hirasawa, 2021 and the HRD-signature genes from Pent et al, 2021.

DETAILED DESCRIPTION OF THE INVENTION

[0054] Before the present invention is described in detail below, it is to be understood that this invention is not limited to the particular methodology, protocols and reagents described herein as these may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to limit the scope of the present invention which will be limited only by the appended claims. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art.

[0055] Several documents are cited throughout the text of this specification. Each of the documents cited herein (including all patents, patent applications, scientific publications, manufacturer's specifications, instructions etc.), whether supra or infra, is hereby incorporated by reference in its entirety. Nothing herein is to be construed as an admission that the invention is not entitled to antedate such disclosure by virtue of prior invention. Some of the documents cited herein are characterized as being "incorporated by reference". In the event of a conflict between the definitions or teachings of such incorporated references and definitions or teachings recited in the present specification, the text of the present specification takes precedence.

[0056] In the following, the elements of the present invention will be described. These elements are listed with specific embodiments, however, it should be understood that they may be combined in any manner and in any number to create additional embodiments. The variously described examples and preferred embodiments should not be construed to limit the present invention to only the explicitly described embodiments. This description should be understood to support and encompass embodiments which combine the explicitly described embodiments with any number of the disclosed and/or preferred elements. Furthermore, any permutations and combinations of all described elements in this application should be considered disclosed by the description of the present application unless the context indicates otherwise.

Definitions

[0057] To practice the present invention, unless otherwise indicated, conventional methods of chemistry, biochemistry, and recombinant DNA techniques are employed which are explained in the literature in the field (cf., e.g., Molecular Cloning: A Laboratory Manual, 2nd Edition, J. Sambrook et al. eds., Cold Spring Harbor Laboratory Press, Cold Spring Harbor 1989).

[0058] In the following, some definitions of terms frequently used in this specification are provided. These terms will, in each instance of its use, in the remainder of the specification have the respectively defined meaning and preferred meanings.

[0059] As used in this specification and the appended claims, the singular forms "a", "an", and "the" include plural referents, unless the content clearly dictates otherwise.

[0060] The term "Chromodomain-helicase-DNA-binding protein 1-like" abbreviated CHD1L refers to a protein that is also termed ALC1. The amino acid sequence of human ALC1 is as specified in SEQ ID NO: 1. The 897 amino acid residues long protein consists of an N-terminal Snf2-like DNA dependent ATPase domain spanning amino acid residues 40 to 513, which contains the conserved helicase motifs critical for catalysis (Flaus et al., 2006). This domain is composed of two RecA like lobes ranging from amino acid residues 48 to 261 and 351 to 513, respectively. The structure of a truncated N-terminal lobe of the ATPase domain has been determined by homology modeling in order to identify putative allosteric binding sites, a minimal coordinate file of this model is provided as FIG. **20** to allow the skilled person to identify model compounds within the allosteric binding pocket defined for the first time by the present inventors. The allosteric binding pocket is spatially separated from that part of ALC1 involved in binding ATP. The ATPase domain is followed by a linker region ranging from amino acid residues 514 to 703, which contains a putative coiled-coil region (amino acid residues 638 to 675), and a C-terminal macrodomain (amino acid residues 704 to 897). The macrodomain has been shown to directly interact with the ATPase domain, thereby inhibiting its catalytic function (Lehmann et al., 2017; Singh et al., 2017). This interaction is released upon poly(ADP-ribose) binding to the macrodomain, leading to an activation of the chromatin remodelling enzyme.

[0061] The term "alkyl" as used in the context of the present invention refers to a saturated straight or branched carbon chain. Preferably, the chain comprises from 1 to 10 carbon atoms, i.e. 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10, e.g. methyl, ethyl propyl (n-propyl or iso-propyl), butyl (n-butyl, iso-butyl, sec-butyl, tert-butyl), pentyl, hexyl, heptyl, octyl, nonyl, decyl. Alkyl groups are optionally substituted.

[0062] The term "heteroalkyl" as used in the context of the present invention refers to a saturated straight or branched carbon chain. Preferably, the chain comprises from 1 to 9 carbon atoms, i.e. 1, 2, 3, 4, 5, 6, 7, 8, or 9, e.g. methyl, ethyl, propyl, iso-propyl, butyl, iso-butyl, sec-butyl, tert-butyl, pentyl, hexyl, heptyl, octyl, nonyl, which is interrupted one or more times, e.g. 1, 2, 3, 4, 5, with the same or different heteroatoms. Preferably, the heteroatoms are selected from O, S, and N, e.g. $—(CH_2)_n—X—(CH_2)_mCH_3$, with n=0, 1, 2, 3, 4, 5, 6, 7, 8, or 9, m=0, 1, 2, 3, 4, 5, 6, 7, 8, or 9 and X═S, O or NR' with R'═H or hydrocarbon (e.g. C_1 to C_6 alkyl). In particular "heteroalkyl" refers to $—O—CH_3$, $—OC_2H_5$, $—CH_2—O—CH_3$, $—CH_2—O—C_2H_5$, $—CH_2—O—C_3H_7$, $—CH_2—O—C_4H_9$, $—CH_2—O—C_5H_{11}$, $—C_2H_4O—CH_3$, $—C_2H_4—O—C_2H_5$, $—C_2H_4—O—C_3H_7$, $—C_2H_4—O—C_4H_9$ etc. Heteroalkyl groups are optionally substituted.

[0063] The term "haloalkyl" refers to a saturated straight or branched carbon chain in which one or more hydrogen atoms are replaced by halogen atoms, e.g. by fluorine, chlorine, bromine or iodine. Preferably, the chain comprises from 1 to 10 carbon atoms, i.e. 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10. In particular, "haloalkyl" refers to $—CH_2F$, $—CHF_2$, $—CF_3$, $—C_2H_4F$, $—C_2H_3F_2$, $—C_2H_2F_3$, $—C_2HF_4$, $—C_2F_5$, $—C_3H_6F$, $—C_3H_5F_2$, $—C_3H_4F_3$, $—C_3H_3F_4$, $—C_3H_2F_5$, $—C_3HF_6$, $—C_3F_7$, $—CH_2Cl$, $—CHCl_2$, $—CCl_3$, $—C_2H_4Cl$, $—C_2H_3Cl_2$, $—C_2H_2Cl_3$, $—C_2HCl_4$, $—C_2Cl_5$, $—C_3H_6Cl$, $—C_3H_5Cl_2$, $—C_3H_4Cl_3$, $—C_3H_3Cl_4$, $—C_3H_2Cl_5$, $—C_3HCl_6$, and $—C_3Cl_7$. Haloalkyl groups are optionally substituted.

[0064] The term "carbocycle" is used in the context of the present invention to refer to mono-, bi or tricyclic "cycloalkyl", "cycloalkenyl", "spiroalkyl" "sprioalkenyl" or "aryl" wherein the ring is formed by carbon atoms. Carbocycle groups are optionally substituted.

[0065] The term "5, 6, or 7 membered carbocycle" is used in the context of the present invention to refer to "cycloalkyl", "cycloalkenyl", "spiroalkyl" or "aryl", preferably "cycloalkyl", or "cycloalkenyl" with 5, 6, or 7 carbon atoms forming a ring or phenyl.

[0066] The term "cycloalkyl" includes cyclopentyl, cyclohexyl, and cycloheptyl. Cycloalkyl groups are optionally substituted.

[0067] The term "cycloalkenyl" includes cyclopentenyl, cyclohexenyl, and cycloheptenyl. Cycloalkenyl groups are optionally substituted.

[0068] The term "aryl" refers to phenyl. Aryl is optionally substituted, e.g. naphthyl.

[0069] The term "heterocycle" is used in the context of the present invention to refer to mono or bicyclic "heterocycloalkyl" or mono- or bicyclic "heteroaryl"; wherein at least one of the carbon atoms are replaced 1, 2, or 3 for the monocyclic heterocycle or 1, 2, 3, or 4 for the bicyclic heterocycle of the same or different heteroatoms, preferably selected from O, N and S.

[0070] The term "5, 6, or 7 membered heterocycle" is used in the context of the present invention to refer to monocyclic "5, 6, or 7 membered heterocycloalkyl" or monocyclic "5, 6, or 7 membered heteroaryl" with 5, 6, or 7 atoms forming a ring.

[0071] The term "5, 6, or 7 membered heterocycloalkyl" refers to a saturated monocycle, wherein at least one of the carbon atoms are replaced by 1, or 2 (for the five membered ring) or 1, 2, or 3 (for the six membered ring) or 1, 2, 3, or 4 (for the seven membered ring) of the same or different heteroatoms, preferably selected from O, N and S. Preferred examples of heterocycloalkyl include 1-(1,2,5,6-tetrahydropyridyl), 1-piperidinyl, 2-piperidinyl, 3-piperidinyl, 4-morpholinyl, 3-morpholinyl, tetrahydrofuran-2-yl, tetrahydrofuran-3-yl, tetrahydrothien-2-yl, tetrahydrothien-3-yl, 1-piperazinyl, or 2-piperazinyl. Heterocycloalkyl groups are optionally substituted.

[0072] The term "heteroaryl" as used in the context of the present invention refers to a 5, 6 or 7-membered aromatic monocyclic ring wherein at least one of the carbon atoms are replaced by 1, 2, or 3 (for the five membered ring) or 1, 2, 3, or 4 (for the six membered ring) of the same or different heteroatoms, preferably selected from O, N and S. Examples of preferred heteroaryls are furanyl, thienyl, oxazolyl, isoxazolyl, 1,2,5-oxadiazolyl, 1,2,3-oxadiazolyl, pyrrolyl, imidazolyl, pyrazolyl, 1,2,3-triazolyl, thiazolyl, isothiazolyl, 1,2,3,-thiadiazolyl, 1,2,5-thiadiazolyl, pyridinyl, pyrimidinyl, pyrazinyl, 1,2,3-triazinyl, 1,2,4-triazinyl, 1,3,5-triazinyl. Heteroaryls groups are optionally substituted.

[0073] If two or more radicals can be selected independently from each other, then the term "independently" means that the radicals may be the same or may be different.

[0074] The term "optionally substituted" in each instance if not further specified refers to halogen (in particular F, Cl, Br, or I), $—NO_2$, —CN, —OR''', —NR'R'', —COOR''', —CONR'R'', —NR'COR'', —NR''COR''', —NR'CONR'R'', $—NR'SO_2E$, —COR'''; $—SO_2NR'R''$, —OOCR''', —CR'''R''''OH, —R'''OH, and -E;

[0075] R' and R'' is each independently selected from the group consisting of hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, heterocycloalkyl, aryl, aralkyl, and heteroaryl or together form a heteroaryl, or heterocycloalkyl;

[0076] R''' and R'''' is each independently selected from the group consisting of hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, heterocycloalkyl, alkoxy, aryl, aralkyl, heteroaryl, and —NR'R'';

[0077] E is selected from the group consisting of alkyl, alkenyl, alkynyl, cycloalkyl, alkoxy, alkoxyalkyl, heterocycloalkyl, an alicyclic system, aryl and heteroaryl; optionally substituted.

[0078] "Pharmaceutically acceptable" means approved by a regulatory agency of the Federal or a state government or listed in the U.S. Pharmacopeia (United States Pharmacopeia-33/National Formulary-28 Reissue, published by the United States Pharmacopeia Convention, Inc., Rockville Md., publication date: April 2010) or other generally recognized pharmacopeia for use in animals, and more particularly in humans.

[0079] The term "pharmaceutically acceptable salt" refers to a salt of a compound of the present invention. Suitable pharmaceutically acceptable salts of the compound of the present invention include acid addition salts which may, for example, be formed by mixing a solution of a compound described herein or a derivative thereof with a solution of a pharmaceutically acceptable acid such as hydrochloric acid, sulfuric acid, fumaric acid, maleic acid, succinic acid, acetic acid, benzoic acid, citric acid, tartaric acid, carbonic acid or phosphoric acid. Furthermore, where the compound of the invention carries an acidic moiety, suitable pharmaceutically acceptable salts thereof may include alkali metal salts (e.g., sodium or potassium salts); alkaline earth metal salts (e.g., calcium or magnesium salts); and salts formed with suitable organic ligands (e.g., ammonium, quaternary ammonium and amine cations formed using counteranions such as halide, hydroxide, carboxylate, sulfate, phosphate, nitrate, alkyl sulfonate and aryl sulfonate). Illustrative examples of pharmaceutically acceptable salts include but are not limited to: acetate, adipate, alginate, ascorbate, aspartate, benzenesulfonate, benzoate, bicarbonate, bisulfate, bitartrate, borate, bromide, butyrate, calcium edetate, camphorate, camphorsulfonate, camsylate, carbonate, chloride, citrate, clavulanate, cyclopentanepropionate, digluconate, dihydrochloride, dodecylsulfate, edetate, edisylate, estolate, esylate, ethanesulfonate, formate, fumarate, gluceptate, glucoheptonate, gluconate, glutamate, glycerophosphate, glycolylarsanilate, hemisulfate, heptanoate, hexanoate, hexylresorcinate, hydrabamine, hydrobromide, hydrochloride, hydroiodide, 2-hydroxyethanesulfonate, hydroxynaphthoate, iodide, isothionate, lactate, lactobionate, laurate, lauryl sulfate, malate, maleate, malonate, mandelate, mesylate, methanesulfonate, methylsulfate, mucate, 2-naphthalenesulfonate, napsylate, nicotinate, nitrate, N-methylglucamine ammonium salt, oleate, oxalate, pamoate (embonate), palmitate, pantothenate, pectinate, persulfate, 3-phenylpropionate, phosphate/diphosphate, picrate, pivalate, polygalacturonate, propionate, salicylate, stearate, sulfate, subacetate, succinate, tannate, tartrate, teoclate, tosylate, triethiodide, undecanoate, valerate, and the like (see, for example, Berge, S. M., et al, "Pharmaceutical Salts", Journal of Pharmaceutical Science, 1977, 66, 1-19). Certain specific compounds of the present invention contain both basic and acidic functionalities that allow the compounds to be converted into either base or acid addition salts.

[0080] The neutral forms of the compounds may be regenerated by contacting the salt with a base or acid and isolating the parent compound in the conventional manner. The parent form of the compound differs from the various salt forms in certain physical properties, such as solubility in polar solvents, but otherwise the salts are equivalent to the parent form of the compound for the purposes of the present invention.

[0081] In addition to salt forms, the present invention provides compounds which are in a prodrug form. Prodrugs of the compounds described herein are those compounds that readily undergo chemical changes under physiological conditions to provide a compound of formula (I) to (IV), and especially a compound shown in FIG. **14**. A prodrug is an active or inactive compound that is modified chemically through in vivo physiological action, such as hydrolysis, metabolism and the like, into a compound of this invention following administration of the prodrug to a patient. Additionally, prodrugs can be converted to the compounds of the present invention by chemical or biochemical methods in an ex vivo environment. For example, prodrugs can be slowly converted to the compounds of the present invention when placed in a transdermal patch reservoir with a suitable enzyme. The suitability and techniques involved in making and using prodrugs are well known by those skilled in the art. For a general discussion of prodrugs involving esters, see Svensson L. A. and Tunek A. (1988) Drug Metabolism Reviews 19(2): 165-194 and Bundgaard H. "Design of Prodrugs", Elsevier Science Ltd. (1985). Examples of a masked carboxylate anion include a variety of esters, such as alkyl (for example, methyl, ethyl), cycloalkyl (for example, cyclohexyl), aralkyl (for example, benzyl, p-methoxybenzyl), and alkylcarbonyloxyalkyl (for example, pivaloyloxymethyl). Amines have been masked as arylcarbonyloxymethyl substituted derivatives which are cleaved by esterases in vivo releasing the free drug and formaldehyde (Bundgaard H. et al. (1989) J. Med. Chem. 32(12): 2503-2507). Also, drugs containing an acidic NH group, such as imidazole, imide, indole and the like, have been masked with N-acyloxymethyl groups (Bundgaard H. "Design of Prodrugs", Elsevier Science Ltd. (1985)). Hydroxy groups have been masked as esters and ethers. EP 0 039 051 A2 discloses Mannich-base hydroxamic acid prodrugs, their preparation and use.

[0082] The compounds of the present invention may also contain unnatural proportions of atomic isotopes at one or more of the atoms that constitute such compounds. For example, the compounds may be radiolabeled with radioactive isotopes, such as for example tritium (^{3}H), iodine-125 (^{125}I) or carbon-14 (^{14}C). All isotopic variations of the compounds of the present invention, whether radioactive or not, are intended to be encompassed within the scope of the present invention.

[0083] As used herein, "para position" when referring to the substituent of an aryl means that the substituent occupies the position opposite to the position at which the aryl is linked to the backbone of the compound.

[0084] As used herein, a "patient" means any mammal or bird that may benefit from a treatment with the compounds described herein. Preferably, a "patient" is selected from the group consisting of laboratory animals, domestic animals, or primates including chimpanzees and human beings. It is particularly preferred that the "patient" is a human being.

[0085] As used herein, "treat", "treating" or "treatment" of a disease or disorder means accomplishing one or more of the following: (a) reducing the severity of the disorder; (b) limiting or preventing development of symptoms characteristic of the disorder(s) being treated; (c) inhibiting worsening of symptoms characteristic of the disorder(s) being treated; (d) limiting or preventing recurrence of the disorder (s) in patients that have previously had the disorder(s); and (e) limiting or preventing recurrence of symptoms in patients that were previously symptomatic for the disorder (s).

[0086] As used herein, "prevent", "preventing", "prevention", or "prophylaxis" of a disease or disorder means preventing that a disorder occurs in a subject for a certain amount of time. For example, if a compound described herein is administered to a subject with the aim of preventing a disease or disorder, said disease or disorder is prevented from occurring at least on the day of administration and preferably also on one or more days (e.g. on 1 to 30 days; or on 2 to 28 days; or on 3 to 21 days; or on 4 to 14 days; or on 5 to 10 days) following the day of administration.

[0087] A "pharmaceutical composition" according to the invention may be present in the form of a composition, wherein the different active ingredients and diluents and/or carriers are admixed with each other, or may take the form of a combined preparation, where the active ingredients are present in partially or totally distinct form. An example for such a combination or combined preparation is a kit-of-parts.

[0088] An "effective amount" is an amount of a therapeutic agent sufficient to achieve the intended purpose. The effective amount of a given therapeutic agent will vary with factors such as the nature of the agent, the route of administration, the size and species of the animal to receive the therapeutic agent, and the purpose of the administration. The effective amount in each individual case may be determined empirically by a skilled artisan according to established methods in the art.

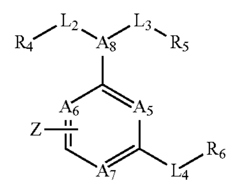
[0089] The term "carrier", as used herein, refers to a diluent, adjuvant, excipient, or vehicle with which the therapeutic agent is administered. Such pharmaceutical carriers can be sterile liquids, such as saline solutions in water and oils, including those of petroleum, animal, vegetable or synthetic origin, such as peanut oil, soybean oil, mineral oil, sesame oil and the like. A saline solution is a preferred carrier when the pharmaceutical composition is administered intravenously. Saline solutions and aqueous dextrose and glycerol solutions can also be employed as liquid carriers, particularly for injectable solutions. Suitable pharmaceutical excipients include starch, glucose, lactose, sucrose, gelatine, malt, rice flour, chalk, silica gel, sodium stearate, glycerol monostearate, talc, sodium chloride, dried skim milk, glycerol, propylene glycol, water, ethanol and the like. The composition, if desired, can also contain minor amounts of wetting or emulsifying agents, or pH buffering agents. These compositions can take the form of solutions, suspensions, emulsions, tablets, pills, capsules, powders, sustained-release formulations and the like. The composition can be formulated as a suppository, with traditional binders and carriers such as triglycerides. The compounds of the invention can be formulated as neutral or salt forms. Pharmaceutically acceptable salts include those formed with free amino groups such as those derived from hydrochloric, phosphoric, acetic, oxalic, tartaric acids, etc., and those formed with free carboxyl groups such as those derived from sodium, potassium, ammonium, calcium, ferric hydroxides, isopropylamine, triethylamine, 2-ethylamino ethanol, histidine, procaine, etc. Examples of suitable pharmaceutical carriers are described in "Remington's Pharmaceutical Sciences" by E. W. Martin. Such compositions will contain a therapeutically effective amount of the compound, preferably in purified form, together with a suitable amount of carrier so as to provide the form for proper administration to the patient. The formulation should suit the mode of administration.

Embodiments of the Invention

[0090] The present inventors have identified and characterized ALC1 inhibitors that appear to be involved in allosteric regulation of the nucleosome sliding activity of ALC1. These compounds specifically bind to an allosteric pocket and are capable of inhibiting activity of ALC1. Compounds that bind to the ATPase site of ALC1 and block the ATPase activity have to compete with ATP for binding to the ATPase site. Since the cellular ATP concentration is in the range of 1 to 10 mM depending on the cellular compartment, very high binding affinities in the low nanomolar range are required to successfully prevent ATP from binding to the ATPase site of ALC1. Allosteric inhibitors of ALC1 do not have this limitation since they do not have to prevent ATP from binding but inhibit ALC1's ATPase activity through a different mechanism. The present inventors have identified compounds that are capable of specifically binding to an allosteric pocket of ALC1. These compounds were also tested for their ability to kill two different tumor cell lines one of which was BRCA deficient.

[0091] Accordingly, in a first aspect the present invention relates to the use of an ALC1i of formula (I):

Formula I



[0092] wherein

[0093] A5 and A8 are each independently selected from N or CH;

[0094] A6 is selected from N or CH, or when A6 takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle, then A6 is C;

[0095] A7 is selected from N or CH, or when A7 takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle, then A7 is C;

[0096] L2 is selected from the group consisting of —CH_2—R4, —CF_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4, —O—R4, —NH—R4, —N═R4;

[0097] L3 is selected from the group consisting of CH_2—R5, —CF_2—R5, —CH_2—CH_2—R5, —CH_2—CF_2—R5, —CH_2—CH_2—CH_2—R5, —O—R5, —NH—R5, —N═R5;

[0098] or

[0099] L2 and L3 together with the A8 to which they are connected form a 5- or 6-membered heterocycle substituted by R4 and/or R5;

[0100] L4 is CH_2, —CF_2—, CH_2—CH_2, CH_2—CH_2—CH_2, 0, N, and NH, or is absent;

[0101] Z is a 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0102] and can be annulated to the central core or connected via a covalent bond

[0103] R4 is 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —CF_3, Me, Et, —OMe, and —SMe, or R4 is hydrogen, methyl, or COOH;

[0104] R5 is a 4-, 5-, 6-, 7-, 8-, 9- or 10-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe;

[0105] R6 is a 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe;

[0106] or R6 is H;

[0107] or

[0108] when A7 takes part in the annulated carbo- or heterocycle Z then A5 and A6 are independently selected from —N or —CH and A8 is selected from —N, —CH, —CH_2—N, —CH_2—CH, or —NH—CH; or pharmaceutically acceptable salt thereof

[0109] for treating or ameliorating a proliferative disease in a patient.

[0110] The ALC1i according to formula (I) preferably shows inhibition >50% at concentrations at or below 250 μM, preferably has an IC_{50} of <250 μM, and more preferably of <25 μM. The IC_{50} is preferably measured in a FRET based nucleosome remodeling assay as described in the Examples.

[0111] The ALC1i according to formula (I) preferably has an EC_{50} of <250 μM, preferably of <50 μM; and more preferably of <10 μM. The EC_{50} is preferably measured in cell proliferation assay with an SRB based readout as described in the Examples.

[0112] In an embodiment of the first aspect it is preferred that:

[0113] A5 and A6 are N; and

[0114] A7 takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2.

[0115] In an embodiment of the first aspect it is preferred that:

[0116] A5 and A6 are N;

[0117] A7 takes part in the annulated carbo- or heterocycle, preferably carbocycle Z, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0118] A8 is —CH_2—N, —CH_2—CH, or —NH—CH, preferably —NH—CH.

[0119] In an embodiment of the first aspect it is preferred that:

[0120] A5 is N and A8 is —N or —CH.

[0121] In an embodiment of the first aspect it is preferred that:

[0122] A5 is N and A8 is —N or —CH; and

[0123] A6 takes part in the annulated carbo- or heterocycle, preferably carbocycle Z.

[0124] In a particularly preferred embodiment A5, A7 and A8 are N.

[0125] In a particularly preferred embodiment A5, A7 and A8 are N and A6 takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2.

[0126] In a particularly preferred embodiment A5, A7 and A8 are N and A6 takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2.

[0127] In an embodiment of the first aspect it is preferred that:

[0128] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4.

[0129] In an embodiment of the first aspect it is preferred that:

[0130] L2 is selected from the group consisting of —CH_2—R4, —CF_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4, —NH—R4; and

[0131] R4 is 6-membered aryl or 5-, 6- or 7-membered heteroaryl, preferably 5- or 6-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —CF_3, Me, Et, —OMe, and —SMe, or R4 is hydrogen, methyl, or COOH.

[0132] In an embodiment of the first aspect it is preferred that:

[0133] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5.

[0134] In an embodiment it is preferred that:

[0135] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, —CH_2—CF_2—R5; and

[0136] R5 is a 4-, 5-, 6-, 7-, 8-, 9- or 10-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0137] In an embodiment of the first aspect it is preferred that:

[0138] L2 and L3 together with the A8 to which they are connected form a 5- or 6-membered heterocycle substituted by R4 and R5.

[0139] In an embodiment of the first aspect it is preferred that:

[0140] L2 and L3 together with the A8 to which they are connected form a 5- or 6-membered heterocycle substituted by R4 and R5;

[0141] R4 is hydrogen, methyl, or COOH; and

[0142] R5 is a 4-, 5-, 6-, 7-membered carbo- or heterocycle, preferably phenyl or 5- or 6-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0143] In an embodiment of the first aspect it is preferred that

[0144] L4 is absent.

[0145] In a particularly preferred embodiment A5, A7 and A8 are N; and

[0146] L4 is absent.

[0147] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0148] L4 is absent.

[0149] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0150] L4 is absent.

[0151] In a particularly preferred embodiment A5, A7 and A8 are N; and

[0152] L4 is absent

[0153] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0154] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0155] L4 is absent

[0156] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0157] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0158] L4 is absent

[0159] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0160] In a particularly preferred embodiment A5, A7 and A8 are N; and

[0161] L4 is absent

[0162] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0163] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0164] L4 is absent; and

[0165] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0166] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0167] L4 is absent; and

[0168] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0169] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0170] L4 is absent; and

[0171] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0172] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl. which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0173] L4 is absent; and

[0174] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0175] In an embodiment of the first aspect it is preferred that:

[0176] R4 is hydrogen, methyl, COOH or tetrazolyl.

[0177] In an embodiment of the first aspect it is preferred that:

[0178] L2 is —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4; and

[0179] R4 is hydrogen, methyl, COOH or tetrazolyl.

[0180] In an embodiment of the first aspect it is preferred that:

[0181] R5 is any 4-, 5-, 6- 7-, 8-, 9-, or 10-membered carbo- or heterocycle, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —CF_3, Me, Et, —OMe, and —SMe.

[0182] In an embodiment of the first aspect it is preferred that:

[0183] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e.

C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0184] In an embodiment of the first aspect it is preferred that:

[0185] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, —CH_2—CF_2—R5; and

[0186] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0187] In an embodiment of the first aspect it is preferred that:

[0188] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe

[0189] or R6 is H.

[0190] In an embodiment of the first aspect it is preferred that:

[0191] L4 is absent; and

[0192] R6 is a 5-, or 6-membered carbo- or heterocycle, preferably 6-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituents selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0193] In above embodiment it is preferred that:

[0194] Z is a 6 membered aryl, or a 5-, 6-membered heteroaryl, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2.

[0195] In an embodiment of the first aspect, it is preferred that:

[0196] A5 is N;

[0197] one of A6 and A7 is CH and the other one is N; or one of A6 and A7 is C and takes part in the annulated carbo- or heterocycle Z and the other one is N;

[0198] A8 is N or CH;

[0199] L2, L3, L4 are independently from each other selected from the group consisting of CH_2, —CF_2—, CH_2—CH_2, CH_2—CH_2—CH_2, 0, N, and NH, or are absent, or L2 and L3 together with the A8 to which they are connected form a 5- or 6-membered heterocycle substituted by R4 and R5;

[0200] Z is any 5-, 6- or 7-membered carbo- or heterocycle and can be annulated to the central core or connected via a covalent bond and optionally substituted with one, two, or three (preferably one) substituent(s)

[0201] selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0202] R4 is any 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —CF_3, Me, Et, —OMe, and —SMe, or R4 is hydrogen, methyl, or COOH;

[0203] R5 is a 4-, 5-, 6- 7-, 8-, 9-, or 10-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituents selected from the group consisting of —Br, —Cl, —F, —I, —CF_3, Me, Et, —OMe, and —SMe;

[0204] R6 is a 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe,

[0205] or R6 is H

[0206] or

[0207] when A7 takes part in the annulated carbo- or heterocycle Z then A5 and A6 are N and A8 is selected from —N, —CH, —CH_2—N, —CH_2—CH, or —NH—CH, preferably —CH_2—N, —CH_2—CH, or —NH—CH;

[0208] wherein R4, R5, and R6.

[0209] In an embodiment of the first aspect it is preferred that:

[0210] A5 is N;

[0211] one of A6 and A7 is C and takes part in the annulated carbo- or heterocycle Z and the other one is N;

[0212] A8 is N;

[0213] L2 is CH_2—CH_2 and L3 is CH_2—CH_2 or CH_2—CF_2; or L2 and L3 together with the A8 to which they are connected form a 5- or 6-membered heterocycle (preferably piperidine or a pyrrolidine) substituted by R4 and R5;

[0214] L4 is absent;

[0215] Z is a 6-membered carbo- or heterocycle annulated to the central core, wherein Z is optionally substituted with one, two, or three (preferably one) substituents selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and NO_2;

[0216] R4 is COOH or CH_2N_4;

[0217] R5 is a 4-, 5-, 6-, 7-, or 10-membered carbo- or heterocycle, optionally substituted with one, two, or three, preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0218] R6 is a 5- or 6-membered carbo- or heterocycle, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe,

[0219] or R6 is H.

[0220] In an embodiment of the first aspect it is preferred that:

[0221] each one of A5, A7 and A8 is N;

[0222] A6 is C and takes part in the annulated carbo- or heterocycle Z, preferably 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe;

[0223] L2 is CH_2—CH_2—R4;

[0224] L3 is CH_2—CH_2—R4 or —CH_2—CF_2—R5;

[0225] or L2 and L3 together with the A8 to which they are connected form a piperidine ring or a pyrrolidine ring, substituted by R4 and R5;

[0226] L4 is absent;

[0227] Z is phenyl or cyclohexyl annulated to the central core, wherein Z is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —OH, Me, —CF_3, —OMe, and —NO_2,

[0228] R4 is COOH or tetrazolyl;

[0229] R5 is phenyl, cyclobutyl, cyclopentyl or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —CF_3, Me, —CH_2—CF_3, and —OMe;

[0230] R6 is a 6-membered carbo- or heterocycle, preferably phenyl or cyclohexyl; more preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of Br, —Cl, —F, Me, —CF_3, —OMe, and —NO_2.

[0231] In a particularly preferred embodiment A5, A7 and A8 are N;

[0232] L2 is —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4;

[0233] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH; and

[0234] L4 is absent.

[0235] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0236] L2 is —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4;

[0237] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH; and

[0238] L4 is absent.

[0239] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0240] L4 is absent.

[0241] In a particularly preferred embodiment A5, A7 and A8 are N;

[0242] L2 is —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4;

[0243] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0244] L4 is absent; and

[0245] R6 is a phenyl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0246] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0247] L2 is —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4;

[0248] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0249] L4 is absent; and

[0250] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0251] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0252] L2 is —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4;

[0253] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0254] L4 is absent; and

[0255] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0256] In a particularly preferred embodiment A5, A7 and A8 are N;

[0257] L2 is —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4;

[0258] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0259] L4 is absent; and

[0260] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0261] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0262] L2 is —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4;

[0263] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0264] L4 is absent; and

[0265] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0266] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0267] L2 is —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4;

[0268] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0269] L4 is absent; and

[0270] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0271] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0272] L2 is —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4;

[0273] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0274] L4 is absent; and

[0275] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0276] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl. which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0277] L2 is —CH_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4;

[0278] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0279] L4 is absent; and

[0280] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0281] In a particularly preferred embodiment A5, A7 and A8 are N;

[0282] L4 is absent; and

[0283] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0284] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0285] L4 is absent; and

[0286] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0287] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0288] L4 is absent; and

[0289] R5 is C_5 to C_7-cycloalkyl, i.e. C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromo-phenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0290] In a particularly preferred embodiment A5, A7 and A8 are N;

[0291] L4 is absent;

[0292] R5 is C_5 to C_7-cycloalkyl, i.e. C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromo-phenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0293] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0294] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0295] L4 is absent;

[0296] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromo-phenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0297] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0298] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0299] L4 is absent;

[0300] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromo-phenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0301] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0302] In a particularly preferred embodiment A5, A7 and A8 are N;

[0303] L4 is absent;

[0304] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromo-phenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0305] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0306] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0307] L4 is absent;

[0308] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0309] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0310] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0311] L4 is absent;

[0312] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0313] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0314] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0315] L4 is absent;

[0316] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0317] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0318] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl. which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0319] L4 is absent;

[0320] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0321] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0322] In a particularly preferred embodiment A5, A7 and A8 are N; and

[0323] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5

[0324] L4 is absent; and

[0325] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0326] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0327] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0328] L4 is absent; and

[0329] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0330] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0331] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0332] L4 is absent; and

[0333] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, i.e. C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0334] In a particularly preferred embodiment A5, A7 and A8 are N;

[0335] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0336] L4 is absent;

[0337] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, i.e. C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0338] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0339] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0340] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0341] L4 is absent; and

[0342] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0343] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0344] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0345] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0346] L4 is absent; and

[0347] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMeR6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0348] In a particularly preferred embodiment A5, A7 and A8 are N;

[0349] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0350] L4 is absent;

[0351] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0352] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0353] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0354] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0355] L4 is absent;

[0356] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0357] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0358] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0359] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0360] L4 is absent;

[0361] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0362] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0363] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0364] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0365] L4 is absent; and

[0366] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe

[0367] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0368] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl. which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0369] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0370] L4 is absent;

[0371] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0372] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0373] In a particularly preferred embodiment A5, A7 and A8 are N; and

[0374] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0375] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0376] L4 is absent; and

[0377] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0378] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0379] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0380] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0381] L4 is absent; and

[0382] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent

(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0383] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0384] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0385] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0386] L4 is absent; and

[0387] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0388] In a particularly preferred embodiment A5, A7 and A8 are N;

[0389] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0390] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0391] L4 is absent;

[0392] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0393] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0394] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0395] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0396] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0397] L4 is absent; and

[0398] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0399] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0400] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0401] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0402] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0403] L4 is absent; and

[0404] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMeR6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0405] In a particularly preferred embodiment A5, A7 and A8 are N;

[0406] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0407] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0408] L4 is absent;

[0409] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0410] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0411] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0412] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0413] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0414] L4 is absent;

[0415] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0416] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0417] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0418] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0419] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0420] L4 is absent;

[0421] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, i.e. C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0422] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0423] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0424] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0425] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0426] L4 is absent; and

[0427] R5 is C_5 to C_7-cycloalkyl C_4 to C_7-cycloalkyl, i.e. C_4-, i.e. C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromo-phenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe

[0428] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0429] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl. which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0430] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0431] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0432] L4 is absent;

[0433] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, i.e. C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromo-phenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0434] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0435] In a particularly preferred embodiment A5, A7 and A8 are N; and

[0436] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5

[0437] L4 is absent; and

[0438] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0439] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, i.e. C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromo-phenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0440] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0441] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0442] L4 is absent; and

[0443] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0444] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromo-phenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0445] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0446] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0447] L4 is absent; and

[0448] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0449] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0450] In a particularly preferred embodiment A5, A7 and A8 are N;

[0451] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0452] L4 is absent;

[0453] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0454] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0455] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0456] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0457] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0458] L4 is absent; and

[0459] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0460] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0461] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0462] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0463] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0464] L4 is absent; and

[0465] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0466] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMeR6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0467] In a particularly preferred embodiment A5, A7 and A8 are N;

[0468] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0469] L4 is absent;

[0470] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0471] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0472] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0473] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0474] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0475] L4 is absent;

[0476] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0477] R5 C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0478] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0479] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0480] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0481] L4 is absent;

[0482] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0483] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0484] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0485] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0486] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0487] L4 is absent; and

[0488] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0489] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe

[0490] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0491] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl. which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0492] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0493] L4 is absent;

[0494] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0495] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, i.e. C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0496] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0497] In a particularly preferred embodiment A5, A7 and A8 are N; and

[0498] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0499] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0500] L4 is absent; and

[0501] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0502] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, i.e. C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0503] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0504] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0505] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0506] L4 is absent; and

[0507] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0508] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, i.e. C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0509] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0510] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0511] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0512] L4 is absent; and

[0513] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0514] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe.

[0515] In a particularly preferred embodiment A5, A7 and A8 are N;

[0516] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0517] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0518] L4 is absent;

[0519] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0520] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0521] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0522] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0523] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0524] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0525] L4 is absent; and

[0526] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0527] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0528] R6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0529] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0530] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0531] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0532] L4 is absent; and

[0533] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0534] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMeR6 is a 6 membered aryl or 5-, 6- or 7-membered heteroaryl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0535] In a particularly preferred embodiment A5, A7 and A8 are N;

[0536] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0537] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0538] L4 is absent;

[0539] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0540] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0541] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0542] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0543] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0544] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0545] L4 is absent;

[0546] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0547] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0548] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0549] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0550] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0551] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0552] L4 is absent;

[0553] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0554] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0555] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0556] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0557] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0558] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0559] L4 is absent; and

[0560] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0561] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe

[0562] R6 is phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0563] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl. which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2; and

[0564] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0565] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0566] L4 is absent;

[0567] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0568] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0569] R6 is a phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe.

[0570] In a particularly preferred embodiment of the above embodiments:

[0571] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0572] In a particularly preferred embodiment of the above embodiments:

[0573] L4 is absent; and

[0574] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0575] In a particularly preferred embodiment A5, A7 and A8 are N;

[0576] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0577] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0578] L4 is absent;

[0579] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0580] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0581] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0582] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0583] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0584] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0585] L4 is absent;

[0586] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0587] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0588] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0589] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0590] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0591] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0592] L4 is absent;

[0593] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0594] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0595] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br

[0596] In a particularly preferred embodiment A5, A7 and A8 are N;

[0597] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0598] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0599] L4 is absent;

[0600] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0601] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0602] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0603] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0604] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0605] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0606] L4 is absent;

[0607] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0608] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0609] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0610] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0611] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0612] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0613] L4 is absent;

[0614] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0615] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0616] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0617] In a particularly preferred embodiment A5, A7 and A8 are N;

[0618] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0619] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0620] L4 is absent;

[0621] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0622] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0623] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0624] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbo- or heterocycle Z, in particular a 5, 6, or 7 membered carbo- or heterocycle Z, preferably carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, which optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0625] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0626] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0627] L4 is absent;

[0628] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0629] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0630] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F,

—I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0631] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0632] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0633] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0634] L4 is absent;

[0635] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0636] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0637] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0638] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl, which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0639] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0640] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0641] L4 is absent;

[0642] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0643] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0644] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0645] In a particularly preferred embodiment A5, A7 and A8 are N and A6 is C and takes part in the annulated carbocycle Z, in particular a 5, 6, or 7 membered carbocycle Z, preferably phenyl. which is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

[0646] L2 is selected from the group consisting of —CH_2—R4, —CH_2—CH_2—R4, and —CH_2—CH_2—CH_2—R4;

[0647] L3 is selected from the group consisting of —CH_2—R5, —CH_2—CH_2—R5, and —CH_2—CF_2—R5;

[0648] L4 is absent;

[0649] R4 is hydrogen, methyl, COOH or tetrazolyl, preferably COOH;

[0650] R5 is C_4 to C_7-cycloalkyl, i.e. C_4-, C_5-, C_6- or C_7-cycloalkyl, C_6 to C_{10}-bicycloalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-bicycloalkyl, C_6 to C_{10}-spiroalkyl, i.e. C_6-, C_7-, C_8-, C_9- or C_{10}-spiroalkyl, phenyl, 5 or 6 membered heteroaryl, adamantyl, optionally substituted with one, two, or three, preferably one substituent (s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; more preferably cyclopentyl, cyclohexyl, phenyl, bromophenyl, bicyclo[2.2.1]heptyl, spiro[3,3]heptyl, or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe, more preferably unsubstituted phenyl or adamantyl substituted with one, or two substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe; and

[0651] R6 is phenyl ortho substituted with a substituent selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, preferably —Br, —Cl, —F, —I, more preferably Br.

[0652] In a particularly preferred embodiment the compounds of the first and further aspect of the invention have the specific structures as indicated in FIG. **5**.

[0653] The use of bifunctional compounds recruiting proteins involved in targeting proteins for degradation by the proteasome has emerged as a potential therapeutic strategy to degrade proteins that are involved in disease processes. This approach has met particular attention in cancer therapy

(Khan S. et al., 2020 and Bushweller J H, 2019). Such bifunctional compounds are generally referred to as PROteolysis TArgeting Chimeras (PROTACs). The inhibitors of ALC1 of the first aspect of the invention specifically bind to ALC1 and are, thus suitable to recruit a protein that is part of the ubiquitination pathway to ALC1.

[0654] Accordingly, in a second aspect the present invention relates to a bifunctional compound comprising the allosteric inhibitor of ALC1 the first or further aspect of the present invention and a compound which recruits a protein that is part of the ubiquitination pathway to ALC1, preferably E3 ubiquitin ligase to ALC1(E3 recruiter), wherein the allosteric inhibitor of ALC1 and the E3 recruiter are covalently linked, optionally through a linker. In the context of the bifunctional compounds of the present invention that ability of the allosteric inhibitors of ALC1 of the first and further aspect of the invention to specifically bind to ALC1 is of particular importance. It is, thus preferred that the allosteric inhibitors of ACL1 bind to full length human ALC1 with an amino acid sequence according to SEQ ID NO: 1 with a KD of 50 PM, more preferably of 10 μM or lower, more preferably of 5 μM or lower, even more preferably of 1 μM, more preferably of 500 nM, more preferably of 200 nM and even more preferably of 100 nM or lower.

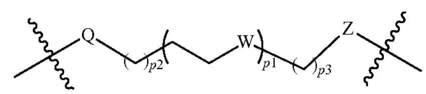
[0655] The protein of the ubiquitination pathway may either be bound by a small molecule or a protein ligand, e.g. an antibody or antibody-like protein, that specifically binds to a protein of the ubiquitination pathway. Such protein ligands have been described, for example in U.S. Pat. No. 7,223,556 Bi. Small molecules compounds that bind to a protein that is part of the ubiquitination pathway are well known in the art and can be used in the bifunctional compounds of the present invention. Examples of such molecules are described in EP 3 131 588, WO 2017/024317, U.S. Pat. Nos. 6,306,663, 7,041,298, US 2016/0176916, US 2016/0235730, US 2016/0235731, US 2016/0243247, WO 2016/105518, WO 2016/077380, WO 2016/105518, WO 2016/077375, WO 2017/007612, and WO 2017/024317.

[0656] In one embodiment the ALC1i is covalently linked to the compound which recruits a protein that is part of the ubiquitination pathway to ALC1. Preferably, the two components are covalently linked to each other through a linker. Suitable linkers have varying length and functionality. Preferably the linker is a carbon chain. In preferred embodiments, the carbon chain optionally comprises one, two, three, or more heteroatoms selected from N, O, and S. In preferred embodiments, the carbon chain comprises only saturated chain carbon atoms. In preferred embodiments, the carbon chain optionally comprises two or more unsaturated chain carbon atoms (e.g., C═C or C≡EC). In certain embodiments, one or more chain carbon atoms in the carbon chain are optionally substituted with one or more substituents, preferably oxo, C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6 alkynyl, C_1-C_3 alkoxy, OH, halogen, NH_2, —NH(C_1-C_3 alkyl), —N(C_1-C_3 alkyl)$_2$, CN, C_3-C_7 cycloalkyl, heterocyclyl, phenyl, and heteroaryl).

[0657] In certain embodiments, the linker comprises at least 5 chain atoms (e.g., C, O, N, and S). In certain embodiments, the Linker comprises less than 20 chain atoms (e.g., C, O, N, and S). In certain embodiments, the Linker comprises 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, or 19 chain atoms (e.g., C, O, N, and S). In certain embodiments, the Linker comprises 5, 7, 9, 11, 13, 15, 17, or 19 chain atoms (e.g., C, O, N, and S). In certain embodiments, the Linker comprises 5, 7, 9, or 11 chain atoms (e.g., C, O, N, and S). In certain embodiments, the Linker comprises 6, 8, 10, 12, 14, 16, or 18 chain atoms (e.g., C, O, N, and S). In certain embodiments, the Linker comprises 6, 8, 10, or 12 chain atoms (e.g., C, O, N, and S).

[0658] In certain embodiments, the Linker is a carbon chain optionally substituted with non-bulky substituents, preferably oxo, C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6 alkynyl, C_1-C_3 alkoxy, OH, halogen, NH_2, —NH(C_1-C_3 alkyl), —N(C_1-C_3 alkyl)$_2$, CN, C_3-C_7 cycloalkyl, and CN).

[0659] In certain embodiments, the Linker is of Formula L(VI):



(VI)

[0660] or an enantiomer, diastereomer, or stereoisomer thereof, wherein

[0661] p1 is an integer selected from 0 to 12;

[0662] p2 is an integer selected from 0 to 12;

[0663] p3 is an integer selected from 1 to 6;

[0664] each W is independently absent, CH_2, O, S, NH or NRS;

[0665] Z is absent, CH_2, O, NH or NRs;

[0666] each R_5 is independently H or C_1-C_3 alkyl, preferably C_1-C_3 alkyl; and

[0667] Q is absent or —CH_2C(O)NH—,

[0668] wherein the linker is covalently bonded to the compound that compound which recruits a protein that is part of the ubiquitination pathway with the bond that is next to Q and to the allosteric inhibitor of ALC1 with the bond that is next to Z, and wherein the total number of chain atoms in the linker is less than 20.

[0669] In a third aspect the present inventions relates to a pharmaceutical composition comprising the ALC1i as specified for the first aspect of the invention or the bifunction compound of the second aspect and at least one pharmaceutically acceptable excipient, preferably for use in treating and amelioration a proliferative disease.

[0670] In a fourth aspect the present inventions relates to an ALC1i for use of the first aspect of the invention or the bifunctional compound of the second aspect of the invention for use in treating or ameliorating a proliferative disease, wherein the treating and ameliorating of the proliferative disease comprises the administration of said ALC1i or of said bifunctional compound and the administration of a Poly(ADP-ribose)-Polymerase inhibitor (PARPi). The present inventors have discovered that the PARP trapping activity of known PARPi's can be enhanced by ALC1i. Due to this surprising synergy the combined use of PARPi's and ALC1i's for use of the first aspect of the invention in the treatment of various proliferative diseases is particularly advantageous.

[0671] The ALC1i may be provided to the physician administering the antiproliferative therapy separately from the PARPi or in a kit of parts. Thus, in the embodiments in which both are to be combined to obtain the benefit of the synergy the ALC1i may be provided with instructions to combine it with a PARPi of alternatively in a fifth aspect the PARPi may be provided with instructions to combine it with a ALC1i. Accordingly, in a fifth aspect the present invention relates to a PARPi for use in treating or ameliorating a proliferative disease in a patient, wherein the treating and ameliorating the proliferative disease comprises the administration of said PARPi and the administration of the ALC1i for use of the first aspect of the invention or of the bifunctional compound of the second aspect of the invention.

[0672] In a preferred embodiment of the ALC1i for use of the fourth aspect of the invention or of the PARPi for use of the fifth aspect of the invention the PARPi lowers PARP activity and/or inhibits PARP1, PARP2 and/or PARP3, preferably PARP2 on chromatin. The latter phenomenon is also referred to as PARP trapping. Thus, in a preferred embodiment PARP1, PARP2 and/or PARP3, preferably PARP2 is trapped.

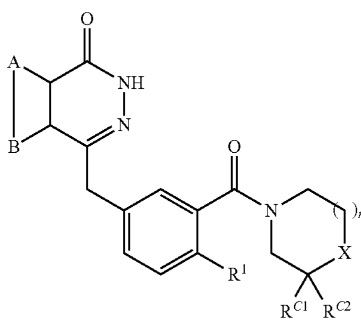
[0673] In a preferred embodiment of the ALC1i for use of the fourth aspect of the invention or of the PARPi for use of the fifth aspect of the invention, the PARPi:

[0674] (i) that lowers PARP activity is selected from small interfering RNA, and

[0675] (ii) that inhibits PARPI is selected from the group consisting of a compound of

[0676] (a) formula (II)

(II)



[0677] and isomers, salts, solvates, chemically protected forms, and prodrugs thereof wherein:

[0678] A and B together represent an optionally substituted, fused aromatic ring;

[0679] X is NR^X or CR^XR^Y;

[0680] if $X{=}NR^X$ then n is 1 or 2 and if $X{=}CR^XR^Y$ then n is 1;

[0681] R^X is selected from the group consisting of H, optionally substituted C_{1-20} alkyl, C_{5-20} aryl, C_{3-20} heterocyclyl, amido, thioamido, ester, acyl, and sulfonyl groups;

[0682] R^Y is selected from H, hydroxy, amino;

[0683] or R^X and R^Y may together form a spiro-C_{3-7} cycloalkyl or heterocyclyl group;

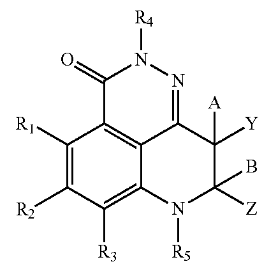
[0684] R^{C1} and R^{C2} are independently selected from the group consisting of hydrogen and C_{1-4} alkyl or when X is CR^XR^Y, R^{C1}, R^{C2}, R^X and R^Y, together with the carbon atoms to which they are attached, may form an optionally substituted fused aromatic ring; and

[0685] R^1 is selected from H and halo;

[0686] and

[0687] (b) formula (III)

(III)



[0688] and isomers, salts, solvates, chemically protected forms, and prodrugs thereof wherein:

[0689] Y and Z are each independently selected from the group consisting of:

[0690] 1. an aryl group optionally substituted with 1, 2, or 3 R_6;

[0691] 2. a heteroaryl group optionally substituted with 1, 2, or 3 R_6;

[0692] 3. a substituent independently selected from the group consisting of hydrogen, alkenyl (e.g. C_{2-6}-alkenyl), alkoxy (e.g. C_{1-6}-alkoxy), alkoxyalkyl (e.g. C_{1-6}-alkoxy-C_{1-6}-alkyl), alkoxycarbonyl (e.g. C_{1-6}-alkoxy-carbonyl), alkoxycarbonylalkyl (e.g. C_{1-6}-alkoxy-carbonyl-C_{1-6}-alkyl), alkyl (e.g. C_{1-6}-alkyl), alkynyl (e.g. C_{2-6}-alkynyl), arylalkyl (e.g. aryl-C_{1-6}-alkyl), cycloalkyl (e.g. C_{3-8}-cycloalkyl), cycloalkylalkyl (e.g. C_{3-8}-cycloalkyl-C_{1-6}-alkyl), haloalkyl (e.g. C_{1-6}-haloalkyl), hydroxyalkylene (e.g. hydroxy-C_{1-6}-alkylene), oxo, heterocycloalkyl (e.g. C_{2-6}-hetero cycloalkyl), heterocycloalkylalkyl (e.g. C_{2-6}-heterocycloalkyl-C_{1-6}-alkyl), alkylcarbonyl (e.g. C_{1-6}-alkyl-carbonyl), arylcarbonyl, heteroarylcarbonyl, alkylsulfonyl (e.g. C_{1-6}-alkyl-sulfonyl), arylsulfonyl, heteroarylsulfonyl, (R_AR_B)alkylene (e.g. (R_AR_B)-C_{1-6}-alkylene), (NR_AR_B)carbonyl, (NR_AR_B)carbonylalkylene (e.g. $NR_AR_B)$carbonyl-C_{1-6}-alkylene), (NR_AR_B)sulfonyl, and (R_AR_B)sulfonylalkylene (e.g. (R_AR_B)sulfonyl-C_{1-6}-alkylene);

[0693] wherein each R_6 is selected from OH, NO_2, CN, Br, Cl, F, I, C_{1-6}-alkyl, C_{3-8}-cycloalkyl, C_{2-8}-heterocycloalkyl; C_{2-6}-alkenyl, alkoxy (e.g. C_{1-6}-alkoxy), alkoxyalkyl (e.g. C_{1-6}-alkoxy-C_{1-6}-alkyl), alkoxycarbonyl (e.g. C_{1-6}-alkoxy-carbonyl), alkoxycarbonylalkyl (e.g. C_{1-6}-alkoxy-carbonyl-C_{1-6}-alkyl), C_{2-6}-alkynyl, aryl, arylalkyl (e.g. aryl-C_{1-6}-alkyl), C_{3-8}-cycloalkylalkyl (e.g. C_{3-8}-cycloalkyl-C_{1-6}-alkyl, haloalkoxy (e.g. C_{1-6}-haloalkoxy), haloalkyl (e.g. C_{1-6}-haloalkyl), hydroxyalkylene (e.g. hydroxy-C_{1-6}-alkylene), oxo, heteroaryl, heteroarylalkoxy (e.g. heteroaryl-C_{1-6}-alkoxy), heteroaryloxy, heteroarylthio, heteroarylalkylthio (e.g. heteroaryl-C_{1-6}-alkylthio), heterocycloalkoxy (e.g. C_{2-6}-heterocycloalkoxy), C_{2-6}-heterocycloalkylthio, heterocyclooxy, heterocyclothio, NR_AR_B, $(R_AR_B)C_{1-6}$-alkylene, (NR_AR_B)carbonyl, (R_AR_B)carbonylalkylene (e.g. $R_AR_B)$carbonyl-C_{1-6}-alkylene), (NR_AR_B)sulfonyl, and (NR_AR_B)sulfonylalkylene (e.g. (NR_AR_B)sulfonyl-C_{1-6}-alkylene);

[0694] R_1, R_2, and R_3 are each independently selected from the group consisting of hydrogen, halogen, alkenyl (e.g. C_{2-6}-alkenyl), alkoxy (e.g.

C_{1-6}-alkoxy), alkoxycarbonyl (e.g. C_{1-6}-alkoxy-carbonyl), alkyl (e.g. C_{1-6}-alkyl), cycloalkyl (e.g. C_{3-8}-cycloalkyl), alkynyl (e.g. C_{2-6}-alkynyl), cyano, haloalkoxy (e.g. C_{1-6}-haloalkoxy), haloalkyl (e.g. C_{1-6}-haloalkyl), hydroxyl, hydroxyalkylene (e.g. hydroxy-C_{1-6}-alkylene), nitro, NR_AR_B, NR_AR_B alkylene (e.g. NR_AR_B C_{1-6}-alkylene), and (R_AR_B)carbonyl;

[0695] A and B are each independently selected from hydrogen, Br, Cl, F, I, OH, C_{1-6}-alkyl, C_{3-8}-cycloalkyl, alkoxy (e.g. C_{1-6}-alkoxy), alkoxyalkyl (e.g. C_{1-6}-alkoxy-C_{1-6}-alkyl), wherein C_{1-6}-alkyl, C_{3-8}-cycloalkyl, alkoxy, alkoxyalkyl are optionally substituted with at least one substituent selected from OH, NO_2, CN, Br, Cl, F, I, C_{1-6}alkyl, and C_{3-8}-cycloalkyl, wherein B is not OH;

[0696] R_A, and R_B are independently selected from the group consisting of hydrogen, alkyl (e.g. C_{1-6}-alkyl), cycloalkyl (e.g. C_{3-8}-cycloalkyl), and alkylcarbonyl (e.g. C_{1-6}-alkyl-carbonyl); or R_A and R_B taken together with the atom to which they are attached form a 3-10 membered heterocycle ring optionally having one to three heteroatoms or hetero functionalities selected from the group consisting

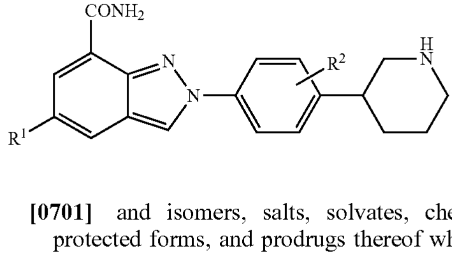
[0697] of —O—, —NH, —N(C_{1-6}-alkyl)-, —NCO(C_{1-6}-alkyl)-, —N(aryl)-, —N(aryl-C_{1-6}-alkyl-), —N(substituted-aryl-C_{1-6}-alkyl-)-, —N(heteroaryl)-, —N(heteroaryl-C_1-C_6-alkyl-)-, —N(substituted-heteroaryl-C_{1-6} alkyl-)-, and —S— or S(O)q-, wherein q is 1 or 2 and the 3-10 membered heterocycle ring is optionally substituted with one or more substituents;

[0698] R_4 and R_5 are each independently selected from the group consisting of hydrogen, alkyl (e.g. C_{1-6}-alkyl), cycloalkyl (e.g. C_{3-8}-cycloalkyl), alkoxyalkyl (e.g. C_{1-6}-alkoxy-C_{1-6}-alkyl), haloalkyl (e.g. C_{1-6}-haloalkyl), hydroxyalkylene (e.g. hydroxy-C_{1-6}-alkylene), and (NR_AR_B)alkylene (e.g. NR_AR_B C_{1-6}-alkylene);

[0699] (iii) that inhibits PARPI and PARP2 is selected from the group consisting of a compound of

[0700] (a) formula (IV)

(IV)



[0701] and isomers, salts, solvates, chemically protected forms, and prodrugs thereof wherein:

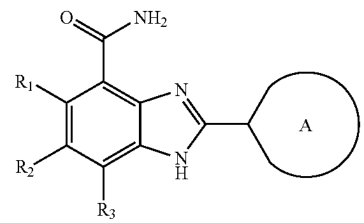
[0702] R_1 is hydrogen or fluorine; and

[0703] R_2 is hydrogen or fluorine;

[0704] and

[0705] (b) formula (V)

(V)



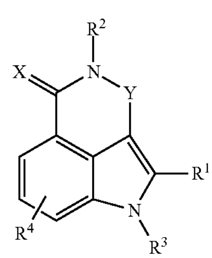
[0706] and isomers, salts, solvates, chemically protected forms, and prodrugs thereof wherein:

[0707] R_1, R_2, and R_3 are independently selected from the group consisting of hydrogen, alkenyl (e.g. C_{1-6}-alkenyl), alkoxy (e.g. C_{1-6}-alkoxy), alkoxycarbonyl (e.g. C_{1-6}-alkoxycarbonyl), alkyl (e.g. C_{1-6}-alkyl), alkynyl (e.g. C_{1-6}-alkynyl), cyano, haloalkoxy (e.g. C_{1-6}-haloalkoxy), haloalkyl (e.g. C_{1-6}-haloalkyl), halogen, hydroxy, hydroxyalkyl (e.g. C_{1-6}-hydroxyalkyl), nitro, NR_AR_B, and (NR_AR_B)carbonyl;

[0708] A is a nonaromatic 4, 5, 6, 7, or 8-membered ring that contains 1 or 2 nitrogen atoms and, optionally, one sulfur or oxygen atom, wherein the nonaromatic ring is optionally substituted with 1, 2, or 3 substituents selected from the group consisting of alkenyl (e.g. C_{1-6}-alkenyl), alkoxy (e.g. C_{1-6}-alkoxy), alkoxyalkyl (e.g. C_{1-6}-alkoxy-C_{1-6}-alkyl), alkoxycarbonyl (e.g. C_{1-6}-alkoxycarbonyl), alkoxycarbonylalkyl (e.g. C_{1-6}-alkoxycarbonyl-C_{1-6}-alkyl), alkyl (e.g. C_{1-6}-alkyl), alkynyl (e.g. C_{1-6}-alkynyl), aryl, arylalkyl (e.g. aryl-C_{1-6}-alkyl), cycloalkyl (e.g. C_{3-8}-cycloalkyl), cycloalkylalkyl (e.g. C_{3-8}-cycloalkyl-C_{1-6}-alkyl), cyano, haloalkoxy (e.g. C_{1-6}-haloalkoxy), haloalkyl (e.g. C_{1-6}-haloalkyl), halogen, heterocycle, heterocyclealkyl (e.g. heterocycle-C_{1-6}-alkyl), heteroaryl, heteroarylalkyl (e.g. heteroaryl-C_{1-6}-alkyl), hydroxy, hydroxyalkyl (e.g. C_{1-6}-hydroxyalkyl), nitro, NR_CR_D, (NR_CR_D)alkyl (e.g. (NR_CR_D)-C_{1-6}-alkyl), (NR_CR_D)carbonyl, (NR_CR_D)carbonylalkyl (e.g. (NR_CR_D)carbonyl-C_{1-6}-alkyl), and (NR_CR_D)sulfonyl; and R_A, R_B, R_C, and R_D are independently selected from the group consisting of hydrogen, alkyl (e.g. C_{1-6}-alkyl), and alkycarbonyl (e.g C_{1-6}-alkylcarbonyl).

[0709] (iv) that inhibits PARPI, PARP2 and PARP3 is a compound of formula (VI)

(VI)



[0710] and isomers, salts, solvates, chemically protected forms, and prodrugs thereof wherein:

[0711] R_1 is: H; halogen; cyano; an optionally substituted alkyl (e.g. C_{1-6}-alkyl), alkenyl (e.g. C_{2-6}-alkenyl), alkynyl (e.g. C_{2-6}-alkynyl), cycloalkyl (e.g. C_{3-8}-cycloalkyl), heterocycloalkyl (e.g. C_{2-8}-heterocycloalkyl), aryl, or heteroaryl group (e.g., unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, and amino, alkoxy (e.g. C_{1-6}-alkoxy), alkyl (e.g. C_{1-6}-alkyl), and aryl groups unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, carboxy, and optionally substituted amino and ether groups (such as O-aryl)); or —C(O)—R_{10}, where R_{10} is: H; an optionally substituted alkyl (e.g. C_{1-6}-alkyl), alkenyl (e.g. C_{1-6}-alkenyl), alkynyl (e.g. C_{1-6}-alkynyl), cycloalkyl (e.g. C_{3-8}-cycloalkyl), heterocycloalkyl (e.g. C_{2-8}-heterocycloalkyl), aryl, or heteroaryl group (e.g., unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, amino, and alkyl (e.g. C_{1-6}-alkyl) and aryl groups unsubstituted or substituted with one or more substituents selected from halo, hydroxy, nitro, and amino); or OR_{100} or $NR_{100}R_{110}$, where R_{100} and R_{110} are each independently H or an optionally substituted alkyl (e.g. C_{1-6}-alkyl), alkenyl (e.g. C_{2-6}-alkenyl), alkynyl (e.g. C_{2-6}-alkynyl), cycloalkyl (e.g. C_{3-8}-cycloalkyl), heterocycloalkyl (e.g. C_{2-6}-heterocycloalkyl), aryl, or heteroaryl group (e.g., unsubstituted or substituted with one or more substituents selected from alkyl (e.g. C_{1-6}-alkyl), alkenyl (e.g. C_{2-6}-alkenyl), alkynyl (e.g. C_{2-6}-alkynyl), cycloalkyl (e.g. C_{3-8}-cycloalkyl), heterocycloalkyl (e.g. C_{2-8}-heterocycloalkyl), aryl, and heteroaryl groups unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, amino, and alkyl (e.g. C_{1-6}-alkyl) and aryl groups unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, and optionally substituted amino groups);

[0712] R_2 is H or alkyl (e.g. C_{1-6}-alkyl);

[0713] R_3 is H or alkyl (e.g. C_{1-6}-alkyl);

[0714] R_4 is H, halogen or alkyl (e.g. C_{1-6}-alkyl);

[0715] X is O or S;

[0716] Y is $(CR_5R_6)(CR_7R_8)_n$ or N—C(R_5), where:

[0717] n is 0 or 1;

[0718] R_5 and R_6 are each independently H or an optionally substituted alkyl (e.g. C_{1-6}-alkyl), alkenyl (e.g. C_{2-6}-alkenyl), alkynyl (e.g. C_{2-6}-alkynyl), cycloalkyl (e.g. C_{3-8}-cycloalkyl), heterocycloalkyl (e.g. C_{2-8}-heterocycloalkyl), aryl, or heteroaryl group (e.g., unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, amino, and lower alkyl (e.g. C_{1-4}-alkyl), lower alkoxy (e.g. C_{1-4}-alkoxy), or aryl groups unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, and amino); and

[0719] R_7 and R_8 are each independently H or an optionally substituted alkyl (e.g. C_{1-6}-alkyl), alkenyl (e.g. C_{2-6}-alkenyl), alkynyl (e.g. C_{2-6}-alkynyl), cycloalkyl (e.g. C_{3-8}-cycloalkyl), heterocycloalkyl (e.g. C_{2-8}-heterocycloalkyl), aryl, or heteroaryl group (e.g., unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, amino, and lower alkyl (e.g. C_{1-4}-alkyl), lower alkoxy (e.g. C_{1-4}-alkoxy), and aryl groups unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, and amino);

[0720] where when R_1, R_4, R_5, R_6, and R_7 are each H, R_8 is not unsubstituted phenyl.

[0721] In a preferred embodiment of the ALC1i or the bifunctional compound for use of the fourth aspect of the invention, or the PARPi for use of the fifth aspect of the invention the PARPi is selected from the group consisting of Olaparib, Talazoparib, Niraparib, Rucaparib, and Veliparib, in particular of Veliparib, Olaparib, and Talazoparib.

[0722] In a preferred embodiment the ALC1i or the bifunctional compound for use of fourth aspect of the invention or the PARPi for use of the fifth aspect of the invention the ALC1i is the inhibitor of ALC1 of any of the first or further aspect of the invention or the bifunctional compound of the second aspect.

[0723] In a preferred embodiment the ALC1i or the bifunctional compound for use of the fourth aspect of the invention or the PARPi for use of the fifth aspect of the invention the tumor disease is selected from hepato cellular carcinoma, breast cancer, ovarian cancer, prostate cancer, and colorectal cancer.

[0724] In a preferred embodiment the ALC1i or the bifunctional compound for use of the fourth aspect of the invention or the PARPi for use of the fifth aspect of the invention (i) the ALC1i potentiates the cancer-cell killing efficacy of the PARPi, (ii) a reduced amount of PARPi is administered, and/or (iii) PARPi resistance is bypassed.

[0725] In a preferred embodiment the ALC1i for use of the fourth aspect of the invention or the PARPi for use of the fifth aspect of the invention the PARPi and ALC1i are administered concomitantly or separately.

[0726] In a sixth aspect the present invention relates to a kit of parts comprising separately packaged a PARPi and an ALC1i or a composition comprising a PARPi and an ALC1i, preferably with instructions for use to treat or ameliorate a proliferative disease.

[0727] BRCA1 and BRCA2 proteins are involved in both promoting homologous recombination (HR)-mediated DNA repair and also controlling the stability of stalled replication forks. Many tumor types, including, e.g., breast cancer, ovarian cancer, prostate cancer, pancreatic carcinomas, fallopian tube cancer, peritoneal cancer, acute myeloid leukemia, and uveal melanoma often have underlying defects in BRCA1 or BRCA2 activity. These defects are often due to germline or somatic mutations in the BRCA1 or BRCA2 genes. These tumors, with underlying defects in HR repair, are typically sensitive to PARP inhibitors.

[0728] However, they can develop PARP inhibitor resistance over the time. To treat such tumors the use of the ALC1i for use of the first aspect of the invention the bifunctional compound of the second aspect of the invention, the pharmaceutical of the third aspect of the invention, the ALC1i or the bifunctional compound for use of the fourth aspect of the invention is particular suitable if used alone or in combination with PARPi. In a preferred embodiment of the ALC1i for use according to the first aspect of the invention the bifunctional compound of the second aspect of the invention, the pharmaceutical of the third aspect of the invention, the ALC1i or the bifunctional compound for use of the fourth aspect of the invention or the PARPi for use of the fifth aspect of the invention the proliferative disease is selected from a cancer.

[0729] As used herein, the term "cancer" refers to cells having the capacity for autonomous growth. Examples of such cells include cells having an abnormal state or condition characterized by rapidly proliferating cell growth. The term is meant to include cancerous growths, e.g., tumors; oncogenic processes, metastatic tissues, and malignantly transformed cells, tissues, or organs, irrespective of histopathologic type or stage of invasiveness. Also included are malignancies of the various organ systems, such as respiratory, cardiovascular, renal, reproductive, hematological, neurological, hepatic, gastrointestinal, and endocrine systems; as well as adenocarcinomas which include malignancies such as most colon cancers, renal-cell carcinoma, prostate cancer and/or testicular tumors, non-small cell carcinoma of the lung, and cancer of the small intestine. Cancer that is "naturally arising" includes any cancer that is not experimentally induced by implantation of cancer cells into a subject, and includes, for example, spontaneously arising cancer, cancer caused by exposure of a patient to a carcinogen(s), cancer resulting from insertion of a transgenic oncogene or knockout of a tumor suppressor gene, and cancer caused by infections, e.g., viral infections. The term "carcinoma" is art recognized and refers to malignancies of epithelial or endocrine tissues. The term also includes carcinosarcomas, which include malignant tumors composed of carcinomatous and sarcomatous tissues. An "adenocarcinoma" refers to a carcinoma derived from glandular tissue or in which the tumor cells form recognizable glandular structures. The term "sarcoma" is art recognized and refers to malignant tumors of mesenchymal derivation. The term "hematopoietic neoplastic disorders" includes diseases involving hyperplastic/neoplastic cells of hematopoietic origin. A hematopoietic neoplastic disorder can arise from myeloid, lymphoid or erythroid lineages, or precursor cells thereof. In some embodiments, the cancer is breast cancer, ovarian cancer, prostate cancer, pancreatic carcinomas, colorectal cancer, hepatocellular carcinoma, uterine cancer, bone cancer (preferably osteosarcoma), gastric cancer, gastroesophageal cancer, non-small cell lung cancer, fallopian tube cancer, peritoneal cancer, acute myeloid leukemia, or uveal melanoma. In some embodiments, the cancer is basal-like cancer, basal-like breast cancer, triple negative breast cancer, high grade cancer, or high grade serous ovarian cancer cell.

[0730] Thus, a cancer that is BRCA1 deficient or BRCA2 deficient refers to a cancer that has one or more cells having abnormal BRCA1 levels or activities, or abnormal BRCA2 levels or activities. These abnormal levels or activities interfere with the normal function of BRCA1 or BRCA2, and can cause a defect in HR-mediated DNA repair or decrease the stability of replication forks.. Preferably the proliferative disease is selected from a BRCA-1 and/or BRCA-2-deficient tumor, and/or the proliferative disease is selected from hepatocellular carcinoma, breast cancer, ovarian cancer, prostate cancer, colorectal cancer, gastric cancer, gastroesophageal cancer, non-small cell lung cancer, or pancreatic cancer.

[0731] Thus in a seventh aspect the ALC1i for use according to the first aspect of the invention the bifunctional compound of the second aspect of the invention, the pharmaceutical of the third aspect of the invention, the ALC1i or the bifunctional compound for use of the fourth aspect of the invention or the PARPi for use of the fifth aspect of the invention the proliferative disease is selected from a cancer or the kit of the sixth aspect of the invention, wherein the proliferative disease, preferably cancer, and more preferably the cancer is selected from a BRCA1 and/or 2-deficient tumor, and a tumor in which expression of PARPi, PARP2, PARP3 and/or ALC1 is increased in comparison to non-tumor cells.

[0732] Preferred examples of cancers treatable according to the various aspects of the invention are ovarian and fallopian tube cancer, breast cancer, pancreatic and gastric cancer, colorectal cancer and pulmonary cancer.

[0733] Preferably, the cancer to be treated in accordance with the various aspects of the invention has relapsed or progressed, preferably after a first line chemotherapy. Thus, preferably, the inhibitor of ALC1 (ALC1i) according to the present invention is for use as second line or third line therapy.

[0734] Preferably, the cancer to be treated is at stage III (locally advanced) or IV (metastatic).

[0735] Preferably, the cancer to be treated in accordance with the various aspects of the invention has underlying defects in DNA damage repair, like HR repair, i.e. is a HR deficient cancer. In one embodiment, the cancer cells have mutations/deletions/insertions in one or more DNA repair genes as listed in Tables 13 and 14.

[0736] In one embodiment, the cancer to be treated or the cancer patients to be treated are selected based on the presence of tumor markers. In the clinical setting cancer/patient selection would include but is not limited to eligible tumor biomarkers, i.e., deleterious mutations including BRCA1, BRCA2, BARD1, BRIP1, ANCA, FANCE, NBN, PALB2, RAD51C, RAD51D, RAD51 and/or RAD51B and/or other gene variants in the HR pathway as listed in Table 13 and Table 14.

[0737] Preferably, the patient to be treated in accordance with the various aspects of the invention is a cancer patient, preferably having breast cancer, ovarian cancer, prostate cancer, pancreatic carcinomas, gastric, gastroesophageal, non-small cell lung cancer, colorectal cancer, hepatocellular carcinoma, uterine cancer, bone cancer (preferably osteosarcoma), fallopian tube cancer, peritoneal cancer, acute myeloid leukemia, or uveal melanoma, and optionally having mutations/deletions/insertions in one or more of the HR genes as listed in Tables 13 and 14.

Experimental Section

Cell Lines Used

[0738] As a BRCA wild-type cell line, MDA-MB-231 cells were used. The cells were established from an aneuploid female human. The cells were extracted from the mammary gland (breast) in the metastatic site as a pleural effusion. (MDA-MB-231 (ATCC® HTB-26™ *Homo sapiens* epithelial mammary gland) It is available from numerous sources including ATCC® HTB-26™

[0739] As a BRCA negative cell line, SUM-149-PT cells were used. The cell line is a triple negative breast cancer (TNBC) cell line, derived from primary human invasive ductual carcinoma metastatic nodule from a 40 year old female. It contains a hemizygous BRCA1 mutation (p.Pro724Leufs*12) and is available from numerous sources including bioIVT.

[0740] As an example for pancreatic cancer, PSN1 cells were used. The human cell line was derived from pancreatic adenocarcinoma tissue. It harbors an amplification of c-myc and activated c-Ki-ras and a loss of one of the two p53 alleles. The cell line is available from numerous sources including MERCK (94060601).

[0741] BxPC-3 cells were used. The cells were extracted from the pancreas tissue of a 61-year-old female with adenocarcinoma. The established cell line does not express CFTR (cystic fibrosis transmembrane conductance regulator), but it expresses mucin, the pancreas cancer specific antigen, and CEA (carcinoembryonic antigen). The cells are available from numerous sources including MERCK (93120816).

[0742] CaCo2 cells were used. The cells are established from colon tissue from a 72-year old female with colorectal adenocarcinoma and is available at ATCC (Caco-2 [Caco2] HTB-37™)

[0743] Capan-1 cells were used. These cells were isolated from the pancreas of a 40 year old male with pancreatic adenocarcinoma. The cells are available at ATCC (Capan-1 HTB-79™)

[0744] 22Rv1 cells were used. The cell line is derived from a xenograft that was serially propagated in mice after castration-induced regression and relapse of the parental CWR22 xenograft that was androgen-dependent. The cells originate from prostate carcinoma and are epithelial. They express PSA (prostate-specific antigen) and have a damaging mutation in BRCA2. The cell line is available at numerous sources including Accegen (ABC-TCO0004).

[0745] DU145 cells were used. The cells were isolated from the brain of a 69-year-old male with prostate cancer, the established cell line has epithelial morphology. It is hypotriploid and has mutations in BRCA1, BRCA2, RAD50, WRN and TP53. The cell line is available at ATCC HTB-81™

[0746] DLD1-wt cells were used. The cells were extracted from the large intestine of an adult male patient with colorectal carcinoma and a cell line was established. They are pseudodiploid(2n=46), express p53 antigen and have damaging mutations in BRCA2, FANCA, and TP53. The cell line is available at numerous sources including MERCK (90102540).

[0747] DLD1-BRCA2−/− cells were used. This cell line originated from the DLD1-wt and is a BRCA2 knockout cell line. It is available at several sources including Creative Biogene (CSC-RT0028).

[0748] HCT116 cells were used. This cell line originates from colon tissue from an adult male with colorectal carcinoma. It has damaging mutations in BRCA2, FANCA and POLD1. This cell line is available at numerous sources including the DSMZ (ACC 581).

[0749] PC-3 cells were used. The cell line originates from a 62-year-old male patient with a grade IV prostatic adenocarcinoma, specifically from a bone metastasis. The cells are epithelial, near triploid and have a damaging mutation in TP53. They are available at numerous sources including MERCK (90112714).

[0750] SW620 cells were used. The cells were isolated from the lymph node of a 51-year-old male with Dukes' C colorectal adenocarcinoma and a cell line was established. The cell line is available at numerous sources including ATCC CCL-227™

[0751] T47D cells were used. The cell line was established from a 54-year-old female human with an infiltrating ductal carcinoma of the breast. The cells were extracted from the mammary gland (breast) as a pleural effusion. They are available at numerous sources including ATCC HTB-133™

[0752] HUH7 cells were used. They are available for example at the cell lines service (CLS.shop). these cells were isolated from a the liver tumor of a 57-year old male.

[0753] HCC1428 cells were used. They are available at ATCC (HCC1428 CRL-2327™) and were derived from the mammary gland of a white female patient with adenocarcinoma.

[0754] HCC1937 cells were used. They were isolated from a 23-year-old female with a primary ductal carcinoma, specifically from a mammary gland. The cell line has a homozygous BRCA1 mutation and is negative for expression of p53 and Her2-neu, the morphology is epithelial. The cells are available at several sources including Amsbio (Cat. No. AMS.EP-CL-0093).

[0755] MDA-MB-436 cells were used. The cell line is derived from a 43-year-old female with breast adenocarcinoma, it was extracted from the mammary gland in the metastatic site as a pleural effusion. The cells are pleomorphic with multinucleated component cells. They are available at numerous sources including Accegen (Cat. No. ABC-TC0655).

[0756] HEPG2 cells were used. The cell line was established from a 15-year-old male patient with a hepatocellular carcinoma. The cells express several major plasma proteins like albumin, fibrinogen, and alpha 2-macroglobulin, the morphology is epithelial-like. The cell line is available at ATCC HB-8065™

[0757] LnCap cells were used. The cells were isolated from the left supraclavicular lymph node of a male with metastatic prostate carcinoma and are available at ATCC (LNCaP clone FGC CRL-1740™)

[0758] U2OS cells were used. They are available at ATCC (U-2 OS HTB-96™) and were isolated from the tibia of a 15 year old female with osteosarcoma.

Synergy with PARPi

[0759] For determination of the ZIP synergy score, a 2-D titration of two compounds is added to cells in the 96h-SRB-survival assay format. The score is calculated by adding the readout of the survival data from the SRB assay of at least 3 replicate plates to an open-source program called Synergy Finder.

[0760] "SynergyFinder (https://synergyfinder.fimm.fi) is a stand-alone web-application for interactive analysis and visualization of drug combination screening data. Since its first release in 2017, SynergyFinder has become a widely used web-tool both for the discovery of novel synergistic drug combinations in pre-clinical model systems (e.g. cell lines or primary patient-derived cells), and for better understanding of mechanisms of combination treatment efficacy or resistance" (Ianevski et al. 2020).

[0761] In the Zero interaction potency (ZIP) model, the drug interaction relationship is determined by comparison of the change in potency of the dose-dependent curves between individual drugs and their combination (https://synergyfinder.fimm.fi/synergy/synfin_docs/). The model is further described in https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4759128/. Results for treatment with ALC1 inhibitors and PARP inhibitors together are shown in FIG. **4**.

Inhibitor Vs. Response (Cell Survival) Assay

[0762] For validation of the small molecule inhibitors, the synthetic lethality of BRCA and ALC1 was addressed using MDA-MB-231 cells as a BRCA wild-type cell line and SUM-149-PT as a BRCA1 deficient cell line. Cells were seeded in 96-well plates (5000 cells/well) and treated with titrations of ALC1 inhibitors starting at 50 μM. As a control for "no-treatment", DMSO was added to the cells. The cells were cultured at 37° C., CO_2 5% for 5 days until they were fixed with 10% TCA for 1 h and stained with sulforhodamine dye for 30 minutes. After washing the cells with 1% Acetic Acid, 10 mM Tris (pH 10.5) solution was used to solubilize the stained cells. The absorbance was measured at 492 nm using the SUNRISE TECAN, the data were normalized to the number of cells at timepoint 0 and to 100% survival (=DMSO control) and analyzed using GraphPad Prism. Survival curves were fitted using "Inhibitor vs. response curves with variable slope (four parameters)". The IC_{50} values for treatment with ALC1 inhibitors are shown in FIG. **7**.

[0763] For further validation of the compounds, colony formation assays were applied. Here, less cells were seeded for the assay (100 cells/well). The cells were treated with ALC1 inhibitor or with a combination of PARPi and an ALC1 inhibitor for 11 days. The cells were fixed and data was analyzed as mentioned above. Results for treatment with ALC1 inhibitors are shown in FIGS. **3** and **7**.

Fret-Based Nucleosome Sliding Assay

[0764] This assay utilizes mid-positioned mononucleosomes that allow for monitoring the sliding activity of the ALC1 remodeling enzyme using a FRET readout. Each nucleosome is labeled with two FRET dyes: the octamer is labeled with Cy5 (Cy5-maleimide coupling to H2B) and one of the DNA ends is labeled with Cy3. The DNA template includes the 147 bp 601 DNA positioning sequence flanked by DNA overhangs on each side. Other nucleosome positioning sequences, both artificial constructs and naturally occurring sequences, even if less efficient than the 601 sequence in positioning nucleosomes, can also be used. The nucleosomes are assembled by salt gradient dialysis using purified, Cy5-labeled histone octamers and purified, Cy3-labeled DNA templates to yield the FRET-labeled mid-positioned nucleosomes. These nucleosomes will start with low FRET and will have a low Cy5 fluorescence signal when excited with the Cy3 excitation maximum wavelength as the two fluorophores are too far apart for efficient FRET. As the ALC1 remodeling reaction proceeds and the remodeling enzymes slides the octamer towards the DNA end, the distance between the two FRET dyes decreases and the signal from Cy5/FRET increases. Hence, increase in FRET can be directly used as readout for sliding.

DNA Template Construction for Mononucleosome Reconstitution

[0765] For reconstitution of mononucleosomes, the non-natural Widom 601 nucleosome positioning sequence was used as a high affinity binding site for the histone octamer (see Lowary, P. T. & Widom, J. New DNA sequence rules for high affinity binding to histone octamer and sequence-directed nucleosome positioning. *J. Mol. Biol.* 276, 19-42 (1998)). Cy3-labeled DNA containing these 601 sequences can be constructed using methods including PCR amplification, restriction digestion of DNA plasmids followed by a Klenow end labeling reaction or other standard molecular biological techniques.

Preparation of ALC1 and Histone Octamers

[0766] Full length human ALC1 (1-897) or truncated versions thereof were expressed and purified as N-terminally 6× His-tagged fusion protein from *E. coli* as published before (Singh, H. R., et al., 2017).

[0767] Human histone proteins were recombinantly expressed in *E. coli* (either using classical IPTG induction or using autoinduction media) and purified from *E. coli* inclusion bodies. The purification scheme includes the extraction/solubilization of histones from inclusion bodies using guanidium chloride, followed by reverse phase chromatography. The purified histones were lyophilized resulting in TFA-salts of the purified histone proteins.

Preparation of Nucleosomes

[0768] To assemble the nucleosomes, template DNA (250 g/ml final concentration) was mixed with purified histone octamers in a high salt buffer at different molar ratios of histone octamers to DNA. The ionic strength of this mixture was then reduced <600 mM NaCl by continuous dialysis against a low salt buffer at 4° C. Finally, the material was dialyzed against TEA-20 (10 mM triethanolamine-Cl pH 7.5, 20 mM NaCl, 0.1 mM EDTA). The best molar ratio, i.e. the ratio that yields full assembly of the DNA into nucleosomes was picked and used for further assemblies.

[0769] Alternatively, the nucleosomes can be assembled by other methods such as deposition of histone octamers onto DNA using polyglutamate or histone chaperones, or by salt step dilution.

Method of Measuring ALC1 Mediated Sliding; ALC1 Nucleosomal Sliding Assay

[0770] Sliding reactions were performed in 384 well plates at RT in 10 mM Tris-HCl, pH 8.1, 75 mM KCl, 1 mM $MgCl_2$, 1.0 mM EGTA, 10% glycerol, 0.5 mM dithiothreitol (DTT), 0.01% TritonX100, 0.02% NP40 and reaction mixtures contained mid-positioned nucleosome, tri-ADP ribose or (ADP-ribose). and ALC1 chromatin remodeling enzyme. Sliding was initiated by addition of ATP followed by shaking the plates for 5 sec at 1450 rpm and sealing them with a foil compatible with fluorescence reading. The FRET signal was immediately recorded using a fluorometer (BMG reader PheraStar FSX, channel A: excitation 520 nm, emission 680 nm; channel B: excitation 520 nm, emission 590 nm) and unless stated otherwise, remodeling proceeded for 30 min.

[0771] The FRET signal was calculated as the signal at 680 nm (emission of Cy5) divided by the signal at 590 nm (emission of Cy3) and multiplied by 10,000. To obtain an apparent rate of ALC1-mediated nucleosome sliding, the increase in FRET as a function over time was plotted and the initial velocities of ALC1-mediated nucleosome sliding were obtained by fitting the resulting kinetic trace by a linear curve fit.

High Throughput Screening (HTS) and IC_{50} Determination for Molecules that Modulate ALC1-Mediated Sliding

[0772] For High Throughput Screening (HTS) and IC_{50} determination for compounds that modulate the sliding activity of ALC1, the nucleosome was incubated for 30 min as described above with ALC1 and triADP-ribose or (ADP-ribose)$_n$ in the presence of the putative ALCi prior to initiating sliding by ATP addition. Then, the rate of sliding (initial velocity) was determined as described above and compared against the rate of sliding in the absence of the putative modulator/compound (% inhibition). For IC_{50} determination, the observed % inhibition (y-axis) was plotted against compound concentrations (x-axis) using GraphPad Prism and fitted using a nonlinear regression model (four parameters). The results of the IC_{50} of inhibition of nucleosome sliding are shown in FIG. **6**. IC_{50} values are presented by the following symbols: +++: IC_{50} <25 μM; ++: IC_{50}=25-100 μM; +: IC_{50}=100-250 μM; and '-': compound is not active, i.e. never observe inhibition >50%.

SAR Analysis of ALC1 Inhibitors

[0773] In general, the ALCi cluster presented here has straightforward SAR clearly indicated by the presence of a large number of molecular match pairs. For example, in the R^6 substitution, there is a clear preference for halogens over either oxygen or methyl substitutions, as demonstrated by the improved potency of ALCi-1002 over ALCi-1009 and ALCi-1055. Additionally, in the R^4 region, there is a preference the tetrazole over the carboxylic acid (no tetrazole containing compound with IC50 >25 μM). In the R^5 region of the molecule, nonaromatic, hydrophobic moieties tend to be preferred for increased potency and selectivity in cells, especially when combined with a tetrazole in the R^4 region (ALCi-1013 vs ALCi-1002). By extrapolating from the currently available SAR, the skilled person should easily be able to develop additional ALC1 inhibitors in the same vein of those presented below.

Synthesis of ALC1 Inhibitors

[0774] The synthesis of compounds of general formula (I) can be carried out according to the general synthesis scheme shown in FIGS. **9** and **10**. The compounds prepared in FIGS. **9** and **10** reflect a basic compound scaffold in which the phenyl groups can be substituted at any position, not only where the R groups are depicted.

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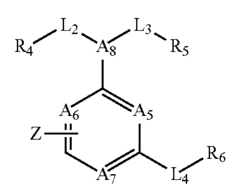
1. A method of treating or ameliorating a proliferative disease in a patient in need thereof comprising administering an inhibitor of ALC1 (ALC1i) according to formula (I)

Formula I



wherein

A5 and A8 are each independently selected from N or CH;

A6 is selected from N or CH, or when A6 takes part in the annulated carbo- or heterocycle Z, then A6 is C;

A7 is selected from N or CH, or when A7 takes part in the annulated carbo- or heterocycle Z, then A7 is C;

L2 is selected from the group consisting of —CH_2—R4, —CF_2—R4, —CH_2—CH_2—R4, —CH_2—CH_2—CH_2—R4, —O—R4, —NH—R4, —N═R4;

L3 is selected from the group consisting of CH_2—R5, —CF_2—R5, —CH_2—CH_2—R5, —CH_2—CF_2—R5, —CH_2—CH_2—CH_2—R5, —O—R5, —NH—R5, —N═R5;

or

L2 and L3 together with the A8 to which they are connected form a 5- or 6-membered heterocycle substituted by R^4 and/or R^5;

L4 is CH_2, —CF_2—, CH_2—CH_2, CH_2—CH_2—CH_2, 0, N, and NH, or is absent;

Z is a 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and —NO_2;

and can be annulated to the central core or connected via a covalent bond

R^4 is 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —CF_3, Me, Et, —OMe, and —SMe, or R^4 is hydrogen, methyl, or COOH;

R^5 is a 4-, 5-, 6-, 7-, 8-, 9- or 10-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe;

R^6 is a 5-, 6- or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe;

or R^6 is H;

or

when A7 takes part in the annulated carbo- or heterocycle Z then A5 and A6 are independently selected from —N or —CH and A8 is selected from —N, —CH, —CH_2—N, —CH_2—CH, or —NH—CH;

or pharmaceutically acceptable salt thereof.

2. The method according to claim **1**, wherein

A5 is N;

one of A6 and A7 is CH and the other one is N; or one of A6 and A7 is C and takes part in the annulated carbo- or heterocycle Z and the other one is N;

A8 is N or CH,

or

A5 is N;

one of A6 and A7 is C and takes part in the annulated 5-, 6- or 7-membered carbo- or heterocycle, preferably 5- or 6 membered aryl or heteroaryl Z and the other one is N;

A8 is N;

L2 is CH_2—CH_2—R4 and L3 is CH_2—CH_2—R5 or CH_2—CF_2—R5 or L2 and L3 together with the A8 to which they are connected form a 5- or 6-membered heterocycle, preferably piperidinyl or a pyrrolidinyl substituted by R4 and R5;

L4 is absent;

Z is any 6-membered carbo- or heterocycle, preferably 6-membered aryl or heteroaryl annulated to the central core, wherein Z is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, —OH, Me, —CF_3, Et, —OMe, —SMe, and NO_2;

R4 is COOH or tetrazolyl;

R5 is any 4-, 5-, 6-, or 7-membered carbo- or heterocycle, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —I, Me, —CF_3, Et, —OMe, and —SMe;

R6 is any 5- or 6-membered carbo- or heterocycle, optionally substituted with one, two, or three (preferably one) substituents selected from the group consisting of —Br, —Cl, —F, —I, —OH, —NO_2, Me, —CF_3, Et, —OMe, and —SMe, or R6 is H.

3. The method according to claim **1**, wherein

each one of A5, A7 and A8 is N;

A6 is C and takes part in the annulated carbo- or heterocycle Z;

L3 is CH_2—CF_2—R5 or each one of L2 and L3 is CH_2—CH_2; or L2 and L3 together with the A8 to which they are connected form a piperidine ring or a pyrrolidine ring, substituted by R4 and R5;

L4 is absent;

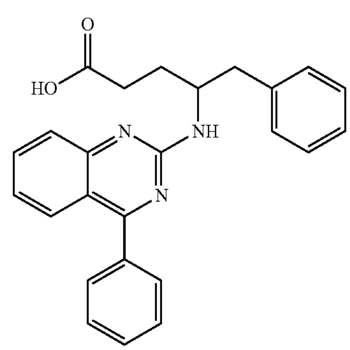
Z is phenyl or cyclohexyl annulated to the central core, wherein Z is optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —F, —OH, Me, —CF_3, —OMe, and —NO_2,

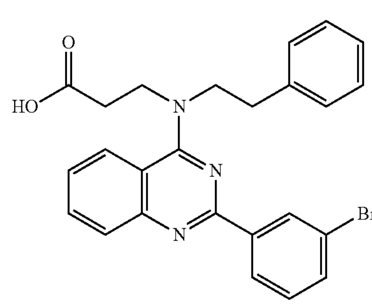
R4 is COOH or tetrazolyl;

R5 is phenyl, cyclobutyl, cyclopentyl or adamantyl, optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of —Br, —Cl, —CF_3, Me, —CH_2—CF_3, and —OMe;

R6 is any 6-membered carbo- or heterocycle (preferably phenyl or cyclohexyl; more preferably phenyl), optionally substituted with one, two, or three, preferably one substituent(s) selected from the group consisting of Br, —Cl, —F, Me, —CF_3, —OMe, and —NO_2.

4. The method according to claim **1**, wherein the ALC1 is selected from the group consisting of:





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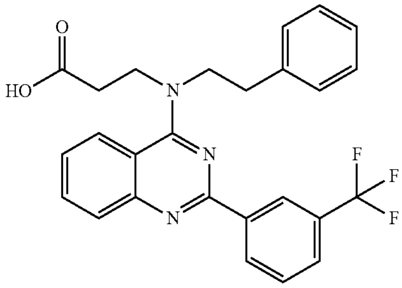

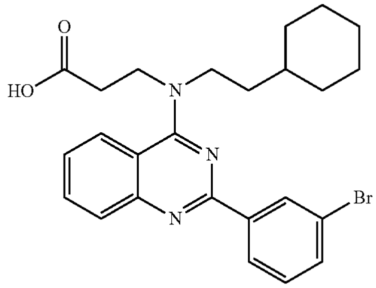

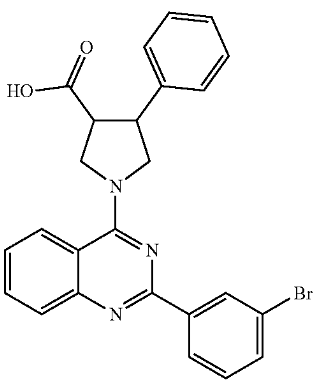

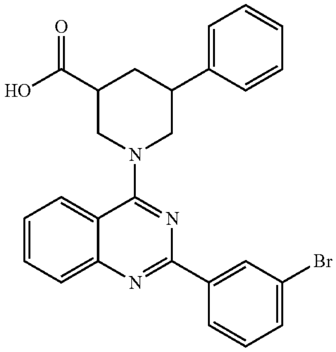

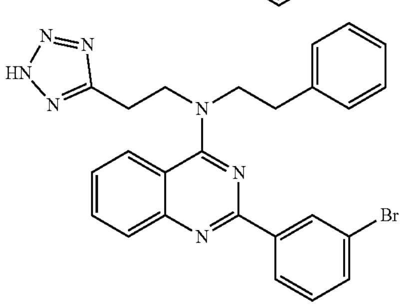

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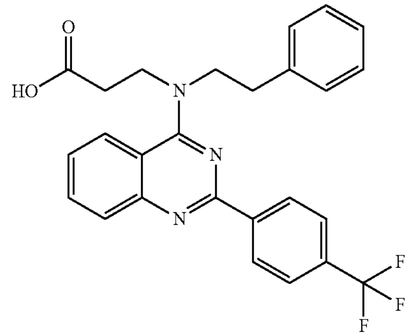

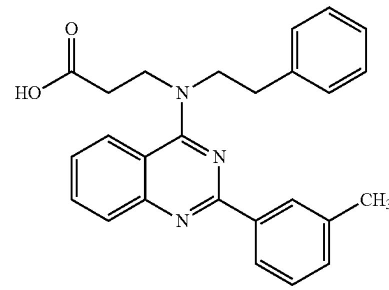

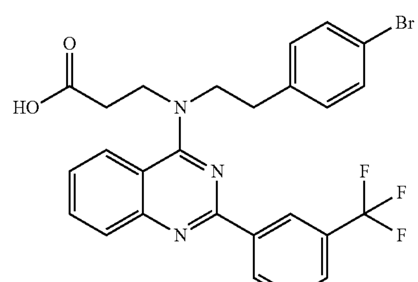

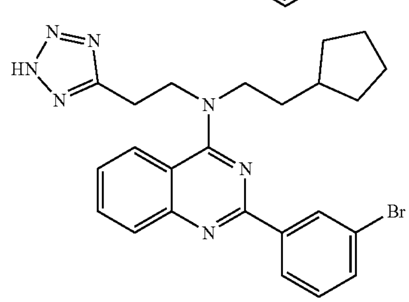

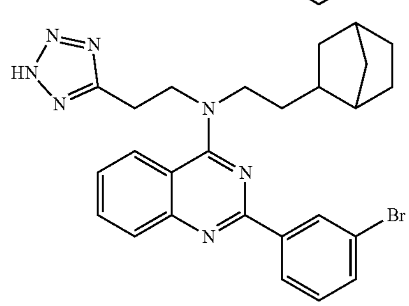

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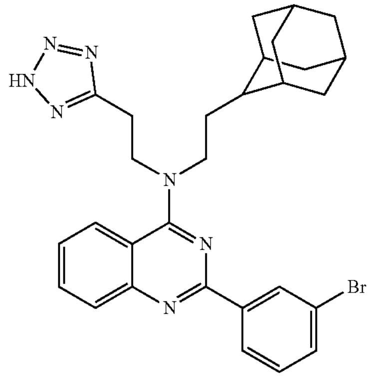

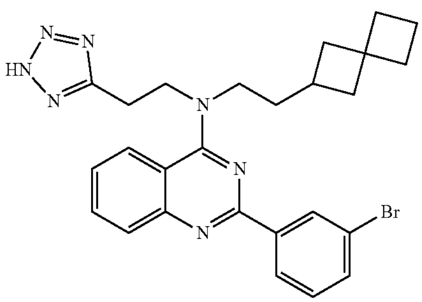

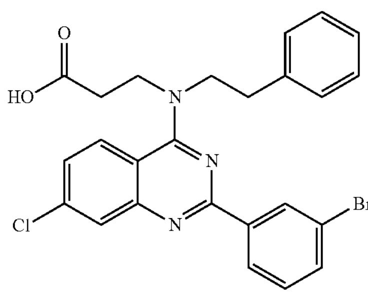

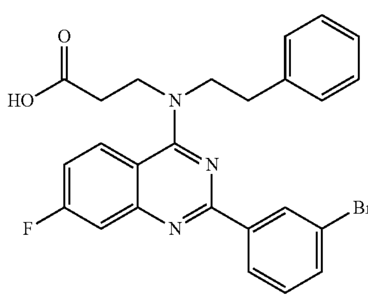

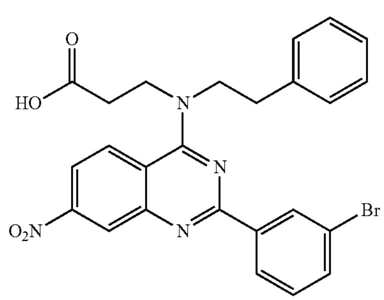

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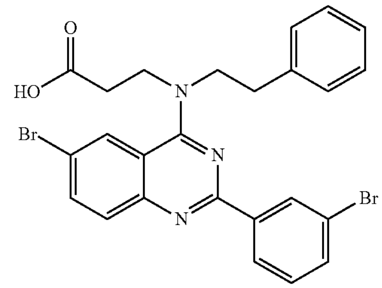

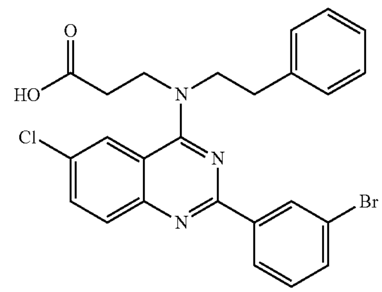

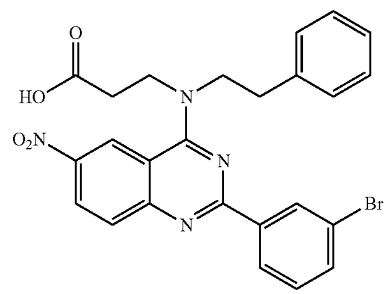

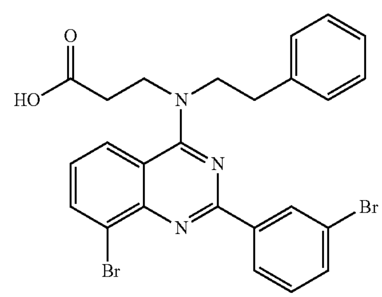

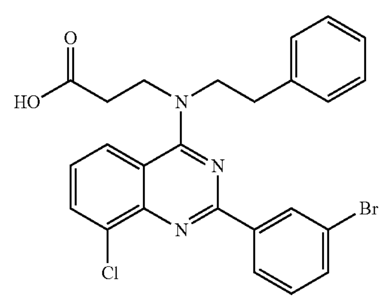

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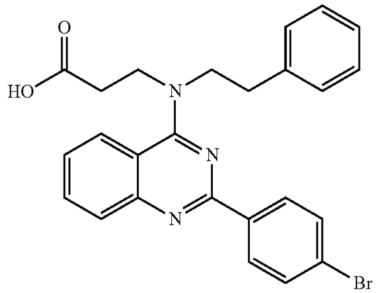

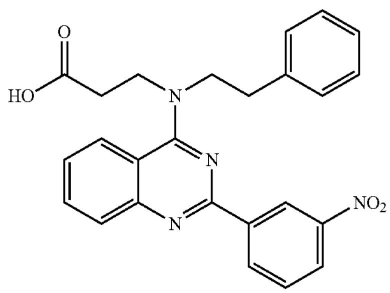

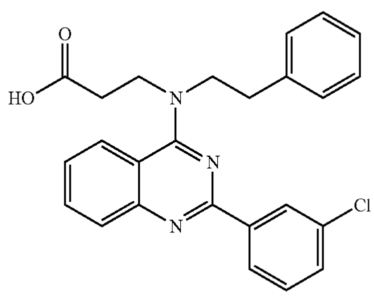

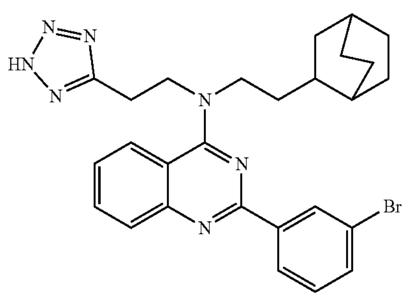

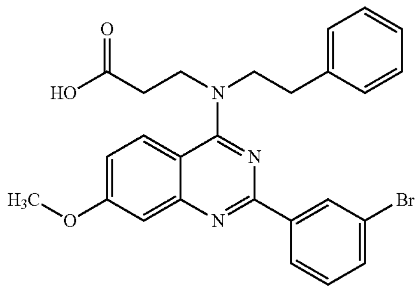

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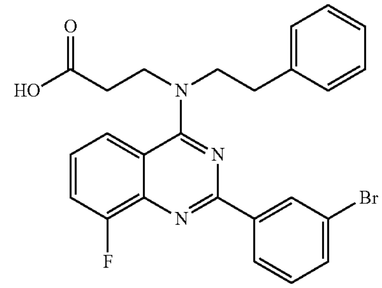

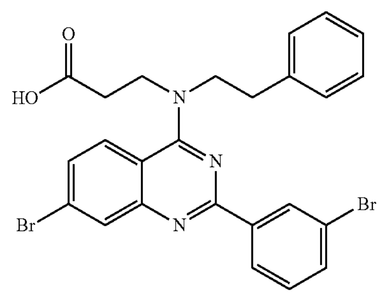

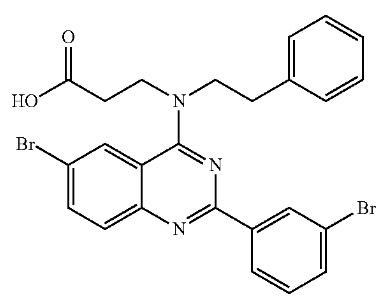

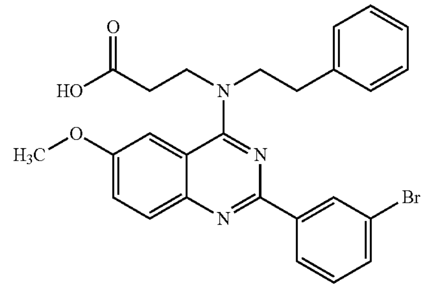

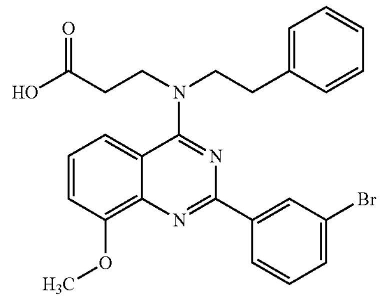

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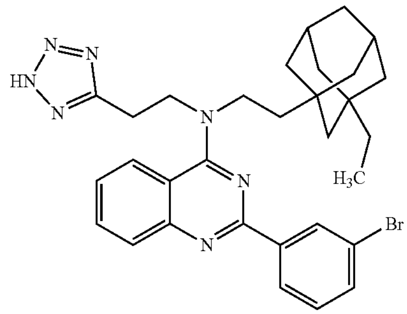

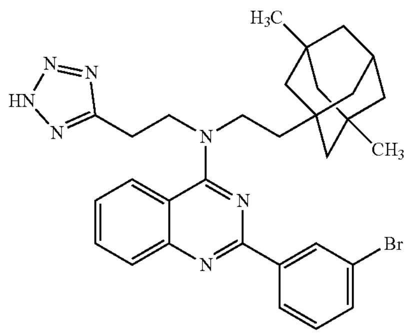

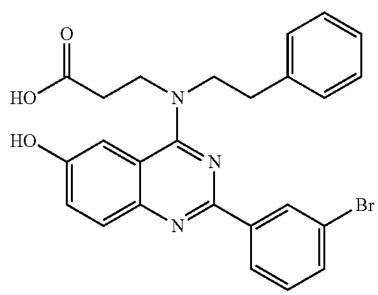

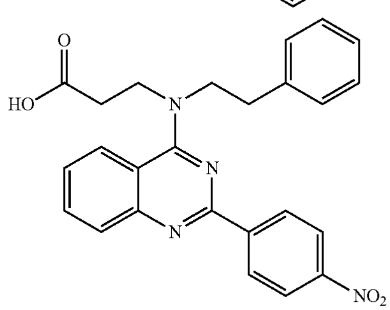

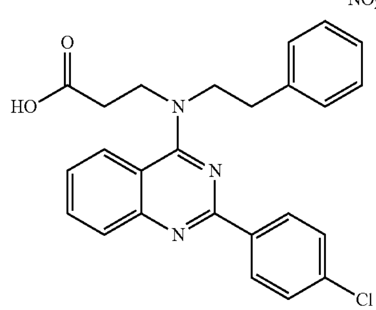

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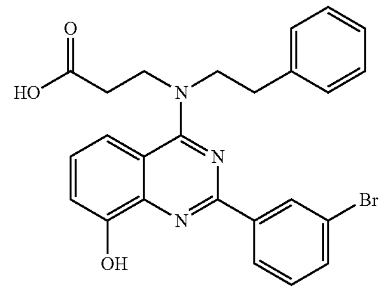

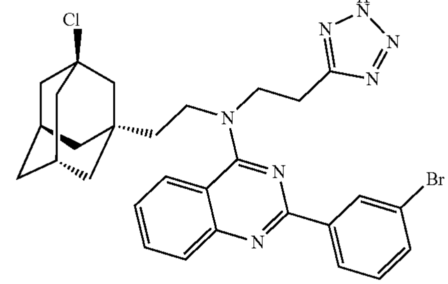

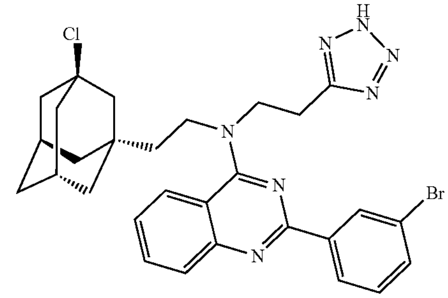

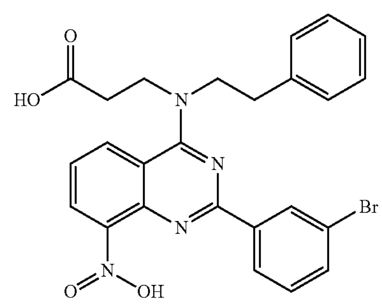

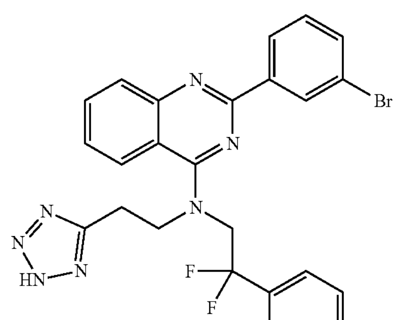

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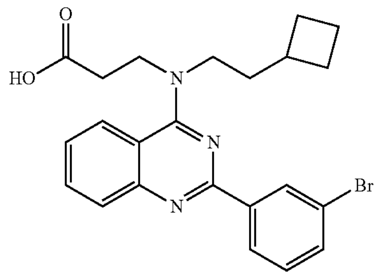

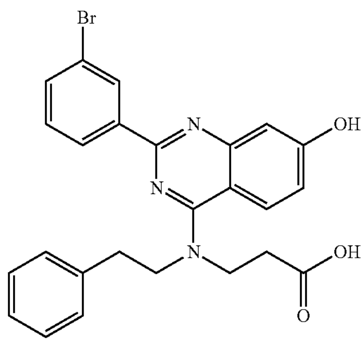

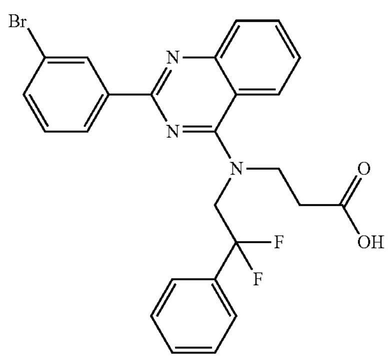

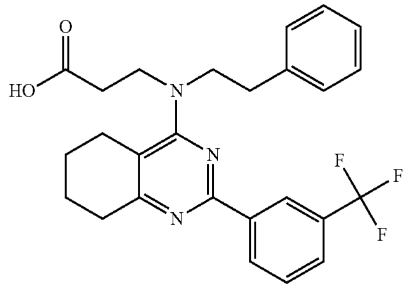

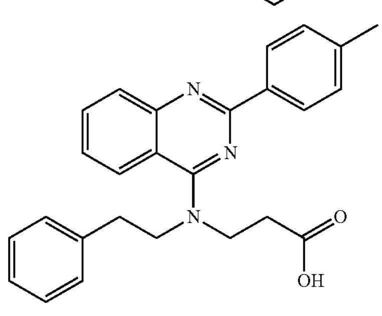

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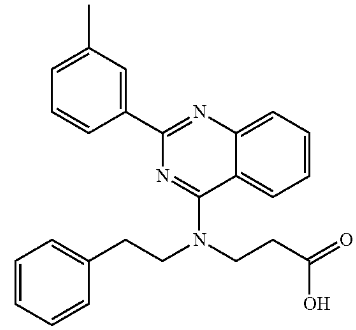

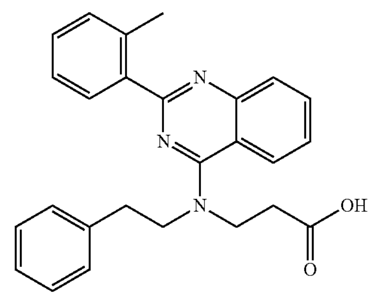

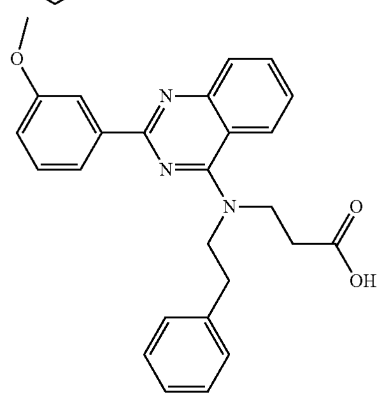

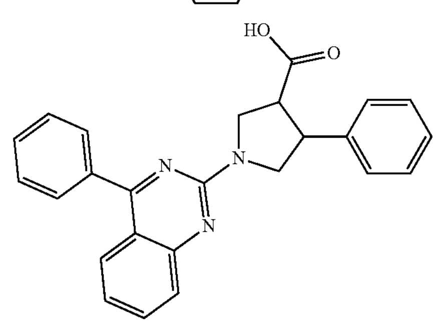

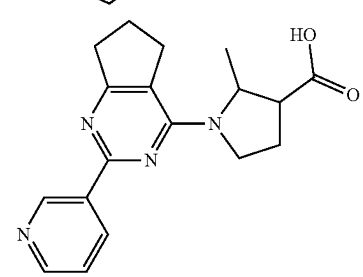

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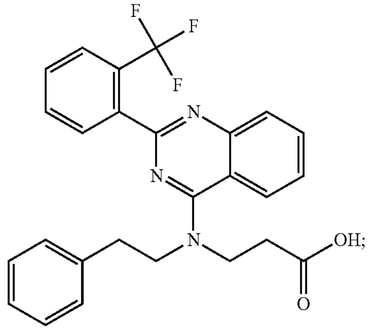
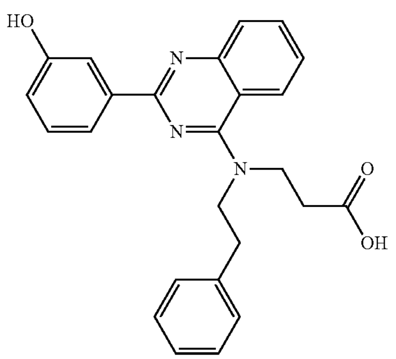


and



5. The method of treating or ameliorating a proliferative disease in a patient in need thereof comprising administering a bifunctional compound comprising the ALC1i according to claim **1** and a compound which recruits E3 ubiquitin ligase to ALC1 (E3 recruiter), wherein the ALC1i and the E3 recruiter are covalently linked.

6. The method of treating a proliferative disease in a patient in need thereof comprising administering a pharmaceutical composition comprising the ALC1i according to claim **1** or the bifunctional compound of claim **5** and a pharmaceutically acceptable excipient.

7. The method according to claim **1**, wherein the treating and ameliorating the proliferative disease comprises the administration of said ALC1i and the administration of a Poly(ADP-ribose)-Polymerase inhibitor (PARPi).

8. (canceled)

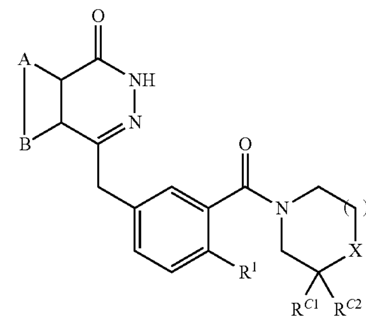
9. The method according to claim **7**, wherein the PARPi lowers PARP activity and/or inhibits PARP1, PARP2 and/or PARP3, wherein preferably the PARPi

(i) that lowers PARP activity is selected from small interfering RNA, and

(ii) that inhibits PARP1 is selected from the group consisting of a compound of

(a) formula (II)

(II)



and isomers, salts, solvates, chemically protected forms, and prodrugs thereof wherein:

A and B together represent an optionally substituted, fused aromatic ring;

X is NR^X or CR^XR^Y;

if $X{=}NR^X$ then n is 1 or 2 and if $X{=}CR^XR^Y$ then n is 1;

R^X is selected from the group consisting of H, optionally substituted C_{1-20} alkyl, C_{5-20} aryl, C_{3-20} heterocyclyl, amido, thioamido, ester, acyl, and sulfonyl groups;

R^Y is selected from H, hydroxy, amino;

or R^X and R^Y may together form a spiro-C_{3-7} cycloalkyl or heterocyclyl group;

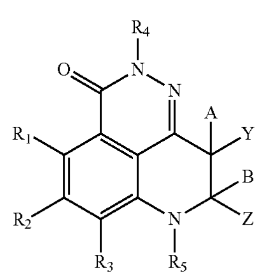
R^{C1} and R^{C2} are independently selected from the group consisting of hydrogen and C_{1-4} alkyl or when X is CR^XR^Y, R^{C1}, R^{C2}, R^X and R^Y, together with the carbon atoms to which they are attached, may form an optionally substituted fused aromatic ring; and

R^1 is selected from H and halo;

and

(b) formula (III)

(III)



and isomers, salts, solvates, chemically protected forms, and prodrugs thereof wherein:

Y and Z are each independently selected from the group consisting of

1. an aryl group optionally substituted with 1, 2, or 3 R_6;
2. a heteroaryl group optionally substituted with 1, 2, or 3 R_6;

3. a substituent independently selected from the group consisting of hydrogen, alkenyl, alkoxy, alkoxyalkyl, alkoxycarbonyl, alkoxycarbonylalkyl, alkyl, alkynyl, arylalkyl, cycloalkyl, cycloalkylalkyl, haloalkyl, hydroxyalkylene, oxo, heterocycloalkyl, heterocycloalkylalkyl, alkylcarbonyl, arylcarbonyl, heteroarylcarbonyl alkylsulfonyl, arylsulfonyl, heteroarylsulfonyl, (R_AR_B)alkylene, (NR_AR_B)carbonyl, (NR_AR_B)carbonylalkylene, (NR_AR_B)sulfonyl, and (R_AR_B)sulfonylalkylene;

wherein each R_6 is selected from OH, NO_2, CN, Br, Cl, F, I, C_{1-6}-alkyl, C_{3-8}-cycloalkyl, C_{2-8}-heterocycloalkyl; C_{2-6}-alkenyl, alkoxy, alkoxyalkyl, alkoxycarbonyl, alkoxycarbonylalkyl, C_{2-6}-alkynyl, aryl, arylalkyl, C_{3-8}-cycloalkylalkyl, haloalkoxy, haloalkyl, hydroxyalkylene, oxo, heteroaryl, heteroarylalkoxy, heteroaryloxy, heteroarylthio, heteroarylalkylthio, heterocycloalkoxy, C_{2-8}-heterocycloalkylthio, heterocyclooxy, heterocyclothio, NR_AR_B, $(R_AR_B)C_{1-6}$-alkylene, (NR_AR_B)carbonyl, (R_AR_B)carbonylalkylene, (NR_AR_B)sulfonyl, and (NR_AR_B)sulfonylalkylene;

R_1, R_2, and R_3 are each independently selected from the group consisting of hydrogen, halogen, alkenyl, alkoxy, alkoxycarbonyl, alkyl, cycloalkyl, alkynyl, cyano, haloalkoxy, haloalkyl, hydroxyl, hydroxyalkylene, nitro, NR_AR_B, NR_AR_B alkylene, and (R_AR_B)carbonyl;

A and B are each independently selected from hydrogen, Br, Cl, F, I, OH, C_{1-6}-alkyl, C_{3-8}-cycloalkyl, alkoxy, alkoxyalkyl wherein C_{1-6}-alkyl, C_{3-8}-cycloalkyl, alkoxy, alkoxyalkyl are optionally substituted with at least one substituent selected from OH, NO_2, CN, Br, Cl, F, I, C_{1-6}-alkyl, and C_{3-8}-cycloalkyl, wherein B is not OH;

R_A, and R_B are independently selected from the group consisting of hydrogen, alkyl, cycloalkyl, and alkylcarbonyl; or R_A and R_B taken together with the atom to which they are attached form a 3-10 membered heterocycle ring optionally having one to three heteroatoms or hetero functionalities selected from the group consisting of —O—, —NH, —N(C_{1-6}-alkyl)-, —NCO(C_{1-6}-alkyl)-, —N(aryl)-, —N(aryl-C_{1-6}-alkyl-), —N(substituted-aryl-C_{1-6}-alkyl-)-, —N(heteroaryl)-, —N(heteroaryl-C_1-C_6-alkyl-)-, —N(substituted-heteroaryl-C_{1-6} alkyl-)-, and —S— or S(O)q-, wherein q is 1 or 2 and the 3-10 membered heterocycle ring is optionally substituted with one or more substituents;

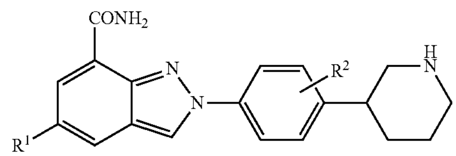
R_4 and R_5 are each independently selected from the group consisting of hydrogen, alkyl, cycloalkyl, alkoxyalkyl, haloalkyl, hydroxyalkylene, and (NR_AR_B)alkylene;

and isomers, salts, solvates, chemically protected forms, and prodrugs thereof;

(iii) that inhibits PARP1 and PARP2 is selected from the group consisting of a compound of

(a) formula (IV)

(IV)



and isomers, salts, solvates, chemically protected forms, and prodrugs thereof wherein:

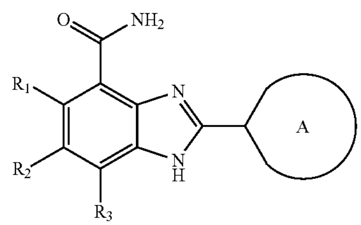
R_1 is hydrogen or fluorine; and

R_2 is hydrogen or fluorine;

and

(b) formula (V)

(V)



and isomers, salts, solvates, chemically protected forms, and prodrugs thereof wherein:

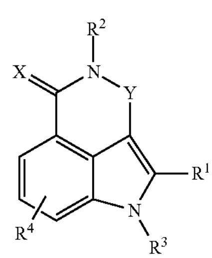
R_1, R_2, and R_3 are independently selected from the group consisting of hydrogen, alkenyl, alkoxy, alkoxycarbonyl, alkyl, alkynyl, cyano, haloalkoxy, haloalkyl, halogen, hydroxy, hydroxyalkyl, nitro, NR_AR_B, and (NR_AR_B)carbonyl;

A is a nonaromatic 4, 5, 6, 7, or 8-membered ring that contains 1 or 2 nitrogen atoms and, optionally, one sulfur or oxygen atom, wherein the nonaromatic ring is optionally substituted with 1, 2, or 3 substituents selected from the group consisting of alkenyl, alkoxy, alkoxyalkyl, alkoxycarbonyl, alkoxycarbonylalkyl, alkyl, alkynyl, aryl, arylalkyl, cycloalkyl, cycloalkylalkyl, cyano, haloalkoxy, haloalkyl, halogen, heterocycle, heterocyclealkyl, heteroaryl, heteroarylalkyl, hydroxy, hydroxyalkyl, nitro, NR_CR_D, (NR_CR_D)alkyl, (NR_CR_D)carbonyl, (NR_CR_D)carbonylalkyl, and (NR_CR_D)sulfonyl; and

R_A, R_B, R_C, and R_D are independently selected from the group consisting of hydrogen, alkyl, and alkycarbonyl

(iii) that inhibits PARP1, PARP2 and PARP3 is a compound of formula (VI)

(VI)



and isomers, salts, solvates, chemically protected forms, and prodrugs thereof wherein:

R_1 is: H; halogen; cyano; an optionally substituted alkyl, alkenyl, alkynyl, cycloalkyl, heterocycloalkyl, aryl, or heteroaryl group (e.g., unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, and amino, alkoxy, alkyl, and aryl groups unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, carboxy, and optionally substituted amino and ether groups (such as O-aryl)); or —C(O)—R_{10}, where R_{10} is: H; an optionally substituted alkyl, alkenyl, alkynyl, cycloalkyl, heterocycloalkyl, aryl, or heteroaryl group (e.g., unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, amino, and alkyl and aryl groups unsubstituted or substituted with one or more substituents selected from halo, hydroxy, nitro, and amino); or OR_{100} or $NR_{100}R_{110}$, where R_{100} and R_{110} are each independently H or an optionally substituted alkyl, alkenyl, alkynyl, cycloalkyl, heterocycloalkyl, aryl, or heteroaryl group (e.g., unsubstituted or substituted with one or more substituents selected from alkyl, alkenyl, alkynyl, cycloalkyl, heterocycloalkyl, aryl, and heteroaryl groups unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, amino, and alkyl and aryl groups unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, and optionally substituted amino groups);

R_2 is H or alkyl;

R_3 is H or alkyl;

R_4 is H, halogen or alkyl;

X is O or S;

Y is $(CR_5R_6)(CR_7R_8)$, or N—$C(R_5)$, where:

n is 0 or 1;

R_5 and R_6 are each independently H or an optionally substituted alkyl, alkenyl, alkynyl, cycloalkyl, heterocycloalkyl, aryl, or heteroaryl group (e.g., unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, amino, and lower alkyl, lower alkoxy, or aryl groups unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, and amino); and

R_7 and R_8 are each independently H or an optionally substituted alkyl, alkenyl, alkynyl, cycloalkyl, heterocycloalkyl, aryl, or heteroaryl group (e.g., unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, amino, and lower alkyl, lower alkoxy, and aryl groups unsubstituted or substituted with one or more substituents selected from halogen, hydroxy, nitro, and amino);

where when R_1, R_4, R_5, R_6, and R_7 are each H, R_8 is not unsubstituted phenyl.

10. The method according to claim **7**, wherein the PARPi is selected from the group consisting of Olaparib, Talazoparib, Niraparib, Rucaparib, and Veliparib, in particular of Veliparib, Olaparib, and Talazoparib.

11. The method according to claim **7**, wherein the proliferative disease is selected from a BRCA-1 and/or BRCA-2-deficient tumor, and a tumor in which expression of PARP-1, PARP-2, PARP-3 and/or ALC1 is increased in comparison to non-tumor cells, and/or wherein the proliferative disease is selected from hepatocellular carcinoma, breast cancer, ovarian cancer, prostate cancer, colorectal cancer.

12. The method according to claim **7**, wherein the PARPi and ALC1i or bifunctional compound are administered concomitantly or separately.

13. A kit of parts comprising separately packaged a PARPi and an ALC1i as specified in claim **1** or a composition comprising a PARPi and an ALC1i as specified in claim **1**, and instructions for use to treat or ameliorate a proliferative disease.

14. The method according to claim **1**, wherein the proliferative disease is selected from a BRCA1 and/or 2-deficient tumor, and a tumor in which expression of PARP1, PARP2, PARP3 and/or ALC1 is increased in comparison to non-tumor cells.

15. A method of treating or ameliorating a proliferative disease in a patient in need thereof comprising administering a bifunctional compound comprising the ALC1i according to claim **1** and a compound which recruits E3 ubiquitin ligase to ALC1 (E3 recruiter), wherein the ALC1i and the E3 recruiter are covalently linked through a linker.

16. A method of treating a proliferative disease in a patient in need thereof comprising administering a pharmaceutical composition comprising the bifunctional compound of claim **5** and a pharmaceutically acceptable excipient.

17. A method of treating a proliferative disease in a patient in need thereof comprising administering a pharmaceutical composition comprising the bifunctional compound of claim **15** and a pharmaceutically acceptable excipient.

18. The method according to claim **9**, wherein the PARPi lowers PARP2 activity and/or inhibits PARP2 on chromatin.

19. A kit of parts comprising separately packaged a PARPi and a bifunctional compound according to claim **5** or a composition comprising a PARPi and a bifunctional compound according to claim **5**, and instructions for use to treat or ameliorate a proliferative disease.

* * * * *